

## Unit 8

# SUPERPOSITION

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55 items: 22 from official papers (147 marks), 33 from the booklet

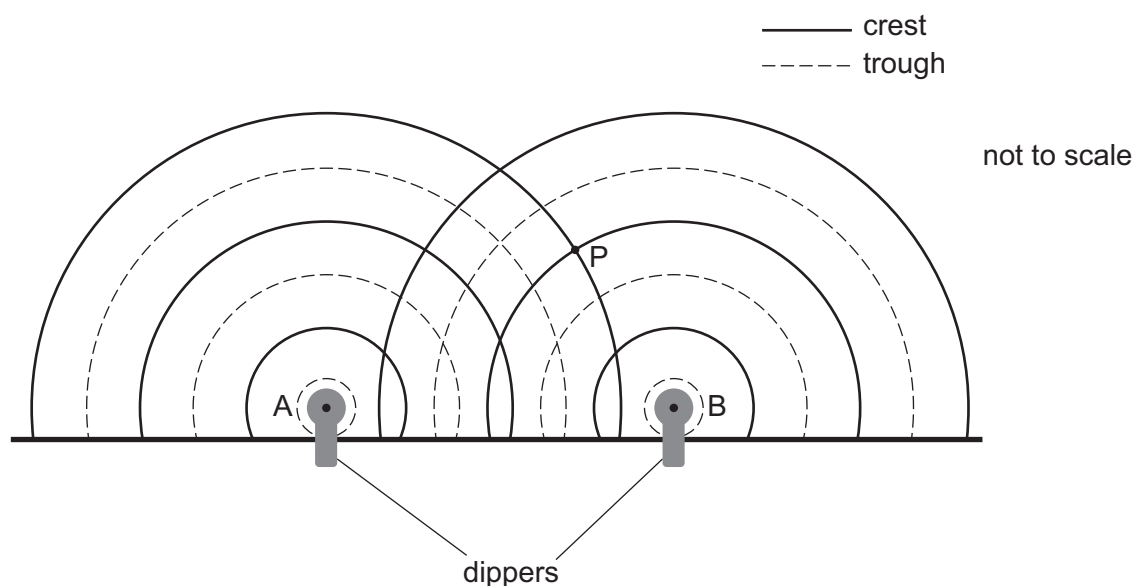
### **Syllabus sections in this unit**

- 8.1 Stationary waves
- 8.2 Diffraction
- 8.3 Interference
- 8.4 The diffraction grating

## Unit 8: Superposition

### 1. M/J 26/P22/Q3/b(ii),c(ii)

- 3 Two oscillating dippers are used to create progressive transverse water waves on the surface of a lake. The waves are shown from above in Figure 3.1.



**Figure 3.1**

Wave crests are represented by a solid line. Wave troughs are represented by a dashed line. The waves superpose and an interference pattern is formed on the surface of the water.

The dippers oscillate in phase with each other and with a frequency of 3.0 Hz.

- (ii) Points A and B are at the centre of the dippers. Point P is shown on Figure 3.1. The distance from A to P is 3.9 m. The distance from B to P is 2.4 m.

Determine the speed of the waves.

speed =  $\text{ms}^{-1}$  [3]

- (c) A ball of mass 0.44 kg is placed at P, where it floats on the surface of the lake. The amplitude of the vertical oscillation of the ball is 15 cm.

- (ii) The frequency of oscillation of the dippers is increased to 4.0 Hz without changing their amplitude of oscillation. The ball remains at the same position on the surface of the lake.

State and explain whether the amplitude of the oscillation of the ball increases, decreases or remains the same.

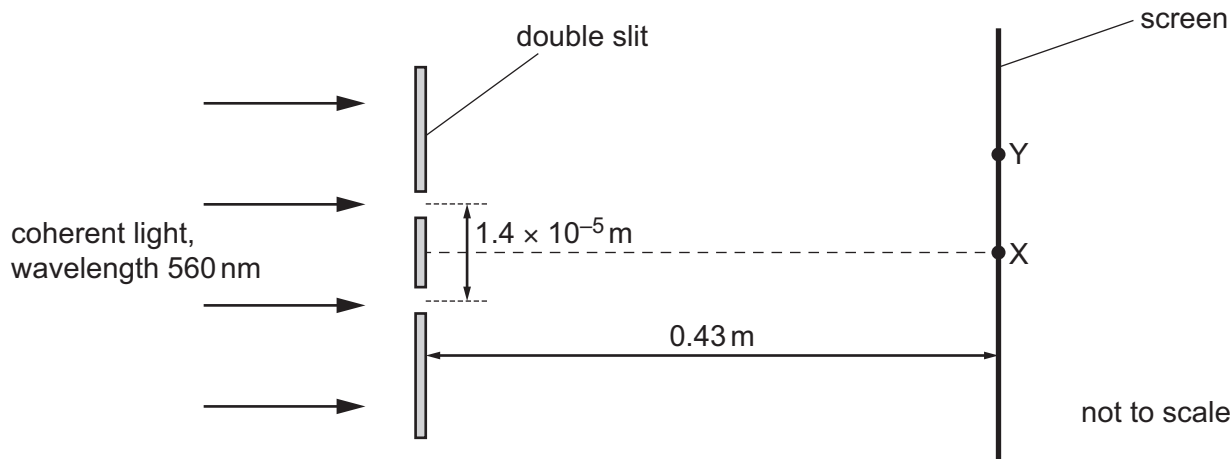
[2]

**2. M/J 26/P23/Q3/a,b,c**

- 3 (a)** State the principle of superposition.

[2]

- (b)** Parallel rays of coherent light of wavelength 560 nm are incident normally in free space on two narrow slits as shown in Figure 3.1.

**Figure 3.1**

Light passing through the double slit reaches a screen that is parallel to the plane of the slits. The distance between the centres of the slits is  $1.4 \times 10^{-5} \text{ m}$ . The distance between the slits and the screen is 0.43 m.

- (i)** State what is meant by coherent.

[1]

- (ii)** State what happens to the light at the point where it passes through the slits.

[1]

- (iii)** At point X on the screen, there is a central bright fringe.

Explain how this bright fringe is formed.

[3]

- (iv)** At point Y on the screen, the dark fringe closest to point X is formed.

Calculate the distance XY.

distance =

m [3]

- (c)** On the screen in Figure 3.1:

- (i)** draw a cross (×) labelled with the letter B at the position of another bright fringe

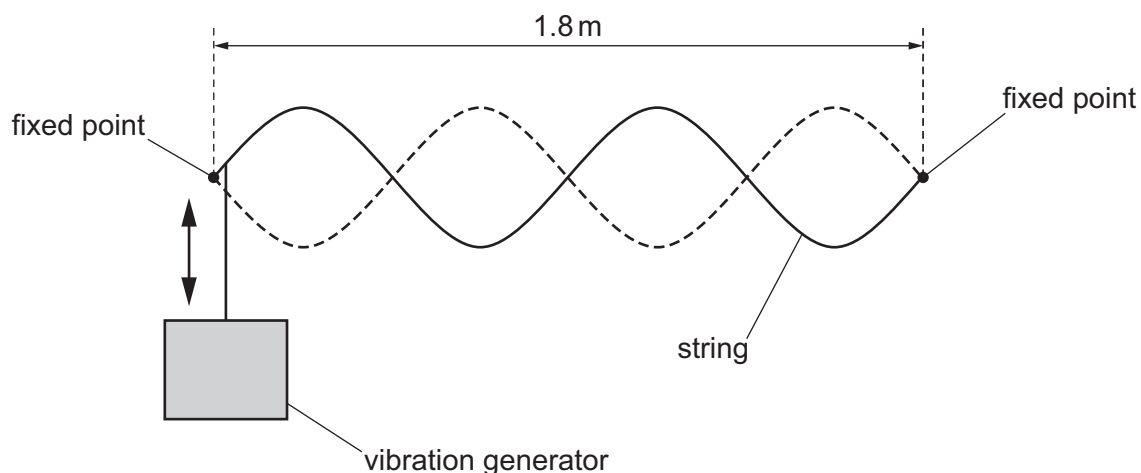
[1]

- (ii)** draw a cross (×) labelled with the letter D at the position of another dark fringe.

[1]

**3. M/J 26/P24/Q3/a,b,c,d**

- 3** A stretched string is attached to two fixed points and a vibration generator as shown in Figure 3.1.

**Figure 3.1**

The vibration generator oscillates vertically and a stationary wave is formed on the string.

- (a) Explain how the stationary wave is formed on the string.

[2]

- (b) On Figure 3.1, draw **one** cross (×) at the position of a node.

[1]

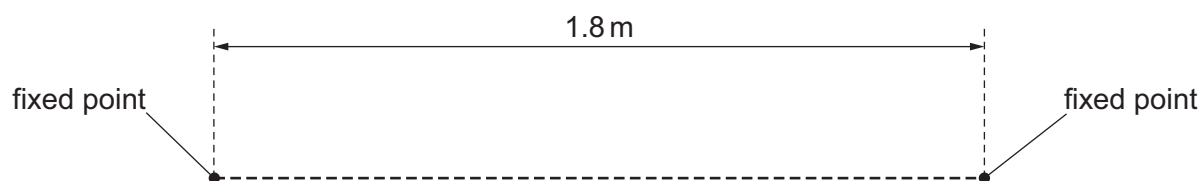
- (c) The distance between the fixed points is 1.8 m. The vibration generator oscillates at a frequency of 26 Hz.

Calculate the speed of progressive waves on the string.

speed =  $\text{ms}^{-1}$  [3]

- (d) The frequency of the vibration generator is now halved and a new stationary wave is formed on the string. The speed of progressive waves on the string is unchanged.

On Figure 3.2, draw the string at an instant when it has maximum displacement.

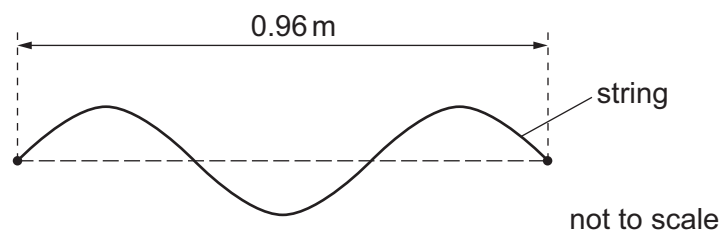
**Figure 3.2**

[2]

**4. MAR 26/P22/Q5/b,c**

- (b) A stretched string is fixed at both ends and made to vibrate with a frequency of 25 Hz. A stationary wave is formed on the string. The distance between the fixed ends of the string is 0.96 m.

Figure 5.1 shows the string at one instant.

**Figure 5.1**

(i) On Figure 5.1, draw a cross (×) at the position of an antinode. [1]

(ii) Show that the speed of a progressive wave on the string is  $16 \text{ ms}^{-1}$ . [3]

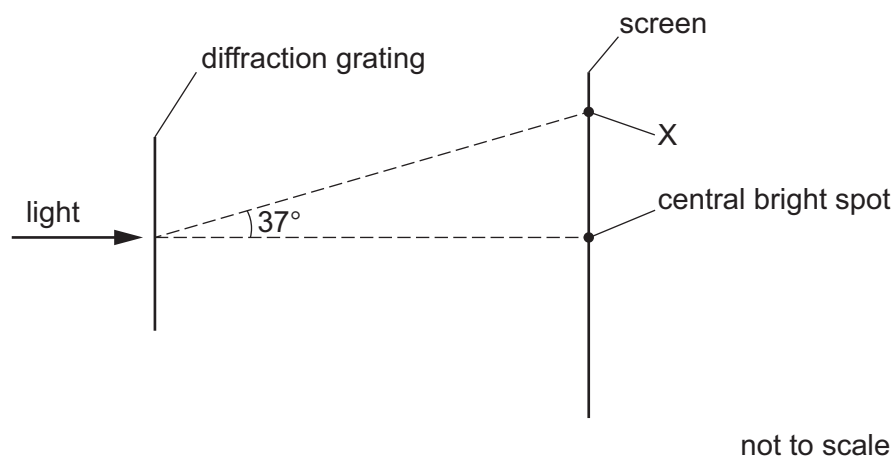
(iii) The frequency of the vibrations is increased to 30 Hz. The speed of the wave on the string does not change.

State and explain whether a stationary wave is formed.

[2]

(c) A beam of light of wavelength  $4.5 \times 10^{-7} \text{ m}$  is incident normally on a diffraction grating with  $6.7 \times 10^5$  lines per metre. The light passes through the grating and forms a pattern of bright spots on a screen that is parallel to the grating.

Figure 5.2 shows two of the bright spots in the pattern.

**Figure 5.2**

The angle formed at the diffraction grating between the central bright spot and the bright spot at position X is  $37^\circ$ .

(i) Calculate the order  $n$  of the bright spot formed at X.

$n =$  [3]

(ii) The diffraction grating is replaced by a grating with fewer lines per metre.

State whether the distance between adjacent bright spots on the screen increases, decreases or remains the same.

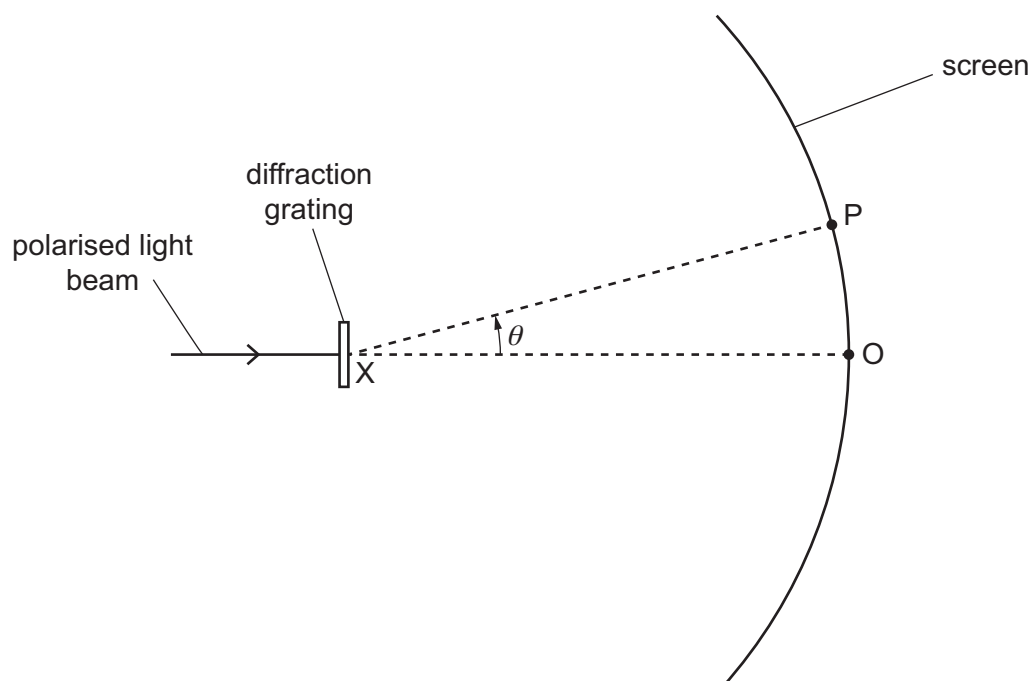
[1]

**5. O/N 25/P21/Q4/a,b**

- 4 (a)** State what is meant by diffraction of a wave.

[2]

- (b)** A beam of vertically polarised light of wavelength 540 nm is incident normally on a diffraction grating, as shown in Fig. 4.1.



**Fig. 4.1** (not to scale)

The diffraction grating has a line spacing of  $5.0 \times 10^{-6}$  m.

The light transmitted by the diffraction grating illuminates a circular screen. The diffraction grating is at the centre X of the circle.

The central bright fringe is formed at point O on the screen and has intensity  $I_0$ .

P is a point on the screen where the line XP is at a variable angle  $\theta$  to the line XO. The intensity  $I$  of light on the screen at P varies with  $\theta$ .

- (i)** Show that the angle  $\theta$  at which the first-order bright fringe is formed is  $6.2^\circ$ .

[2]

- (ii)** Determine the value of  $\theta$  at which the second-order bright fringe is formed.

$\theta =$  ° [1]

- (iii)** On Fig. 4.2, sketch the variation of the intensity  $I$  with  $\theta$  for values of  $\theta$  from  $-15^\circ$  to  $+15^\circ$ .

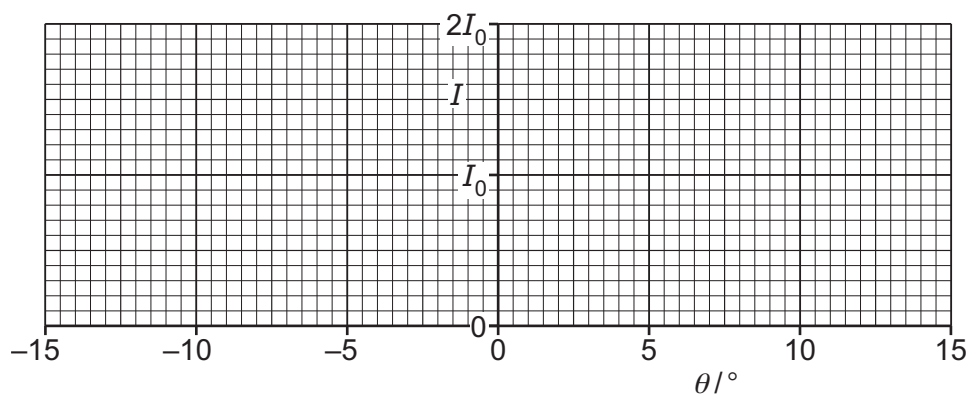


Fig. 4.2

[3]

## 6. O/N 25/P22/Q4/a,b,c

- 4 A laser emits visible light of a single frequency in a vacuum. The light is incident normally on a double slit and then forms a pattern of bright and dark fringes on a screen, as shown in Fig. 4.1.

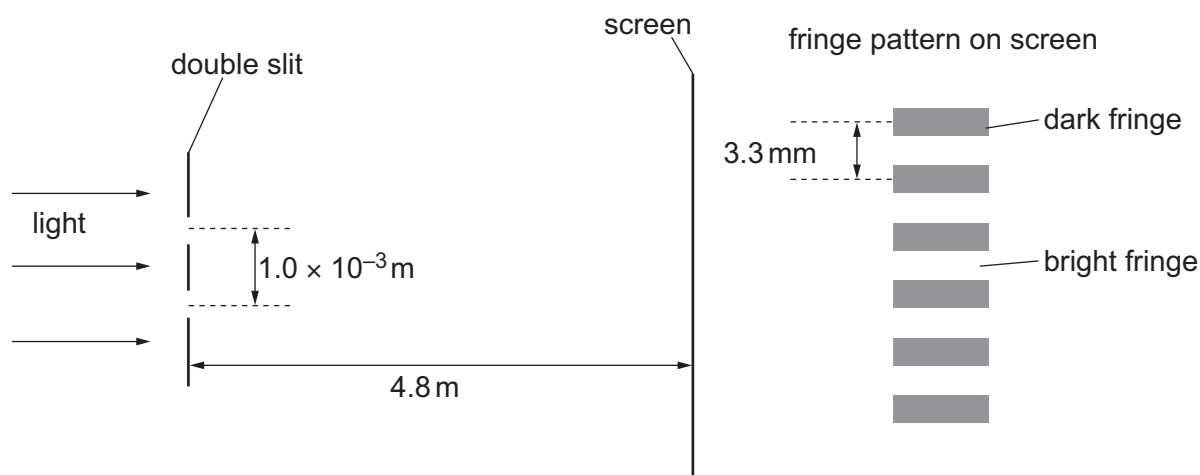


Fig. 4.1 (not to scale)

The separation of the slits is  $1.0 \times 10^{-3} \text{ m}$ . The distance from the slits to the screen is  $4.8 \text{ m}$ . The distance between the centres of adjacent dark fringes on the screen is  $3.3 \text{ mm}$ .

- (a) Explain how the pattern of bright and dark fringes is formed.

[3]

- (b) Calculate the frequency of the light emitted by the laser.

frequency = \_\_\_\_\_ Hz [4]

- (c) The double slit is removed. A second laser is placed beside the first laser. The second laser produces visible light of a different frequency from that of the first laser. The beams of light from the two lasers overlap on the screen.

Explain why a steady pattern of bright and dark fringes is **not** formed on the screen.

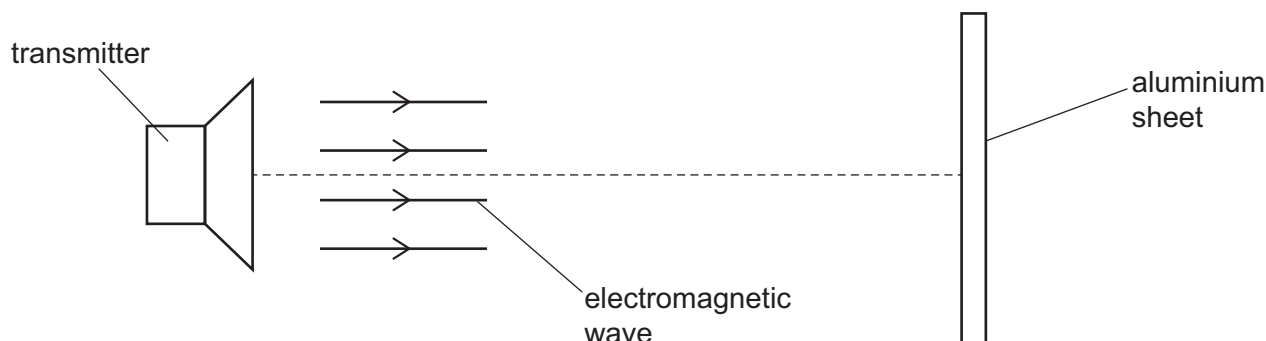
[1]

**7. O/N 25/P24/Q4/a,b(i,iv)**

- 4 (a) State the principle of superposition.

[2]

- (b) An electromagnetic wave of wavelength  $0.026\text{ m}$  in free space is incident normally on an aluminium sheet, as shown in Fig. 4.1.

**Fig. 4.1**

The wave reflects at the aluminium sheet and a stationary wave is formed in the region between the transmitter and the sheet.

- (i) Explain how the stationary wave, including its nodes and antinodes, is formed.

[3]

- (iv) Determine the distance between a node and an adjacent antinode.

distance =

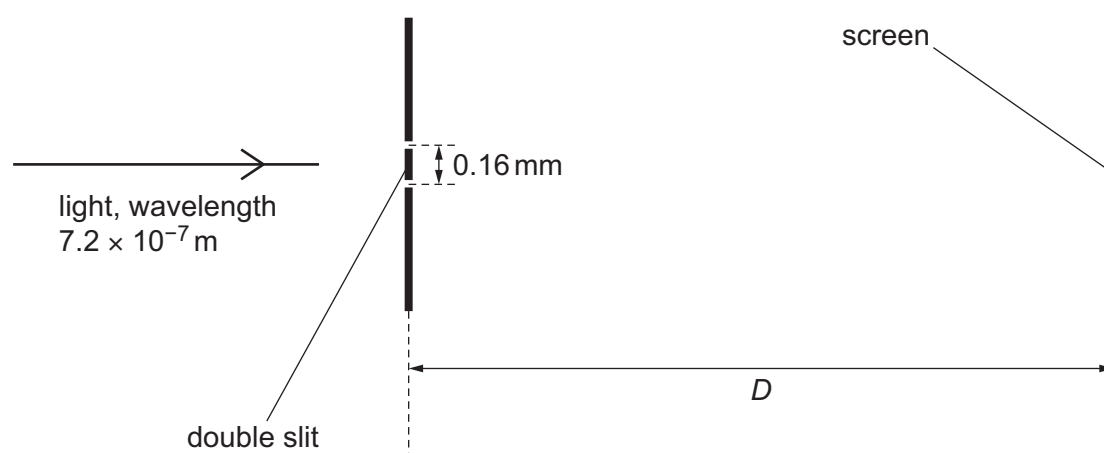
m [1]

**8. M/J 25/P21/Q4/a,b**

- 4 (a) State the principle of superposition.

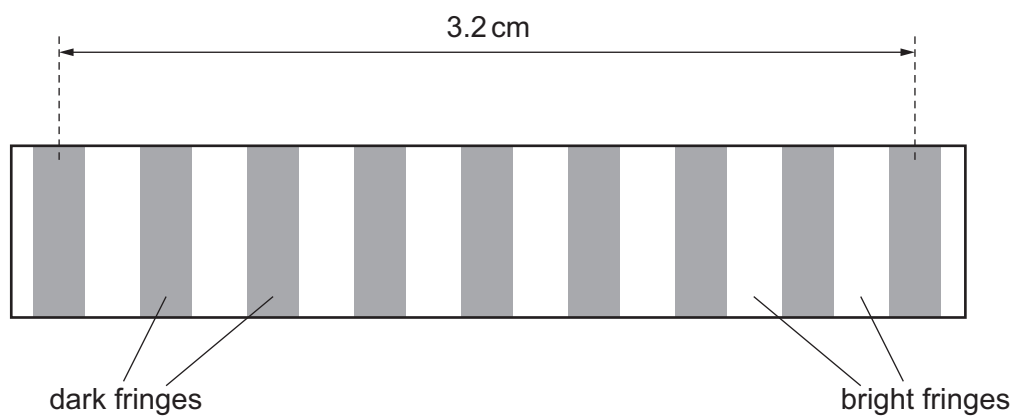
[2]

- (b) Light of wavelength  $7.2 \times 10^{-7}\text{ m}$  is incident normally on a double slit, as shown in Fig. 4.1.

**Fig. 4.1 (not to scale)**

A screen is at a distance  $D$  from the double slit. The double slit and the screen are parallel.

The separation of the slits in the double-slit arrangement is  $0.16\text{ mm}$ . The resulting interference pattern on the screen contains nine dark fringes, as shown in Fig. 4.2.

**Fig. 4.2** (not to scale)

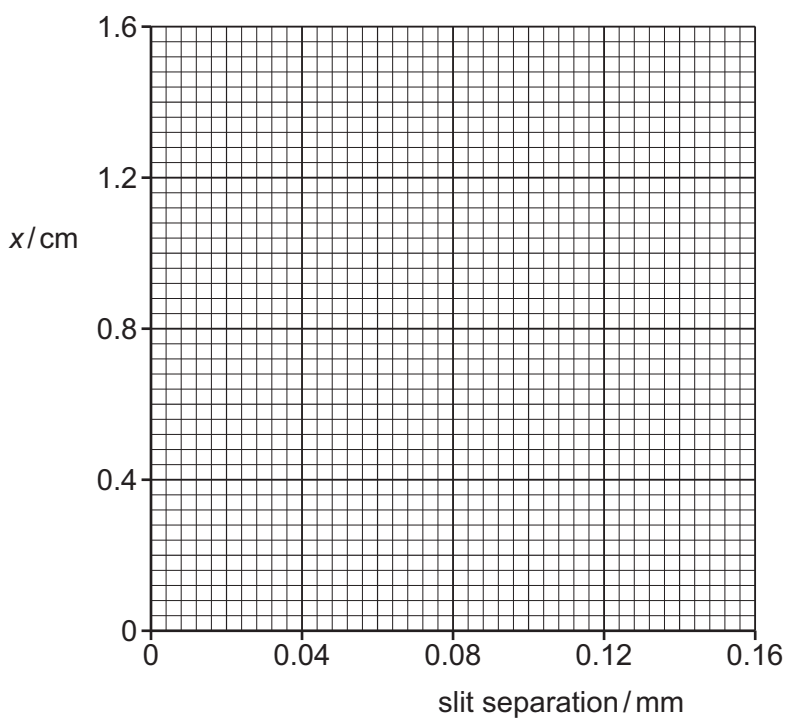
The distance between the centres of the first and ninth dark fringes is 3.2 cm.

- (i) Calculate  $D$ .

$$D = \quad \text{m [3]}$$

- (ii) The slit separation is now gradually decreased from 0.16 mm to 0.04 mm. The distance between the centres of adjacent dark fringes is  $x$ .

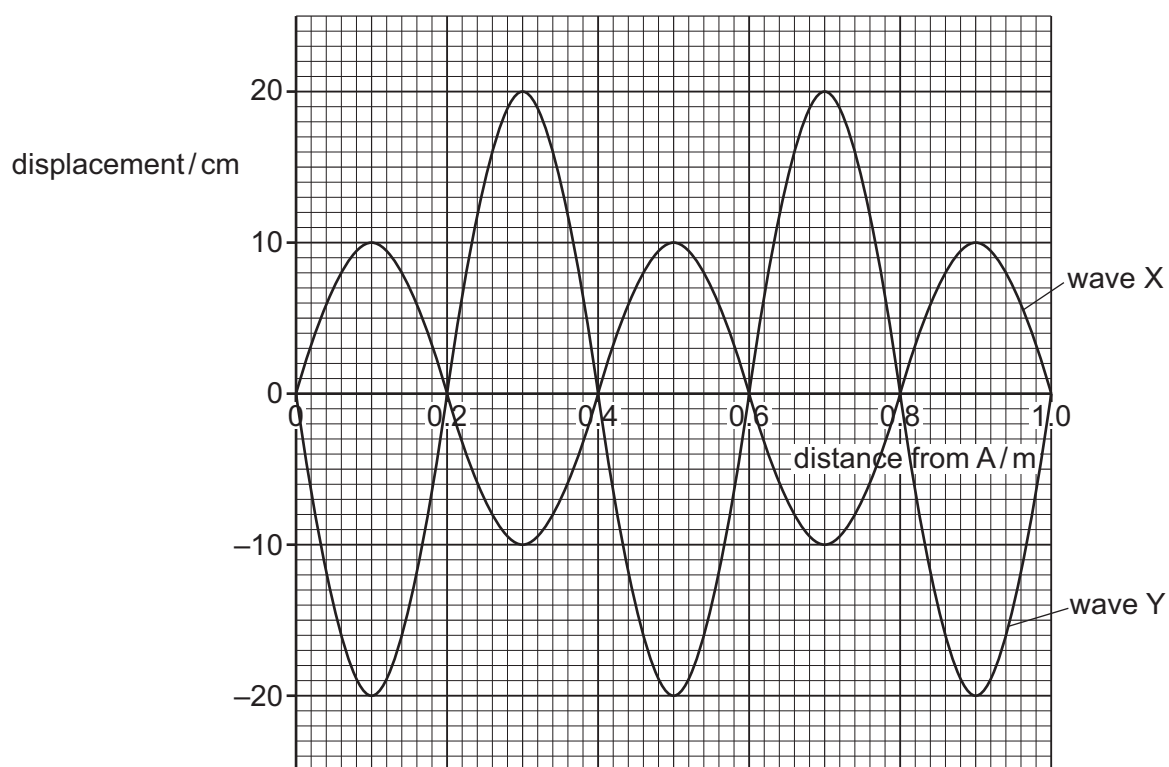
On Fig. 4.3, sketch the variation of  $x$  with slit separation.

**Fig. 4.3**

[3]

**9. M/J 25/P22/Q3/b(ii),c**

- 3 Two progressive water waves X and Y travel along a straight line from point A to point B. The variation of displacement of the waves with distance from A at an instant in time is shown in Fig. 3.1.

**Fig. 3.1**

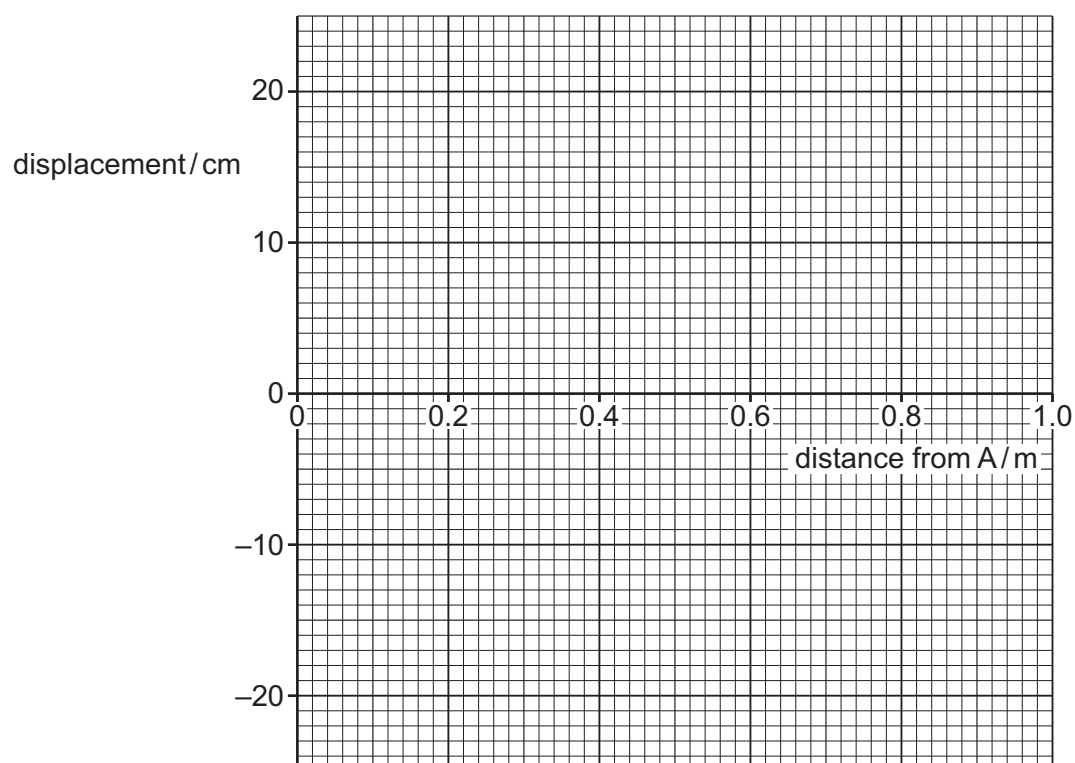
(b) Both waves have frequency 16 Hz.

(ii) State and explain whether X and Y are coherent.

[1]

(c) Wave X and wave Y superpose to form a resultant wave.

On Fig. 3.2, sketch the variation of displacement of the resultant wave with distance from A at the instant of time shown in Fig. 3.1.



**Fig. 3.2**

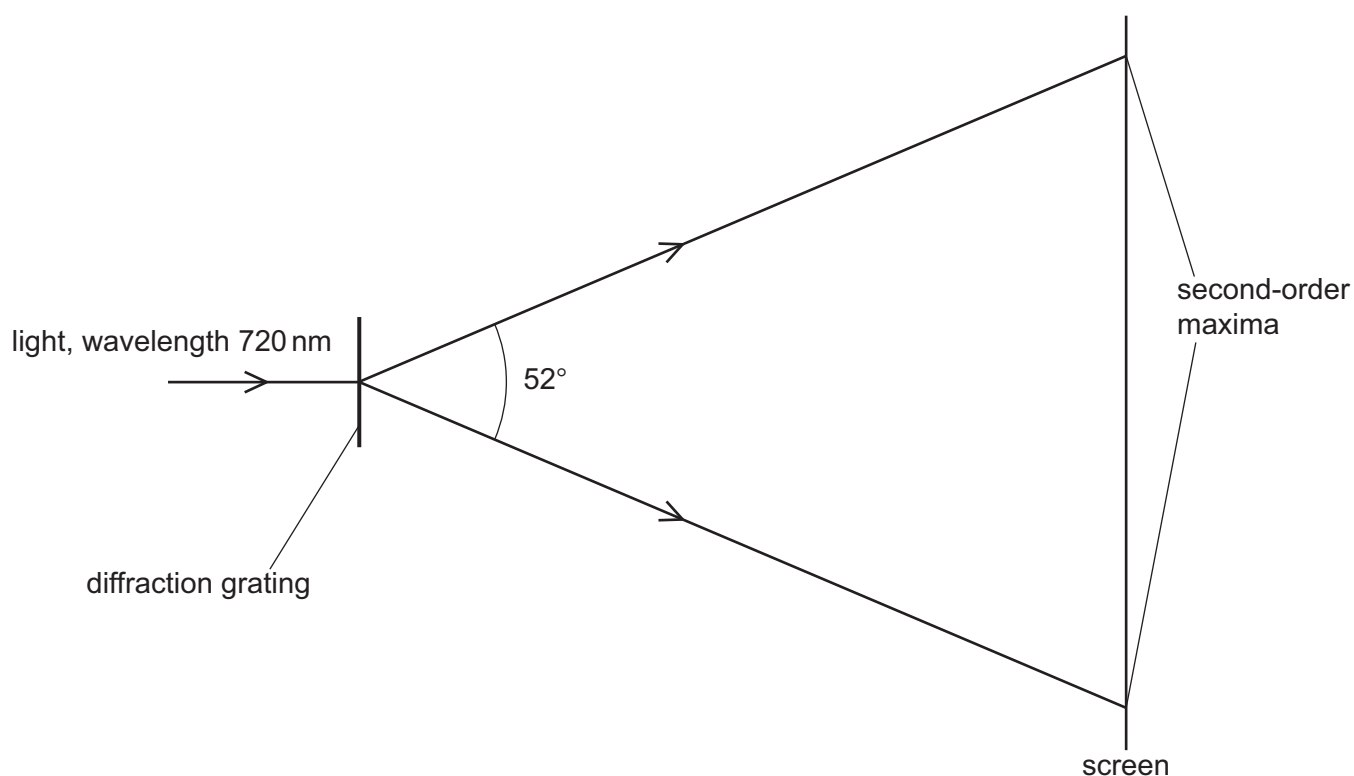
[2]

**10. M/J 25/P23/Q6/a,b(ii,iii)**

- 6 (a)** State what is meant by diffraction.

[1]

- (b)** Light of wavelength 720 nm in a vacuum is incident normally on a diffraction grating as shown in Fig. 6.1.

**Fig. 6.1** (not to scale)

A screen is parallel to the grating. An interference pattern is seen on the screen and the angle between the **second**-order maxima is  $52^\circ$ .

- (ii)** Calculate the number of lines per unit length in the diffraction grating.

$$\text{number per unit length} = \quad \text{m}^{-1} \quad [3]$$

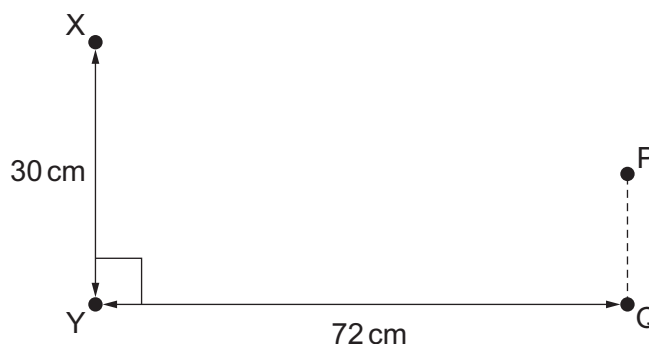
- (iii)** The light in Fig. 6.1 is now replaced with light of a different wavelength  $\lambda$ . It is observed that the third-order maxima of this light are at the same positions as the second-order maxima of the light in Fig. 6.1.

Calculate, in nm, the wavelength  $\lambda$ .

$$\lambda = \quad \text{nm} \quad [2]$$

**11. M/J 25/P24/Q4/b(ii,iii,iv)**

- (b) Two identical microwave sources X and Y emit waves in phase. The sources are separated by a distance of 30 cm, as shown in Fig. 4.1.



**Fig. 4.1** (not to scale)

The intensity of the microwaves is to be investigated at points P and Q.

Line PQ is parallel to line XY. Distance XP is equal to distance YP. Distance YQ is 72 cm and angle XYQ is  $90^\circ$ .

The wavelength of the microwaves is 4.0 cm.

- (ii) Show that the difference between the path lengths XQ and YQ is 6 cm.

[1]

- (iii) State and explain what may be deduced about the intensity of the microwaves at point Q.

[3]

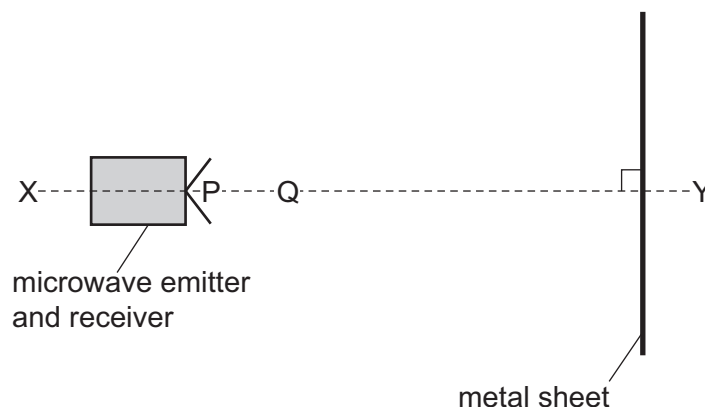
- (iv) A microwave detector is positioned at P and connected to a cathode-ray oscilloscope (CRO). The controls of the CRO are adjusted so that a waveform is shown on the screen.

Describe the changes to the amplitude of the waveform as the detector is moved from P to Q.

[2]

**12. MAR 25/P22/Q4/a,b**

- 4** A device containing a microwave emitter and receiver is placed in front of a large metal sheet in a vacuum as shown in Fig. 4.1.



**Fig. 4.1** (not to scale)

The line XY is perpendicular to the metal sheet. The device emits microwaves of frequency 6.3 GHz.

- (a)** When the device is at position P, a stationary wave is formed between the device and the sheet.

Explain how the stationary wave, including the nodes and the antinodes, is formed.

[4]

- (b) (i)** Calculate the wavelength of the microwaves.

wavelength = m [2]

- (ii)** At point P the receiver detects a maximum amplitude of the stationary wave.

The device is moved slowly from point P along the line XY and the receiver detects a series of minimum and maximum amplitudes. The first time a minimum amplitude is detected by the receiver is when the device is at point Q.

Determine the distance between P and Q.

distance = m [1]

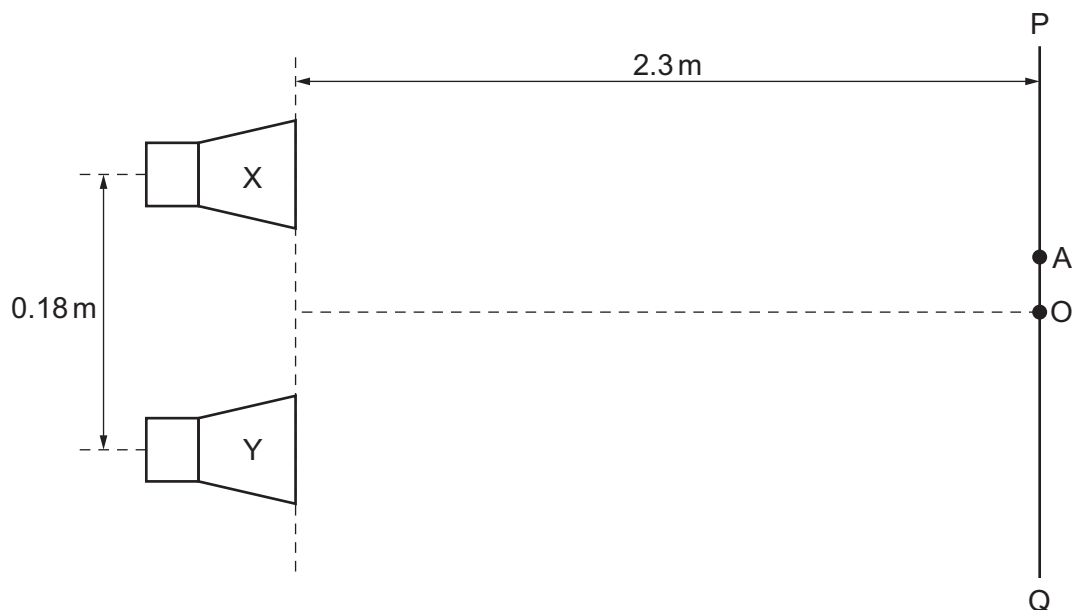
- (iii)** The intensity of the microwaves emitted by the device is increased. The frequency of the microwaves is unchanged. The device is moved slowly along the line XY from point Q until the next maximum amplitude is detected at point R.

State and explain whether the distance QR is greater than, less than or the same as distance PQ.

[1]

**13. O/N 24/P21/Q6/a,b(ii,iii)**

- 6** Two coherent sources X and Y of microwaves of frequency  $2.5 \times 10^{10}$  Hz are a distance of 0.18 m apart in a vacuum, as shown in Fig. 6.1.



**Fig. 6.1** (not to scale)

There is a phase difference of  $90^\circ$  between the waves emitted at the two sources.

A microwave detector moves along the line PQ, which is parallel to the line joining the two sources and 2.3 m away from it.

Point O is on the line PQ at a position that is equidistant from the two sources.

Point A is the position on line PQ where the intensity of the microwaves is the greatest.

- (a) (i)** Explain why the position of greatest intensity is **not** at point O.

[2]

- (ii)** On Fig. 6.1, draw a cross (×) to show the position of the point on line PQ where the intensity minimum that is the closest to point O occurs. Label this point B. [2]

- (ii)** For point A on line PQ, determine the difference in the distances  $\Delta x$  travelled by the microwaves from X and the microwaves from Y.

$$\Delta x = \quad \text{m} \quad [1]$$

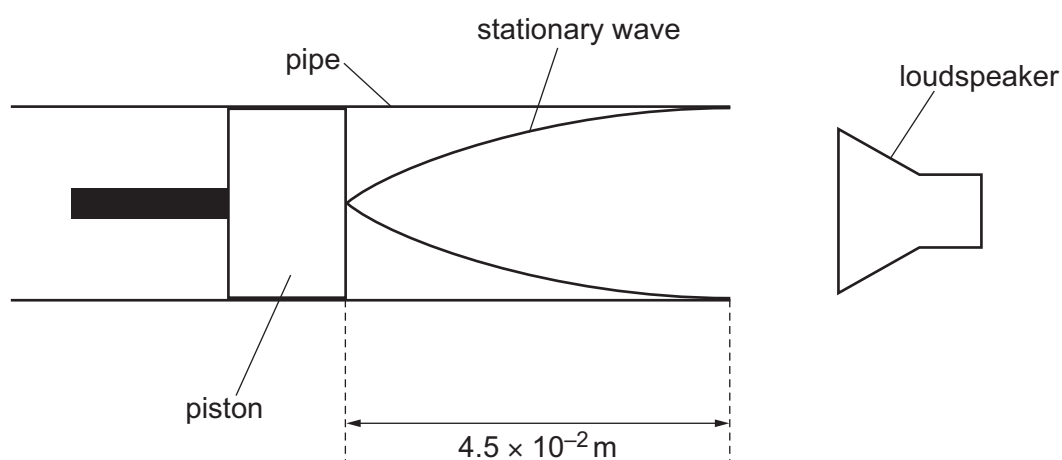
- (iii)** Use the formula for the double-slit interference of light to calculate the distance between adjacent intensity maxima on line PQ.

$$\text{distance} = \quad \text{m} \quad [2]$$

**14. O/N 24/P22/Q4/b**

- (b) A pipe is open at one end and closed at the other with a piston. The piston can slide freely and is at a distance of  $4.5 \times 10^{-2} \text{ m}$  from the open end of the pipe.

A loudspeaker is positioned near the open end of the pipe and emits a sound wave of a single constant frequency. A stationary wave is formed in the pipe, as illustrated in Fig. 4.1.

**Fig. 4.1**

- (i) On Fig. 4.1, draw a letter A at the position of an antinode. [1]

- (ii) The speed of sound in air is  $340 \text{ m s}^{-1}$ .

Determine the frequency of the sound wave.

frequency = \_\_\_\_\_ Hz [3]

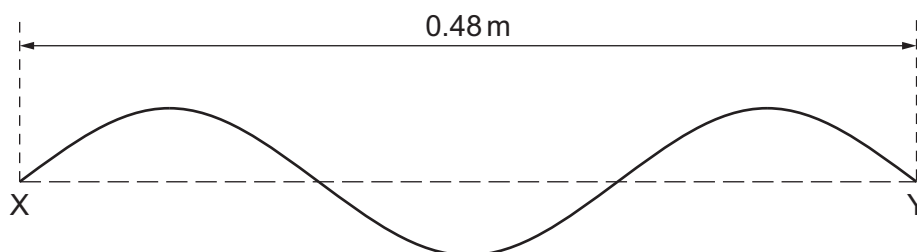
- (iii) The piston is moved to the left. The frequency of the sound wave emitted by the loudspeaker is then changed so that a stationary wave is formed with same number of antinodes as in Fig. 4.1.

State and explain the change that is made to the frequency of the sound wave.

[2]

**15. O/N 24/P23/Q5/a(i,ii)**

- 5 (a) A stationary wave is formed on a string XY that has a length of 0.48 m. Fig. 5.1 shows the string at one instant in time.

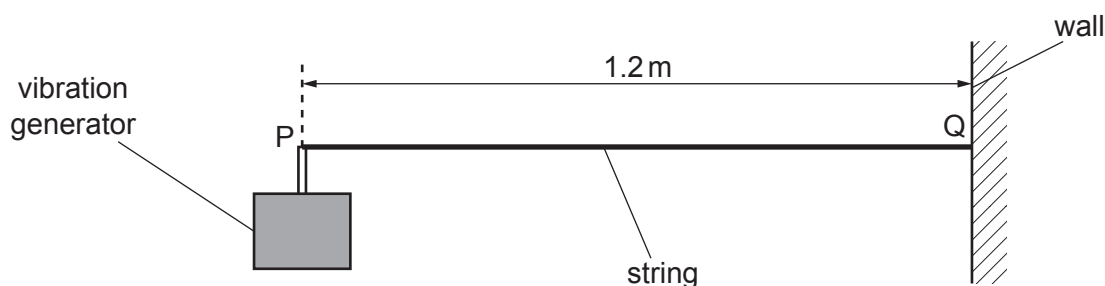
**Fig. 5.1**

The speed of the wave on the string is  $1400 \text{ m s}^{-1}$ .

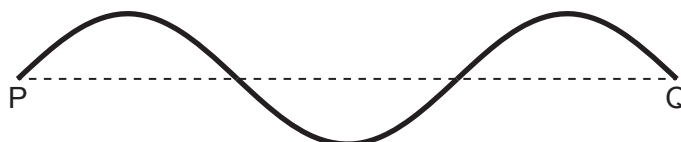
- (i) On Fig. 5.1, draw a cross (×) at **one** position that is a node and another cross at **one** position that is an antinode. Label the node N and the antinode A. [1]
- (ii) Show that the wavelength of the wave produced is 0.32 m. Explain your reasoning. [1]

**16. M/J 24/P21/Q5/a,b,c,d**

- 5 A stretched string PQ has length 1.2 m. One end of the string is attached to a vibration generator and the other end is attached to a wall, as shown in Fig. 5.1.

**Fig. 5.1**

The vibration generator is switched on and a stationary wave is formed on the string. The string is shown at one instant of time in Fig. 5.2.

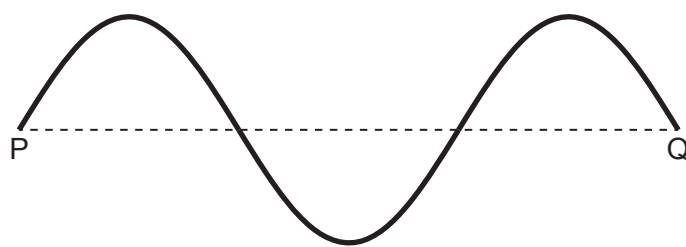
**Fig. 5.2**

- (a) Explain how a stationary wave is formed between the vibration generator and the wall. [2]

- (b) Calculate the wavelength of the stationary wave shown in Fig. 5.2.

wavelength = \_\_\_\_\_ m [1]

- (c) Fig. 5.3 shows the stationary wave at time  $t = 0$  when all points on the wave are at their maximum displacements.

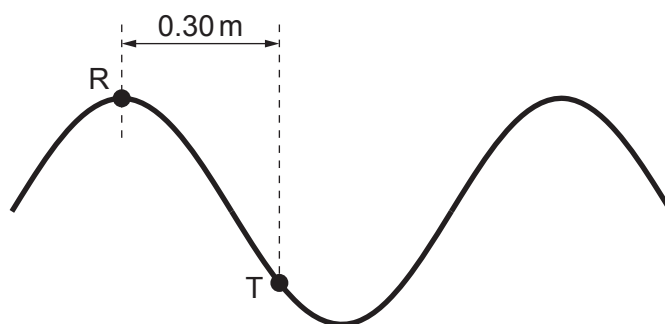
**Fig. 5.3**

The period of the wave is 0.16 s.

On Fig. 5.3, sketch the shape of the stationary wave at time  $t = 0.24$  s.

[2]

- (d) Points R and T on the string are a horizontal distance of 0.30 m apart and in the positions shown in Fig. 5.4.

**Fig. 5.4**

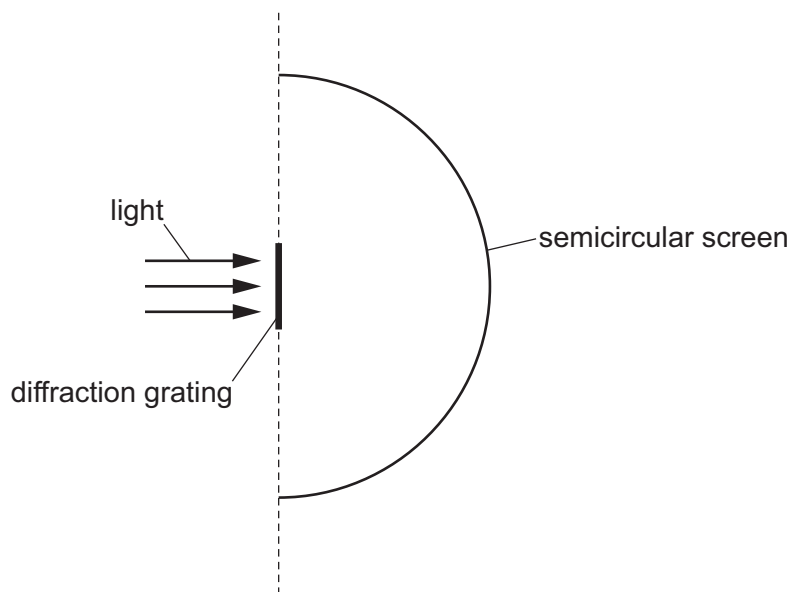
State the phase difference between the oscillations of points R and T.

phase difference =

° [1]

**17. M/J 24/P22/Q6/a,b(ii)**

- 6** Light of a single frequency is incident normally on a diffraction grating. An interference pattern of bright and dark fringes forms on the semicircular screen shown in Fig. 6.1.



**Fig. 6.1** (not to scale)

The light has wavelength 520 nm.

The separation of the lines in the grating is  $3.8 \times 10^{-6}$  m.

- (a)** Determine the total number of bright fringes formed on the screen.

number of bright fringes =

[3]



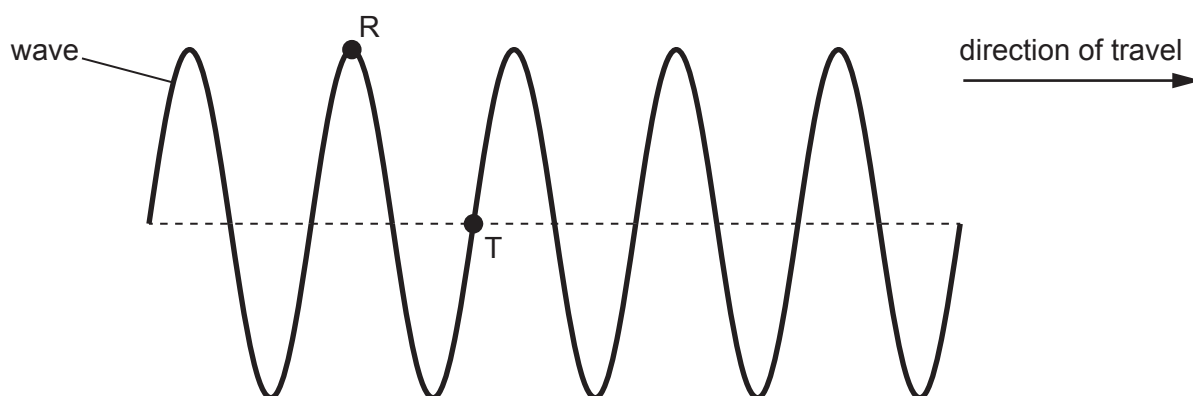
- (b)** The light is replaced with red light of a single frequency.

- (ii)** State and explain the effect of this change on the number of bright fringes formed on the screen. A calculation is not required.

[2]

**18. M/J 24/P23/Q4/d**

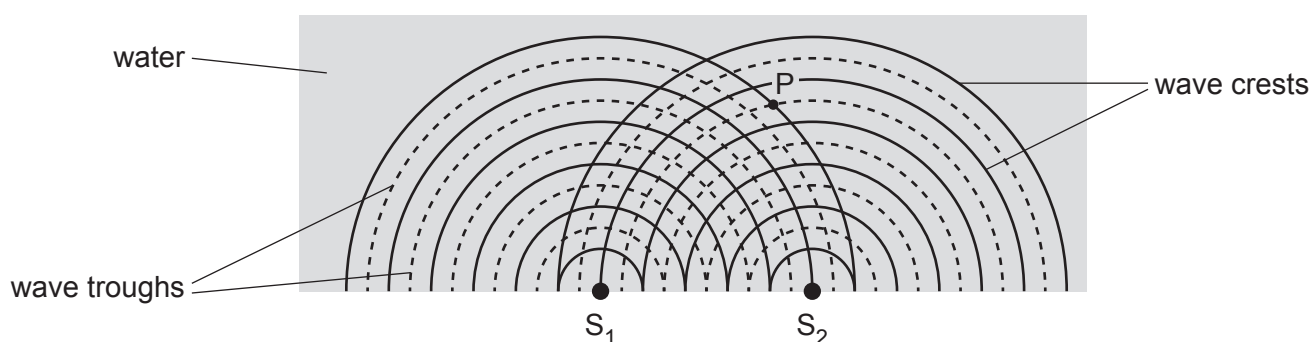
- 4 A progressive transverse wave travelling from left to right is shown at an instant in time in Fig. 4.1.

**Fig. 4.1**

R and T are points on the wave.

- (d) The wave is a water wave produced by a dipper  $S_1$  attached to a vibrator in a ripple tank.

An identical dipper  $S_2$  is attached to the same vibrator. The two dippers produce an interference pattern on the water in the tank, as shown in Fig. 4.3.

**Fig. 4.3** (not to scale)

The wave crests from each source are represented by solid lines on Fig. 4.3 and the wave troughs are represented by dashed lines.

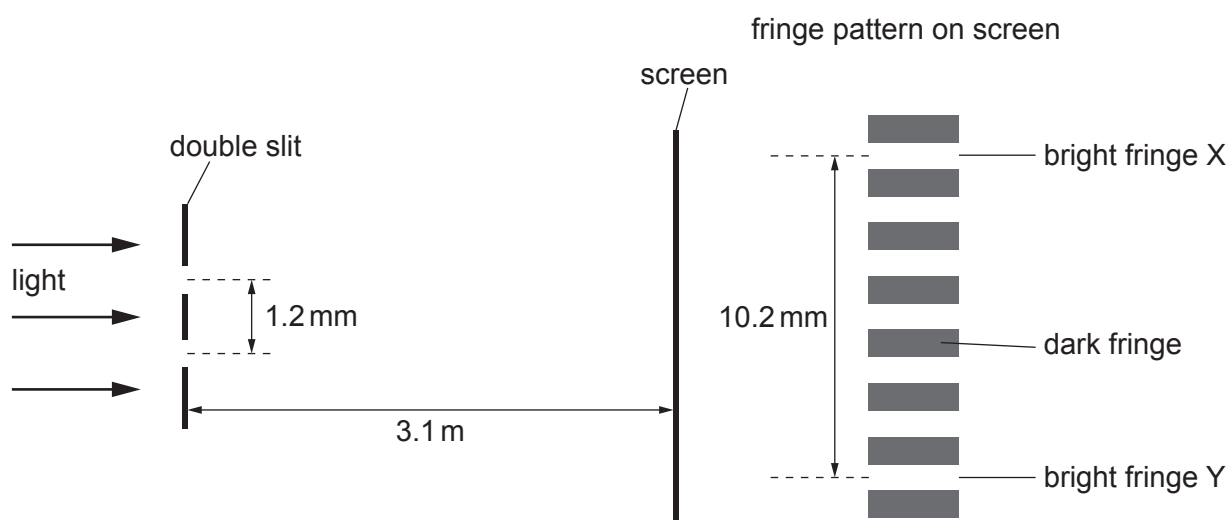
At point P in Fig. 4.3, the wave from  $S_1$  has the same amplitude  $A$  as the wave from  $S_2$ .

Describe and explain the amplitude of the resultant wave at point P.

[3]

**19. MAR 24/P22/Q6/a,b**

- 6 (a) Coherent visible light of a single frequency is incident normally on a double slit. This produces a pattern of bright and dark interference fringes on a screen, as illustrated in Fig. 6.1.



**Fig. 6.1** (not to scale)

There are seven bright fringes.

- (i) Explain how the pattern of bright and dark interference fringes is formed.

[3]

- (ii) The distance between the centres of bright fringe X and bright fringe Y in the pattern is 10.2 mm. The slit spacing is 1.2 mm. The distance from the slits to the screen is 3.1 m.

Calculate the wavelength of the light incident on the slits.

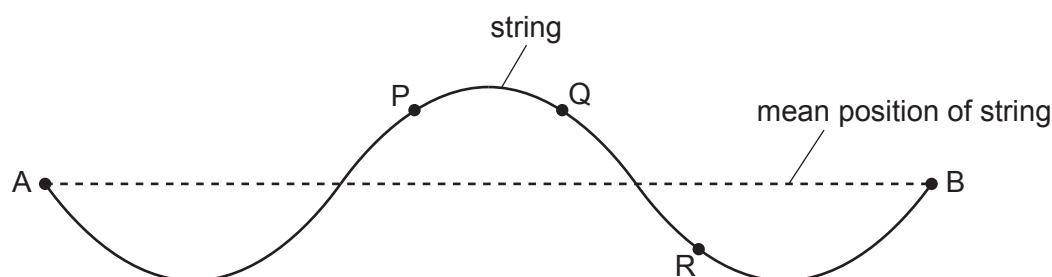
wavelength = m [3]

- (iii) The light is replaced by different visible light with a shorter wavelength.

State how the new fringe separation will compare to the original fringe separation.

[1]

- (b) A stationary wave is formed on a stretched string AB, as shown in Fig. 6.2.



**Fig. 6.2**

P, Q and R are points on the string.

- (i) On Fig. 6.2, draw a cross (×) to show the position of a node.

[1]

- (ii) State the phase difference between P and Q.

phase difference = ° [1]

- (iii) State the phase difference between P and R.

phase difference = ° [1]

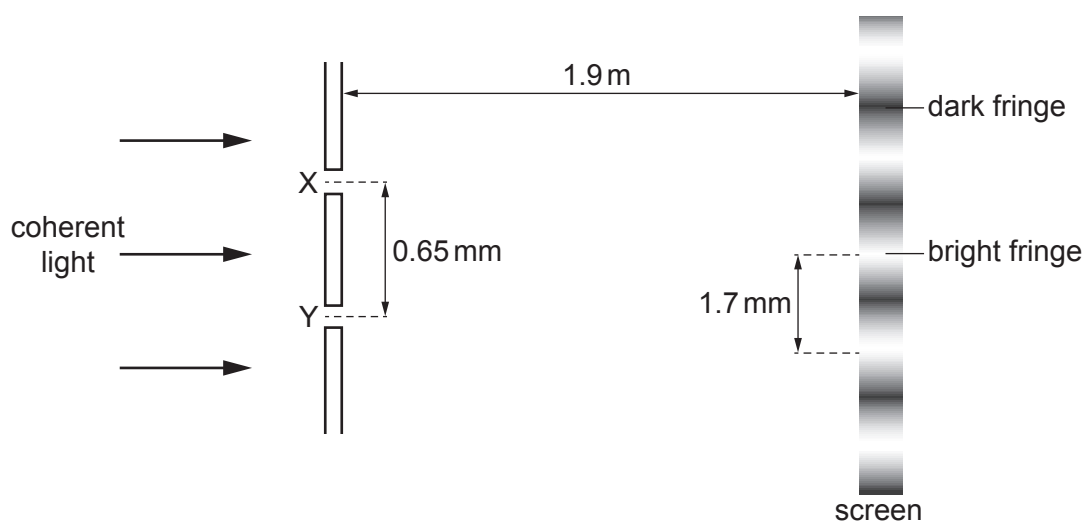
**20. O/N 23/P21/Q4/a,b,c,d**

- 4 (a)** State the principle of superposition.

[2]

- (b)** Coherent light is incident normally on two identical slits X and Y. The diffracted light emerging from the slits superposes to produce an interference pattern on a screen positioned at a distance of 1.9 m from the slits.

Fig. 4.1 shows the arrangement and the central part of the interference pattern of bright and dark fringes formed on the screen.



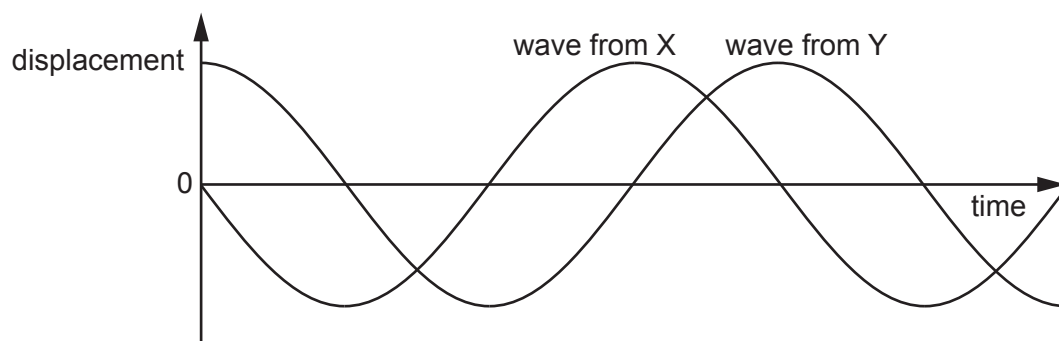
**Fig. 4.1** (not to scale)

The separation of the slits is 0.65 mm. The distance between the centres of adjacent bright fringes is 1.7 mm.

Calculate the wavelength  $\lambda$  of the light.

$\lambda =$  m [3]

- (c)** Light waves from slits X and Y in **(b)** arrive at a point between adjacent bright fringes on the screen. Fig. 4.2 shows the variation of displacement with time for the waves arriving at the point where they meet.



**Fig. 4.2**

A student makes two statements about the waves at this point:

Statement 1: 'The phase difference between the waves is  $90^\circ$ .'

Statement 2: 'The amplitude of the resultant wave is zero.'

(i) Explain how statement 1 is correct.

[1]

(ii) State and explain whether statement 2 is correct.

[1]

(d) The width of each slit in (b) is decreased by the same amount. There is no change to the separation of the slits.

Describe and explain the effect, if any, of this change on the appearance of the interference pattern.

[2]

## 21. O/N 23/P22/Q5/b,c

(b) State what is meant by the diffraction of a wave.

[2]

(c) A beam of light of wavelength  $4.3 \times 10^{-7} \text{ m}$  is incident normally on a diffraction grating in air, as shown in Fig. 5.3.

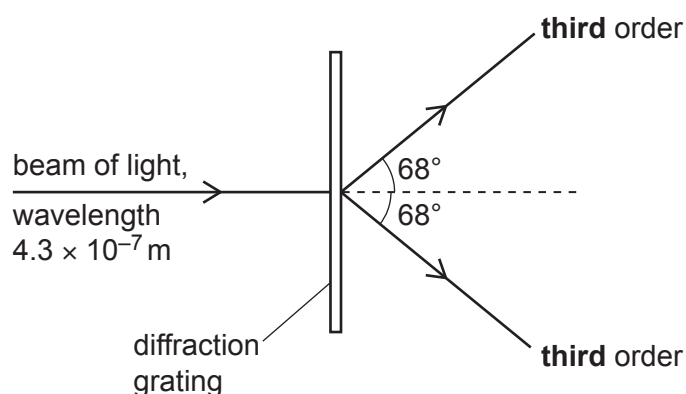


Fig. 5.3 (not to scale)

The **third**-order diffraction maximum of the light is at an angle of  $68^\circ$  to the direction of the incident light beam.

(i) Calculate the line spacing  $d$  of the diffraction grating.

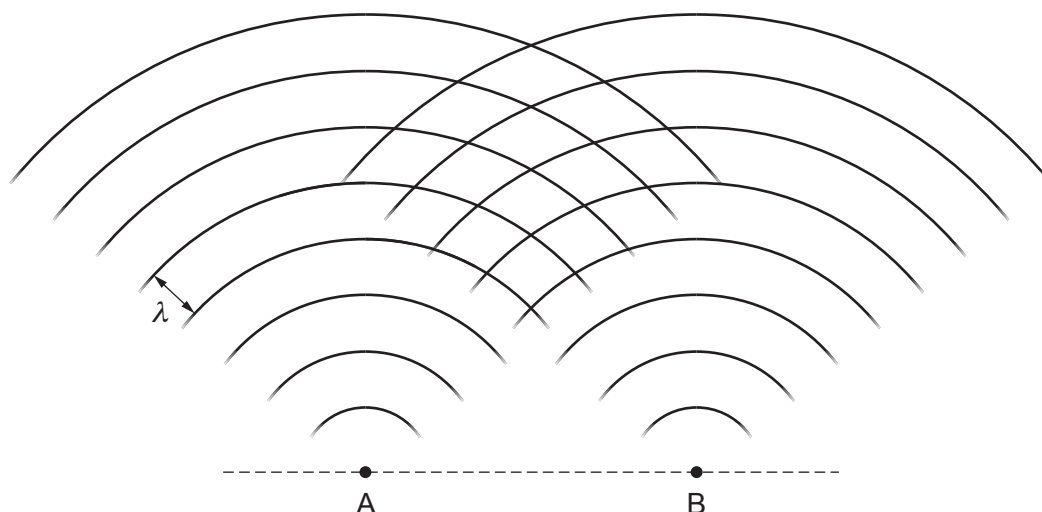
$d =$  m [2]

(ii) Determine a different wavelength of **visible** light that will also produce a diffraction maximum at an angle of  $68^\circ$ .

wavelength = m [2]

**22. O/N 23/P23/Q5/a,b(i)**

- 5 Two point sources, A and B, produce coherent electromagnetic waves. The waves from A and B are emitted in phase and have wavelength  $\lambda$ , as shown in Fig. 5.1.



**Fig. 5.1** (not to scale)

The lines on Fig. 5.1 represent wavefronts. All the points on a wavefront are in phase.

- (a) On Fig. 5.1, mark with a cross (×):

- (i) the position of an interference maximum (label this cross Y) [1]
- (ii) the position of an interference minimum (label this cross Z). [1]

- (b) The waves in air have a wavelength of  $2.9 \times 10^{-5}$  m.

An interference pattern is detected along a line parallel to AB and at a perpendicular distance of 140 m from AB. The spacing between adjacent interference maxima is 1.2 cm.

- (i) Calculate the separation  $a$  of the sources A and B.

$a =$  m [3]

**23. M/J 23/P21/Q5**

booklet (scan + OCR)

(a) An electromagnetic wave in a vacuum has a wavelength of  $8.4 \times 10^{-6} \text{ m}$ .

(i) State the name of the principal region of the electromagnetic spectrum for the wave.

..... [1]

(ii) Calculate the frequency, in THz, of the wave.

frequency = ..... THz [2]

(b) An arrangement that uses a double slit to demonstrate the interference of light from a laser is shown in Fig. 5.1.

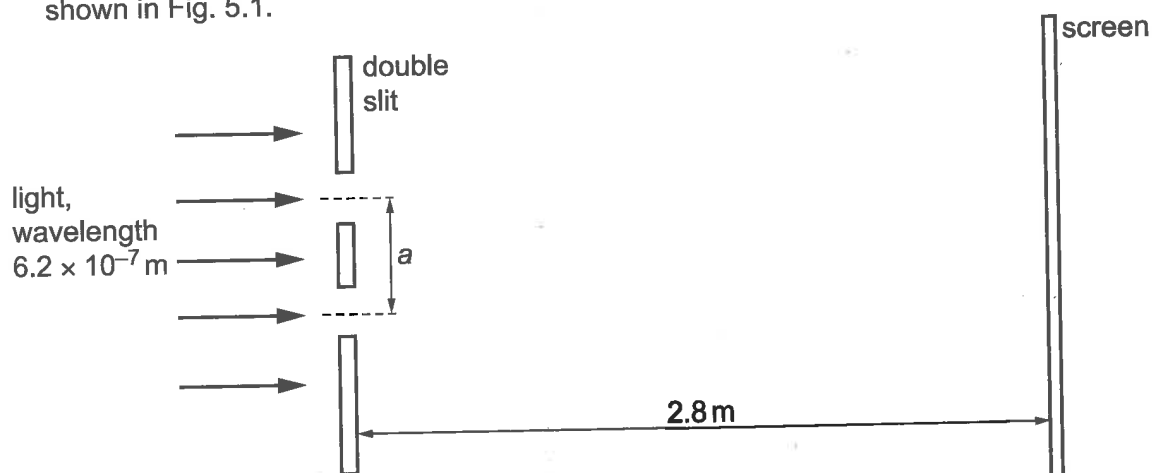


Fig. 5.1 (not to scale)

The light from the laser has a wavelength of  $6.2 \times 10^{-7} \text{ m}$  and is incident normally on the slits. The separation of the two slits is  $a$ . The slits and screen are parallel and separated by a distance of 2.8 m.

An interference pattern of bright fringes and dark fringes is formed on the screen. The distance on the screen across 8 bright fringes is 22 mm, as illustrated in Fig. 5.2.

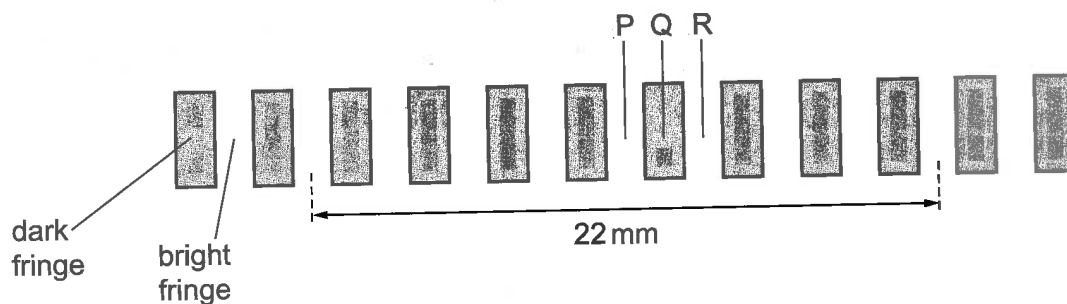


Fig. 5.2

- (i) The light waves emerging from the two slits are coherent.

State what is meant by coherent.

..... [1]

- (ii) Calculate the separation  $a$  of the slits.

$a = \dots\dots\dots$  m [3]

- (c) Fringe P is the central bright fringe of the interference pattern in (b). Fringe Q and fringe R are the nearest dark fringe and the nearest bright fringe respectively to the right of fringe P, as shown in Fig. 5.2.

- (i) Calculate the difference in the distances (the path difference) from each slit to the centre of fringe Q.

difference in the distances = ..... m [1]

- (ii) State the phase difference between the light waves meeting at the centre of fringe R.

phase difference = .....° [1]

#### 24. MAR 23/P22/Q5

booklet (scan + OCR)

- (a) A microphone and cathode-ray oscilloscope (CRO) are used to analyse a sound wave of frequency 5000 Hz. The trace that is displayed on the screen of the CRO is shown in Fig. 5.1.

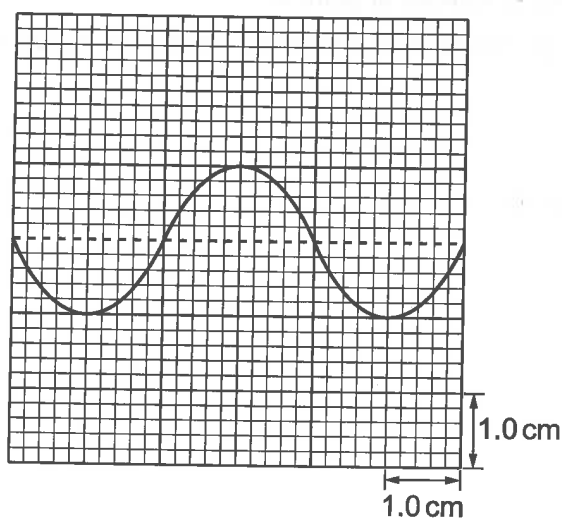


Fig. 5.1

- (i) Determine the time-base setting, in  $\text{s cm}^{-1}$ , of the CRO.

time-base setting = .....  $\text{s cm}^{-1}$  [2]

- (ii) The intensity of the sound detected by the microphone is now increased from its initial value of  $I$  to a new value of  $3I$ . The frequency of the sound is unchanged. Assume that the amplitude of the trace on the CRO screen is proportional to the amplitude of the sound wave.

On Fig. 5.1, sketch the new trace shown on the screen of the CRO. [3]

- (b) An arrangement for demonstrating interference using light is shown in Fig. 5.2.

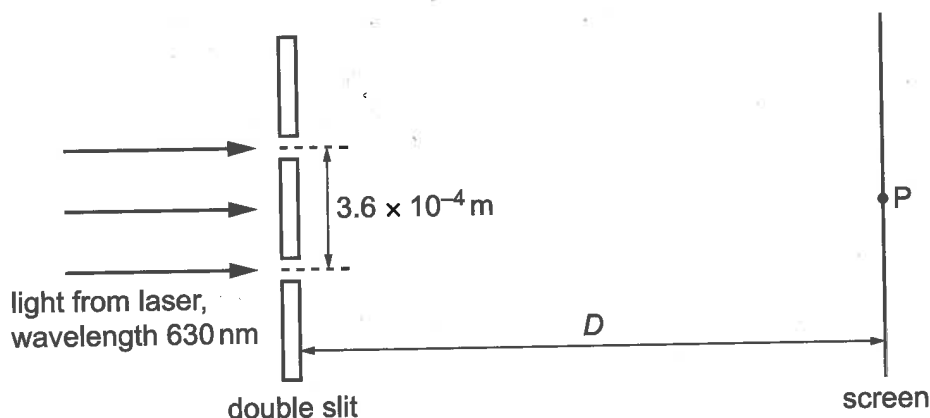


Fig. 5.2 (not to scale)

The wavelength of the light from the laser is 630 nm. The light is incident normally on the double slit. The separation of the two slits is  $3.6 \times 10^{-4} \text{ m}$ . The perpendicular distance between the double slit and the screen is  $D$ .

Coherent light waves from the slits form an interference pattern of bright and dark fringes on the screen. The distance between the centres of two adjacent bright fringes is  $4.0 \times 10^{-3} \text{ m}$ . The central bright fringe is formed at point P.

- (i) Explain why a bright fringe is produced by the waves meeting at point P.

..... [1]

- (ii) Calculate distance  $D$ .

$D = \dots\dots\dots \text{m}$  [3]

- (c) The wavelength  $\lambda$  of the light in (b) is now varied. This causes a variation in the distance  $x$  between the centres of two adjacent bright fringes on the screen. The distance  $D$  and the separation of the two slits are unchanged.

On Fig. 5.3, sketch a graph to show the variation of  $x$  with  $\lambda$  from  $\lambda = 400\text{ nm}$  to  $\lambda = 700\text{ nm}$ . Numerical values of  $x$  are not required.

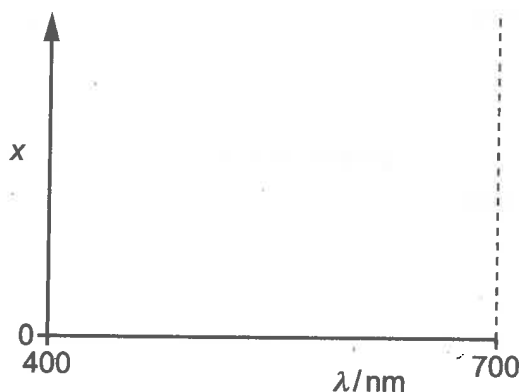


Fig. 5.3

[11]

### 25. M/J 22/P22/Q5

booklet (scan + OCR)

Light from a laser is used to produce an interference pattern on a screen, as shown in Fig. 5.1.

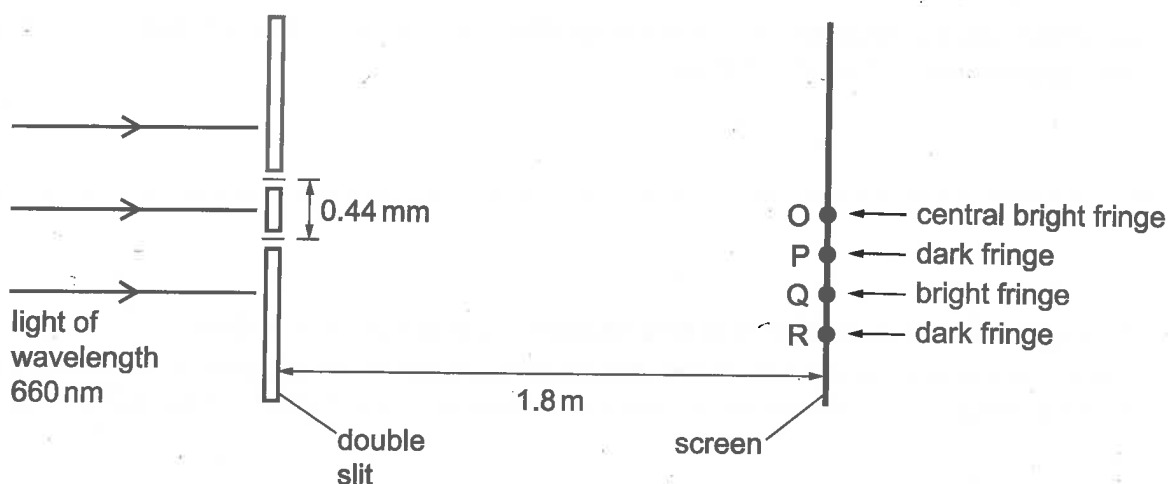


Fig. 5.1 (not to scale)

The light of wavelength  $660\text{ nm}$  is incident normally on two slits that have a separation of  $0.44\text{ mm}$ . The double slit is parallel to the screen. The perpendicular distance between the double slit and the screen is  $1.8\text{ m}$ .

The central bright fringe on the screen is formed at point O. The next dark fringe below point O is formed at point P. The next bright fringe and the next dark fringe below point P are formed at points Q and R respectively.

- (a) The light waves from the two slits are coherent.

State what is meant by coherent.

[1]

- (b) For the two light waves superposing at R, calculate:

- (i) the difference in their path lengths, in nm, from the slits

path difference = ..... nm [1]

(ii) their phase difference.

phase difference = ..... ° [1]

(c) Calculate the distance OQ.

distance OQ = ..... m [3]

(d) The intensity of the light incident on the double slit is increased without changing the frequency.

Describe how the appearance of the fringes after this change is different from, and similar to, their appearance before the change.

.....  
.....  
..... [3]

(e) The light of wavelength 660 nm is now replaced by blue light from a laser.

State and explain the change, if any, that must be made to the separation of the two slits so that the fringe separation on the screen is the same as it was for light of wavelength 660 nm.

.....  
..... [2]

## 26. M/J 22/P23/Q5

booklet (scan + OCR)

- (a) Parallel light rays from the Sun are incident normally on a magnifying glass. The magnifying glass directs the light to an area A of radius  $r$ , as shown in Fig. 5.1.

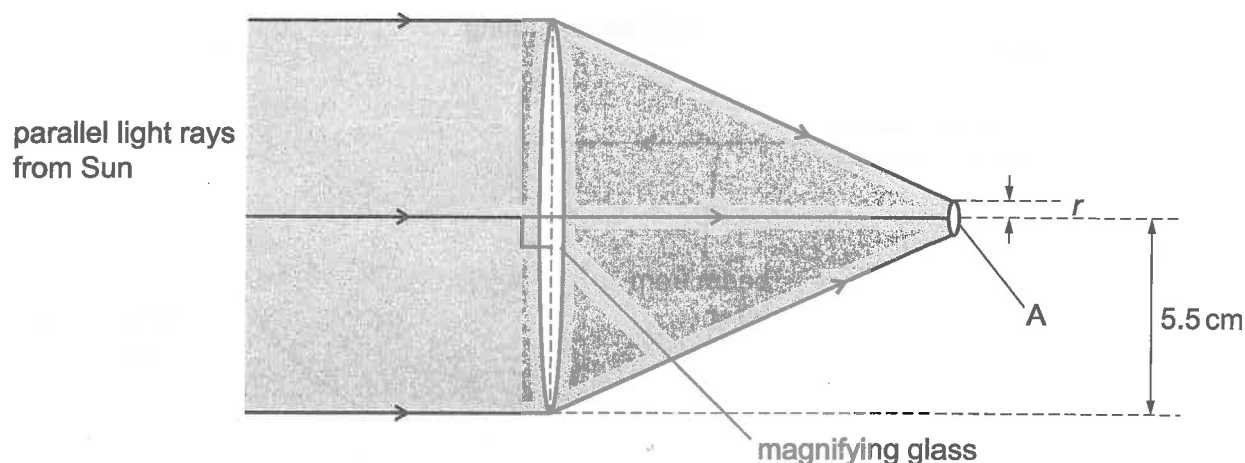


Fig. 5.1 (not to scale)

The magnifying glass is circular in cross-section with a radius of 5.5 cm. The intensity of the light from the Sun incident on the magnifying glass is  $1.3 \text{ kW m}^{-2}$ .

Assume that all of the light incident on the magnifying glass is transmitted through it.

- (i) Calculate the power of the light from the Sun incident on the magnifying glass.

power = ..... W [2]

- (ii) The value of  $r$  is 1.5 mm.

Calculate the intensity of the light on area A.

intensity = .....  $\text{W m}^{-2}$  [1]

- (b) A laser emits a beam of electromagnetic waves of frequency  $3.7 \times 10^{15} \text{ Hz}$  in a vacuum.

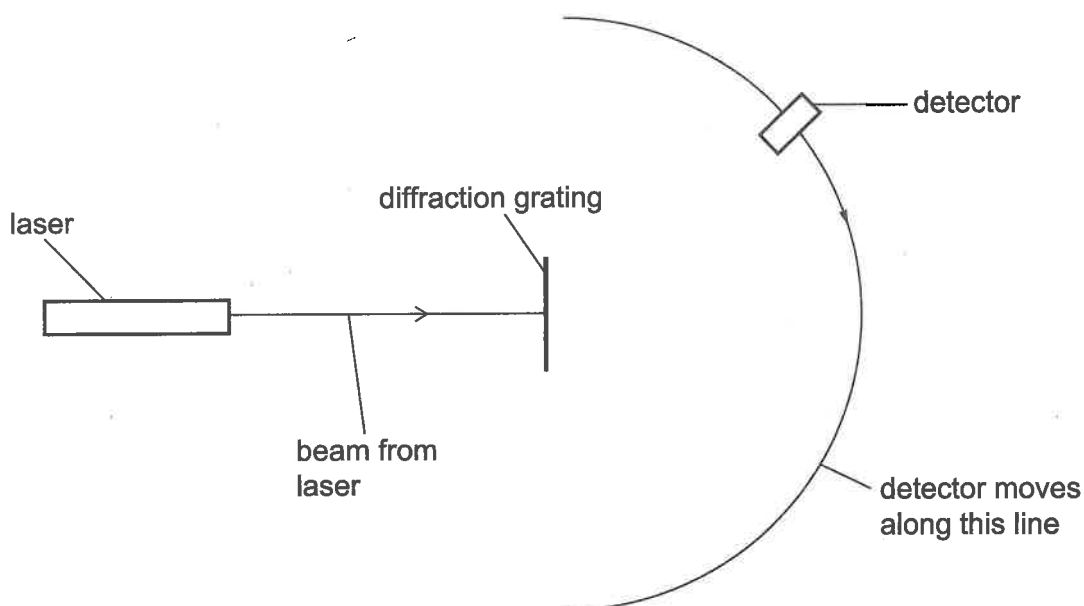
- (i) Show that the wavelength of the waves is  $8.1 \times 10^{-8} \text{ m}$ .

[2]

- (ii) State the region of the electromagnetic spectrum to which these waves belong.

..... [1]

- (iii) The beam from the laser now passes through a diffraction grating with 2400 lines per millimetre. A detector sensitive to the waves emitted by the laser is moved through an arc of  $180^\circ$  in order to detect the maxima produced by the waves passing through the grating, as shown in Fig. 5.2.

**Fig. 5.2**

Calculate the number of maxima detected as the detector moves through  $180^\circ$  along the line shown in Fig. 5.2. Show your working.

number of maxima detected = ..... [4]

- (iv) The laser is now replaced with one that emits electromagnetic waves with a wavelength of 300 nm.

Explain, without calculation, what happens to the number of maxima now detected. Assume that the detector is also sensitive to this wavelength of electromagnetic waves.

.....  
 ..... [2]

### 27. MAR 22/P22/Q5

booklet (scan + OCR)

- (a) (i) State the conditions required for the formation of a stationary wave.

.....  
 ..... [2]

- (ii) State the phase difference between any two vibrating particles in a stationary wave between two adjacent nodes.

phase difference = ..... $^\circ$  [1]

- (b) A motorcycle is travelling at  $13.0 \text{ ms}^{-1}$  along a straight road. The rider of the motorcycle sees a pedestrian standing in the road directly ahead and operates a horn to emit a warning sound. The pedestrian hears the warning sound from the horn at a frequency of  $543 \text{ Hz}$ . The speed of the sound in the air is  $334 \text{ ms}^{-1}$ .
- (i) Calculate the frequency, to three significant figures, of the sound emitted by the horn.

frequency = ..... Hz [2]

- (ii) The motorcycle rider passes the stationary pedestrian and then moves directly away from her. As the rider moves away, he operates the horn for a second time. The pedestrian now hears sound that is increasing in frequency.

State the variation, if any, in the speed of the motorcycle when the rider operates the horn for the second time.

..... [1]

- (c) A beam of vertically polarised monochromatic light is incident normally on a polarising filter, as shown in Fig. 5.1.

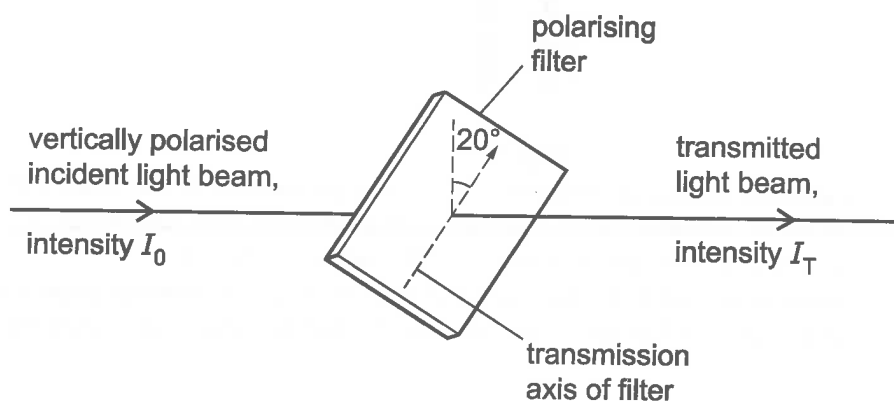


Fig. 5.1

The filter is positioned with its transmission axis at an angle of  $20^\circ$  to the vertical. The incident light has intensity  $I_0$  and the transmitted light has intensity  $I_T$ .

- (i) By considering the ratio  $\frac{I_T}{I_0}$ , calculate the ratio

$$\frac{\text{amplitude of transmitted light}}{\text{amplitude of incident light}}.$$

Show your working.

ratio = ..... [3]

- (ii) The filter is now rotated, about the direction of the light beam, from its starting position shown in Fig. 5.1. The direction of rotation is such that the angle of the transmission axis to the vertical initially increases.

Calculate the minimum angle through which the filter must be rotated so that the intensity of the transmitted light returns to the value that it had when the filter was at its starting position.

angle = ..... ° [1]

## 28. O/N 21/P22/Q5

booklet (scan + OCR)

A tube is initially fully submerged in water. The axis of the tube is kept vertical as the tube is slowly raised out of the water, as shown in Fig. 5.1.

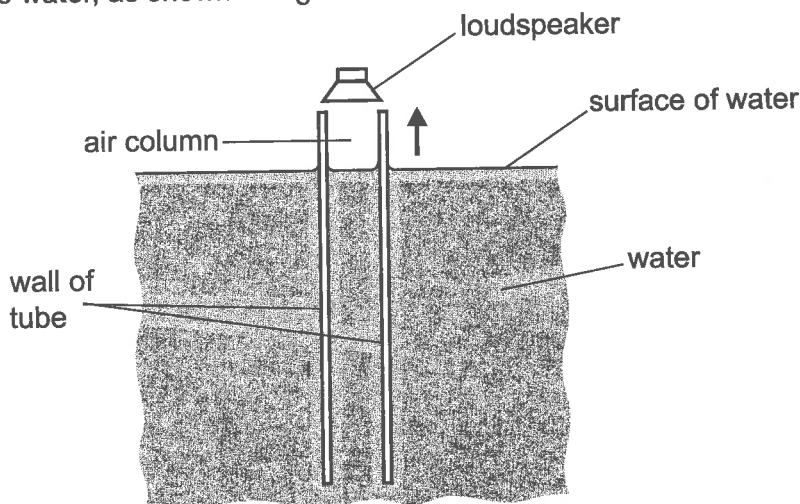


Fig. 5.1

A loudspeaker producing sound of frequency 530 Hz is positioned at the open top end of the tube as it is raised. The water surface inside the tube is always level with the water surface outside the tube. The speed of the sound in the air column in the tube is  $340 \text{ m s}^{-1}$ .

- (a) Describe a simple way that a student, without requiring any additional equipment, can detect when a stationary wave is formed in the air column as the tube is being raised.

..... [1]

- (b) Determine the height of the top end of the tube above the surface of the water when a stationary wave is first produced in the tube. Assume that an antinode is formed level with the top of the tube.

height = ..... m [3]

- (c) Determine the distance moved by the tube between the positions at which the first and second stationary waves are formed.

distance = ..... m [1]

**29. O/N 21/P23/Q5**

booklet (scan + OCR)

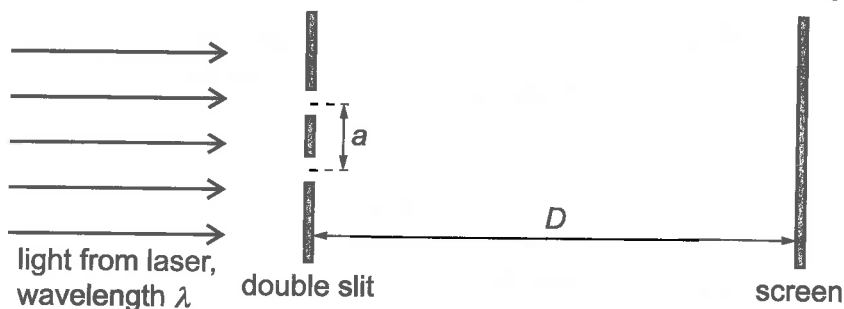
(a) For a progressive wave on a stretched string, state what is meant by *amplitude*.

..... [1]

(b) Light from a laser has a wavelength of 690 nm in a vacuum.  
Calculate the period of the light wave.

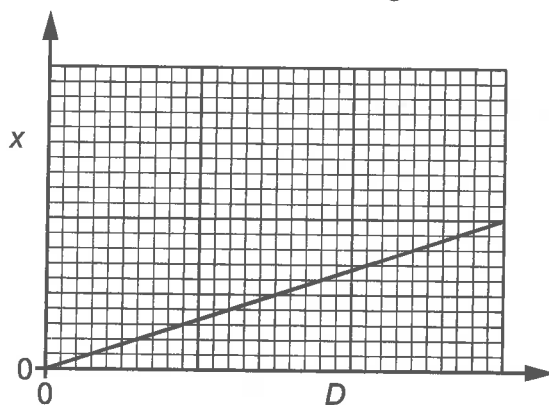
period = ..... s [3]

(c) A two-source interference experiment uses the arrangement shown in Fig. 5.1.



**Fig. 5.1** (not to scale)

Light from a laser is incident normally on a double slit. A screen is parallel to the double slit. Interference fringes are seen on the screen at distance  $D$  from the double slit. The separation of the centres of the slits is  $a$ . The light has wavelength  $\lambda$ . The separation  $x$  of the centres of adjacent bright fringes is measured for different values of distance  $D$ . The variation with  $D$  of  $x$  is shown in Fig. 5.2.



**Fig. 5.2**

The gradient of the graph is  $G$ .

(i) Determine an expression, in terms of  $G$  and  $\lambda$ , for the separation  $a$  of the slits.

$a = \dots\dots\dots$  [2]

(ii) The experiment is repeated with slits of separation  $2a$ . The wavelength of the light is unchanged.

On Fig. 5.2, sketch a graph to show the results of this experiment.

[2]

**30. M/J 21/P22/Q4**

booklet (scan + OCR)

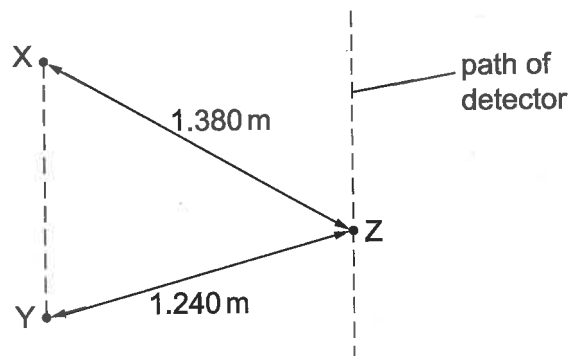
- (a) For a progressive wave, state what is meant by its *period*.

..... [1]

- (b) State the principle of superposition.

..... [2]

- (c) Electromagnetic waves of wavelength  $0.040\text{ m}$  are emitted in phase from two sources X and Y and travel in a vacuum. The arrangement of the sources is shown in Fig. 4.1.



**Fig. 4.1** (not to scale)

A detector moves along a path that is parallel to the line XY. A pattern of intensity maxima and minima is detected.

Distance XZ is  $1.380\text{ m}$  and distance YZ is  $1.240\text{ m}$ .

- (i) State the name of the region of the electromagnetic spectrum that contains the waves from X and Y.

..... [1]

- (ii) Calculate the period, in ps, of the waves.

period = ..... ps [3]

- (iii) Show that the path difference at point Z between the waves from X and Y is  $3.5\lambda$ , where  $\lambda$  is the wavelength of the waves.

[1]

- (iv) Calculate the phase difference between the waves at point Z.

phase difference = .....° [1]

- (v) The waves from X alone have the same amplitude at point Z as the waves from Y alone.  
State the intensity of the waves at point Z.

..... [1]

- (vi) The frequencies of the waves from X and Y are both decreased to the same lower value.  
The waves stay within the same region of the electromagnetic spectrum.  
Describe the effect of this change on the pattern of intensity maxima and minima along  
the path of the detector.

..... [1]

**31. M/J 21/P21/Q4**

booklet (scan + OCR)

- (a) For a progressive wave, state what is meant by *wavelength*.

..... [1]

- (b) A light wave from a laser has a wavelength of 460 nm in a vacuum.  
Calculate the period of the wave.

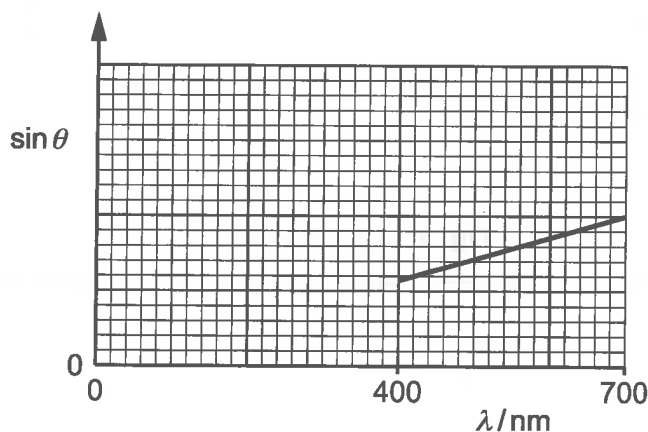
period = ..... s [3]

- (c) The light from the laser is incident normally on a diffraction grating.  
Describe the diffraction of the light waves at the grating.

.....

..... [2]

- (d) A diffraction grating is used with different wavelengths of visible light. The angle  $\theta$  of the **fourth-order** maximum from the zero-order (central) maximum is measured for each wavelength. The variation with wavelength  $\lambda$  of  $\sin \theta$  is shown in Fig. 4.1.

**Fig. 4.1**

- (i) The gradient of the graph is  $G$ .  
Determine an expression, in terms of  $G$ , for the distance  $d$  between the centres of two adjacent slits in the diffraction grating.

$d =$  ..... [2]

- (ii) On Fig. 4.1, sketch a graph to show the results that would be obtained for the **second-order** maxima. [2]

**32. M/J 21/P23/Q4**

booklet (scan + OCR)

- (a)**
- State the principle of superposition.

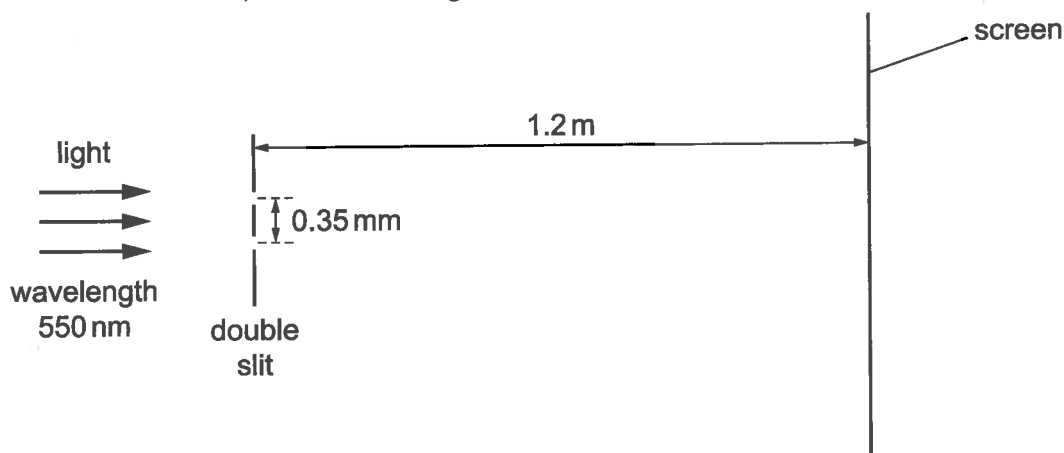
.....

..... [2]

- (b)**
- Two waves, with intensities
- $I$
- and
- $4I$
- , superpose. The waves have the same frequency. Determine, in terms of
- $I$
- , the maximum possible intensity of the resulting wave.

maximum intensity = .....  $I$  [2]

- (c)**
- Coherent light of wavelength
- $550\text{ nm}$
- is incident normally on a double slit of slit separation
- $0.35\text{ mm}$
- . A series of bright and dark fringes forms on a screen placed a distance of
- $1.2\text{ m}$
- from the double slit, as shown in Fig. 4.1. The screen is parallel to the double slit.

**Fig. 4.1** (not to scale)

- (i)**
- Determine the distance between the centres of adjacent bright fringes on the screen.

distance = ..... m [3]

- (ii)**
- The light of wavelength
- $550\text{ nm}$
- is replaced with red light of a single frequency. State and explain the change, if any, in the distance between the centres of adjacent bright fringes.

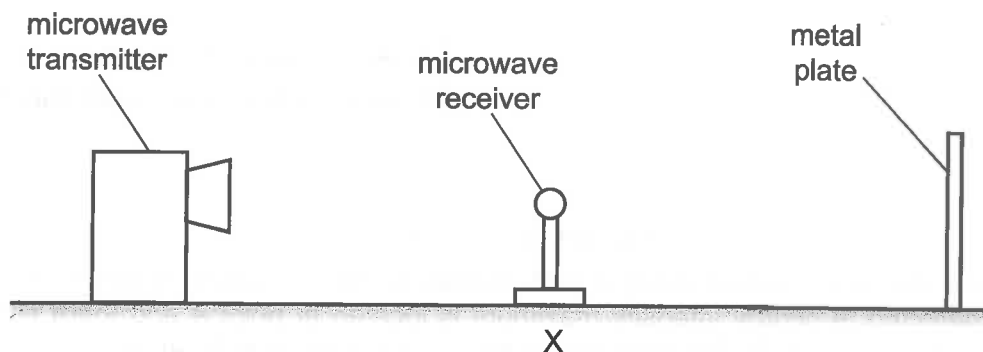
..... [1]

**33. MAR 21/P22/Q4**

booklet (scan + OCR)

**(a) State the principle of superposition.**

..... [2]

**(b) A transmitter produces microwaves that travel in air towards a metal plate, as shown in Fig. 4.1.****Fig. 4.1**

The microwaves have a wavelength of 0.040 m. A stationary wave is formed between the transmitter and the plate.

**(i) Explain the function of the metal plate.**

..... [1]

**(ii) Calculate the frequency, in GHz, of the microwaves.**

frequency = ..... GHz [3]

**(iii) A microwave receiver is initially placed at position X where it detects an intensity minimum. The receiver is then slowly moved away from X directly towards the plate.**

1. Determine the shortest distance from X of the receiver when it detects another intensity minimum.

distance = ..... m

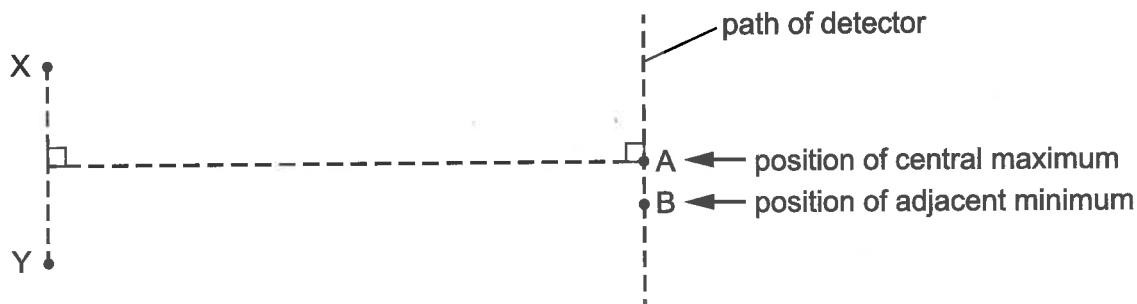
2. Determine the number of intensity maxima that are detected by the receiver as it moves from X to a position that is 9.1 cm away from X.

number = ..... [2]

**34. O/N 20/P22/Q5**

booklet (scan + OCR)

Microwaves with the same wavelength and amplitude are emitted in phase from two sources X and Y, as shown in Fig. 5.1.

**Fig. 5.1** (not to scale)

A microwave detector is moved along a path parallel to the line joining X and Y. An interference pattern is detected. A central intensity maximum is located at point A and there is an adjacent intensity minimum at point B. The microwaves have a wavelength of 0.040 m.

(a) Calculate the frequency, in GHz, of the microwaves.

frequency = ..... GHz [3]

(b) For the waves arriving at point B, determine:

(i) the path difference

path difference = ..... m [1]

(ii) the phase difference.

phase difference = .....° [1]

(c) The amplitudes of the waves from the sources are changed. This causes a change in the amplitude of the waves arriving at point A. At this point, the amplitude of the wave arriving from source X is doubled and the amplitude of the wave arriving from source Y is also doubled. Describe the effect, if any, on the intensity of the central maximum at point A.

..... [2]  
 (d) Describe the effect, if any, on the positions of the central intensity maximum and the adjacent intensity minimum due to the following separate changes.

(i) The separation of the sources X and Y is increased.

..... [1]  
 (ii) The phase difference between the microwaves emitted by the sources X and Y changes to  $180^\circ$ .

..... [1]

### 35. O/N 20/P23/Q5

booklet (scan + OCR)

(a) A sound wave is detected by a microphone that is connected to a cathode-ray oscilloscope (CRO). The trace on the screen of the CRO is shown in Fig. 5.1.

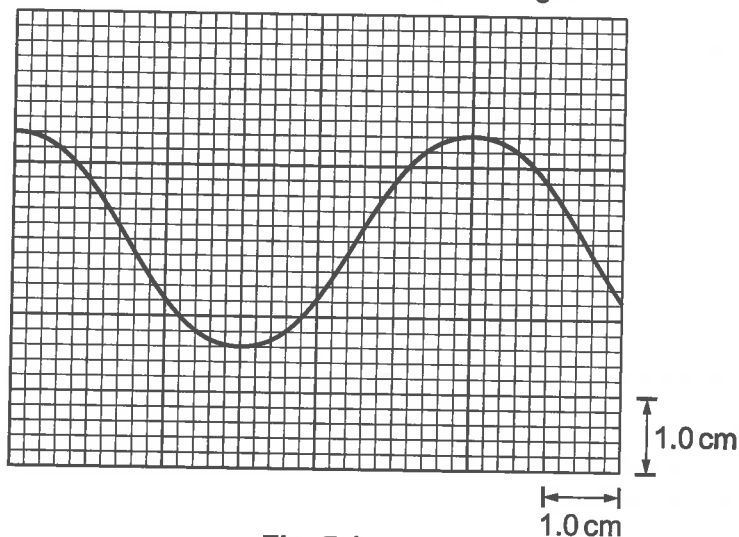


Fig. 5.1

The time-base setting of the CRO is  $2.0 \times 10^{-5} \text{ s cm}^{-1}$ .

(i) Determine the frequency of the sound wave.

frequency = ..... Hz [2]

(ii) The intensity of the sound wave is now doubled. The frequency is unchanged. Assume that the amplitude of the trace is proportional to the amplitude of the sound wave. On Fig. 5.1, sketch the new trace shown on the screen. [2]

(iii) The time-base is now switched off. Describe the trace seen on the screen.

..... [1]

- (b) A beam of light of a single wavelength is incident normally on a diffraction grating, as illustrated in Fig. 5.2.

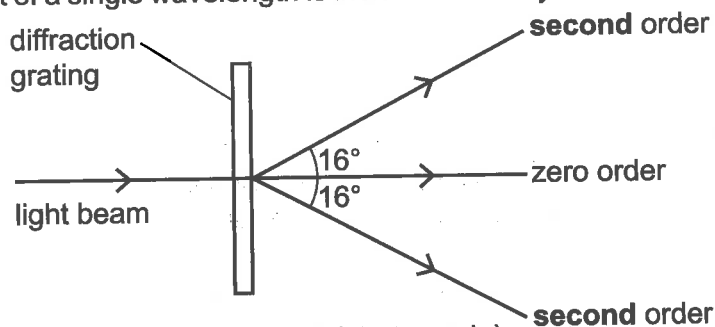


Fig. 5.2 (not to scale)

Fig. 5.2 does not show all of the emerging beams from the grating. The angle between the **second-order** emerging beam and the central zero-order beam is  $16^\circ$ . The grating has a line spacing of  $3.4 \times 10^{-6} \text{ m}$ .

- (i) Calculate the wavelength of the light.

wavelength = ..... m [2]

- (ii) Determine the highest order of emerging beam from the grating.

highest order = ..... [2]

### 36. M/J 20/P22/Q4

booklet (scan + OCR)

- (a) State the difference between progressive waves and stationary waves in terms of the transfer of energy along the wave.

..... [1]

- (b) A progressive wave travels from left to right along a stretched string. Fig. 4.1 shows part of the string at one instant.

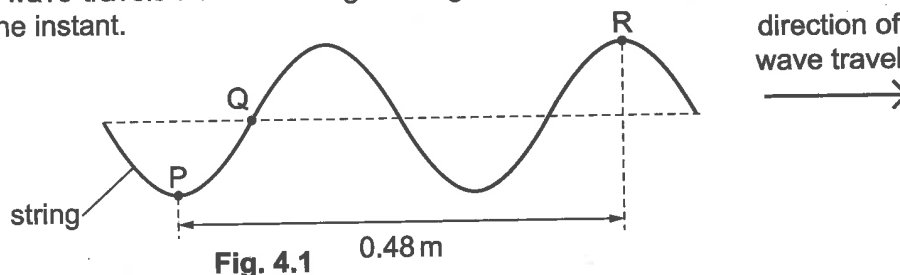


Fig. 4.1

P, Q and R are three different points on the string. The distance between P and R is 0.48 m. The wave has a period of 0.020 s.

- (i) Use Fig. 4.1 to determine the wavelength of the wave.

wavelength = ..... m [1]

(ii) Calculate the speed of the wave.

speed = .....  $\text{ms}^{-1}$  [2]

(iii) Determine the phase difference between points Q and R.

phase difference = .....  $^{\circ}$  [1]

(iv) Fig. 4.1 shows the position of the string at time  $t = 0$ . Describe how the displacement of point Q on the string varies with time from  $t = 0$  to  $t = 0.010\text{ s}$ .

.....  
 ..... [2]

(c) A stationary wave is formed on a different string that is stretched between two fixed points X and Y. Fig. 4.2 shows the position of the string when each point is at its maximum displacement.

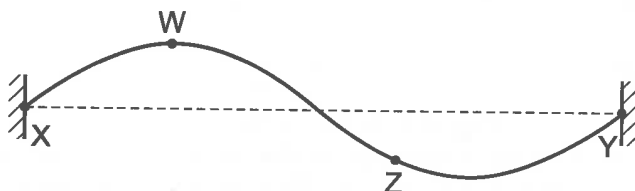


Fig. 4.2

(i) Explain what is meant by a *node* of a stationary wave.

..... [1]

(ii) State the number of antinodes of the wave shown in Fig. 4.2.

number = ..... [1]

(iii) State the phase difference between points W and Z on the string.

phase difference = .....  $^{\circ}$  [1]

(iv) A new stationary wave is now formed on the string. The new wave has a frequency that is half of the frequency of the wave shown in Fig. 4.2. The speed of the wave is unchanged.

On Fig. 4.3, draw a position of the string, for this new wave, when each point is at its maximum displacement.

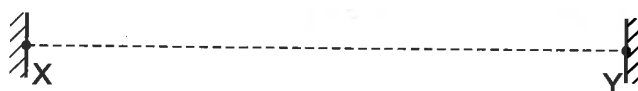


Fig. 4.3

[1]

**37. M/J 20/P21/Q4**

booklet (scan + OCR)

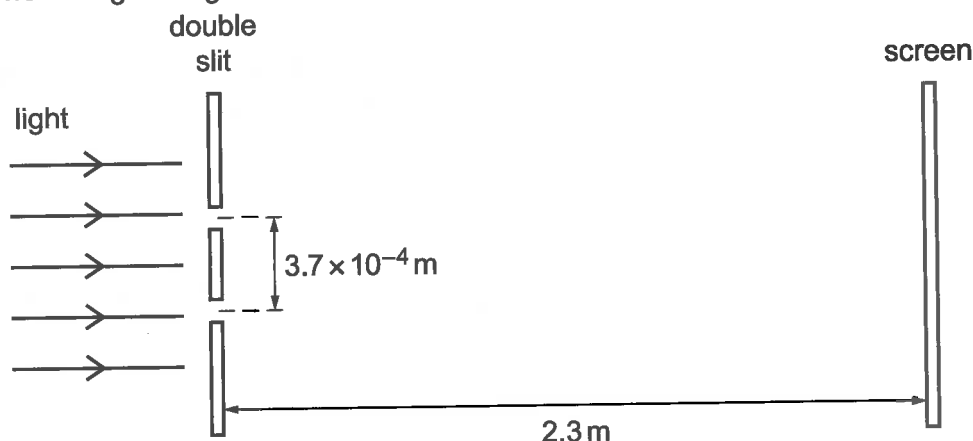
- (a) (i) By reference to the direction of propagation of energy, state what is meant by a *longitudinal wave*.

..... [1]

- (ii) State the principle of superposition.

..... [2]

- (b) The wavelength of light from a laser is determined using the apparatus shown in Fig. 4.1.



**Fig. 4.1** (not to scale)

The light from the laser is incident normally on the plane of the double slit.

The separation of the two slits is  $3.7 \times 10^{-4} \text{ m}$ . The screen is parallel to the plane of the double slit. The distance between the screen and the double slit is  $2.3 \text{ m}$ .

A pattern of bright fringes and dark fringes is seen on the screen. The separation of adjacent bright fringes on the screen is  $4.3 \times 10^{-3} \text{ m}$ .

- (i) Calculate the wavelength, in nm, of the light.

wavelength = ..... nm [3]

- (ii) The intensity of the light passing through each slit was initially the same. The intensity of the light through **one** of the slits is now reduced. Compare the appearance of the fringes before and after the change of intensity.

..... [2]

**38. MAR 20/P22/Q4**

booklet (scan + OCR)

The booklet also files this question under Unit 3

**(a)** For a progressive wave, state what is meant by:**(i)** the *wavelength*

..... [1]

**(ii)** the *amplitude*.

..... [1]

**(b)** A beam of red laser light is incident normally on a diffraction grating.**(i)** Diffraction of the light waves occurs at each slit of the grating. The light waves emerging from the slits are coherent.

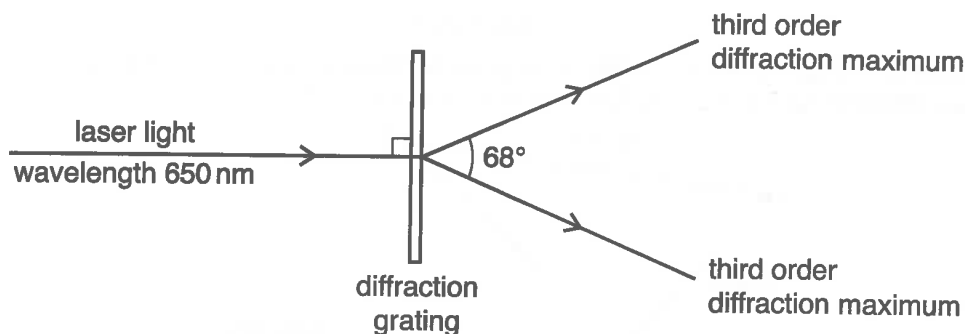
Explain what is meant by:

**1.** *diffraction*

..... [1]

**2.** *coherent*.

..... [1]

**(ii)** The wavelength of the laser light is 650 nm. The angle between the **third** order diffraction maxima is  $68^\circ$ , as illustrated in Fig. 4.1.**Fig. 4.1** (not to scale)Calculate the separation  $d$  between the centres of adjacent slits of the grating. $d =$  ..... m [3]**(iii)** The red laser light is replaced with blue laser light.

State and explain the change, if any, to the angle between the third order diffraction maxima.

.....

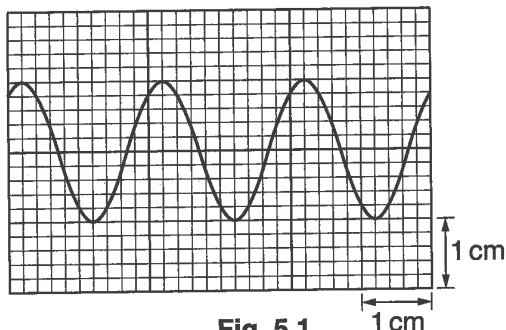
..... [2]

**39. O/N 19/P22/Q5**

booklet (scan + OCR)

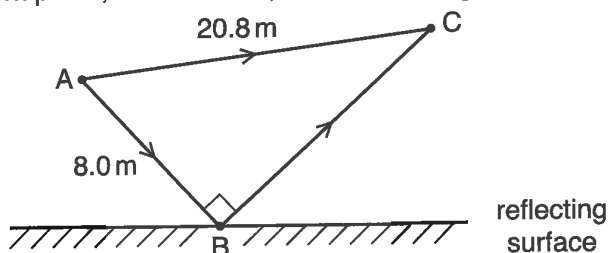
**(a)** State what is meant by the *wavelength* of a progressive wave.

..... [1]

**(b)** A cathode-ray oscilloscope (CRO) is used to analyse a sound wave. The screen of the CRO is shown in Fig. 5.1.**Fig. 5.1**

The time-base setting of the CRO is  $2.5 \text{ ms cm}^{-1}$ .  
Determine the frequency of the sound wave.

frequency = ..... Hz [2]

**(c)** The source emitting the sound in **(b)** is at point A. Waves travel from the source to point C along two different paths, AC and ABC, as shown in Fig. 5.2.**Fig. 5.2** (not to scale)

Distance AB is 8.0 m and distance AC is 20.8 m. Angle ABC is  $90^\circ$ . Assume that there is no phase change of the sound wave due to the reflection at point B. The wavelength of the waves is 1.6 m.

**(i)** Show that the waves meeting at C have a path difference of 6.4 m.

[1]

**(ii)** Explain why an intensity maximum is detected at point C.

.....

..... [2]

**(iii)** Determine the difference between the times taken for the sound to travel from the source to point C along the two different paths.

time difference = ..... s [2]

- (iv) The wavelength of the sound is gradually increased. Calculate the wavelength of the sound when an intensity maximum is next detected at point C.

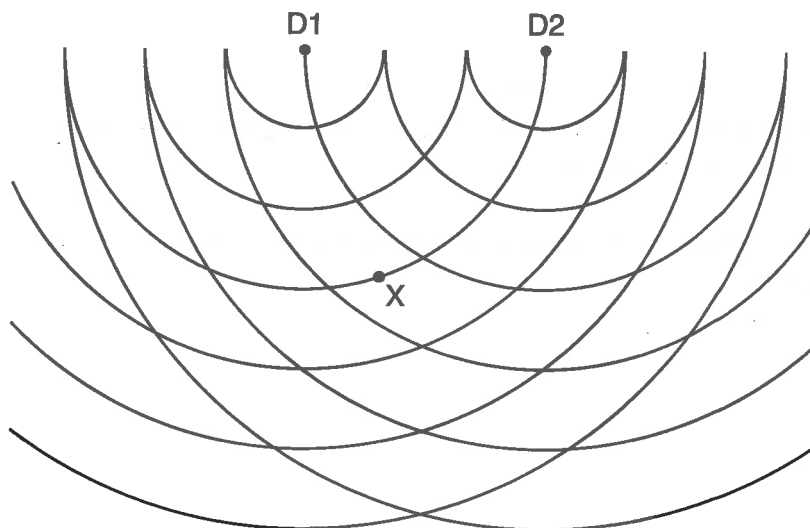
wavelength = ..... m [1]

**40. O/N 19/P21/Q5**

booklet (scan + OCR)

A ripple tank is used to demonstrate the interference of water waves.

Two dippers D1 and D2 produce coherent waves that have circular wavefronts, as illustrated in Fig. 5.1.



**Fig. 5.1**

The lines in the diagram represent crests. The waves have a wavelength of 6.0 cm.

- (a) One condition that is required for an observable interference pattern is that the waves must be coherent.

- (i) Describe how the apparatus is arranged to ensure that the waves from the dippers are coherent.

..... [1]

- (ii) State one other condition that must be satisfied by the waves in order for the interference pattern to be observable.

..... [1]

- (b) Light from a lamp above the ripple tank shines through the water onto a screen below the tank. Describe one way of seeing the illuminated pattern more clearly.

..... [1]

- (c) The speed of the waves is  $0.40 \text{ ms}^{-1}$ . Calculate the period of the waves.

period = ..... s [2]

- (d)** Fig. 5.1 shows a point X that lies on a crest of the wave from D1 and midway between two adjacent crests of the wave from D2.

For the waves at point X, state:

- (i)** the path difference, in cm

path difference = ..... cm [1]

- (ii)** the phase difference.

phase difference = ..... ° [1]

- (e)** On Fig. 5.1, draw **one** line, at least 4 cm long, which joins points where only maxima of the interference pattern are observed. [1]

**41. O/N 19/P23/Q5**

booklet (scan + OCR)

- (a) Light waves emerging from the slits of a diffraction grating are coherent and produce an interference pattern.

Explain what is meant by:

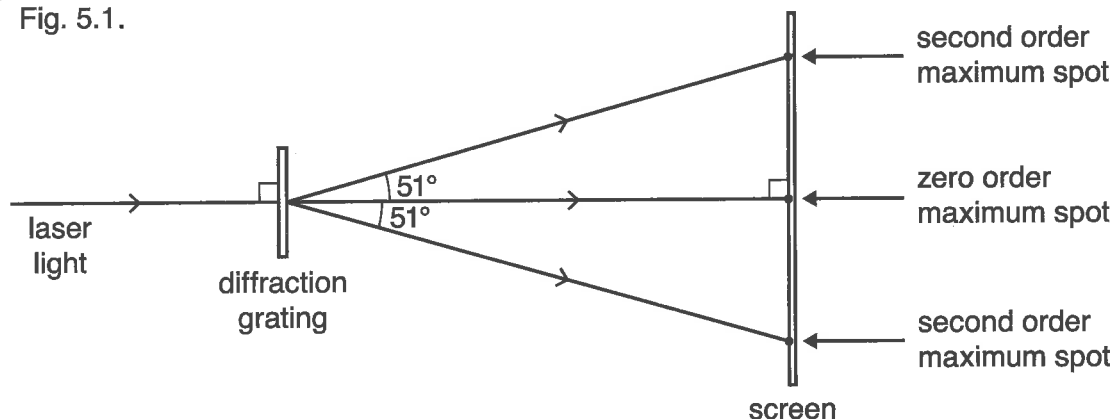
(i) *coherence*

..... [1]

(ii) *interference*.

..... [1]

- (b) A narrow beam of light from a laser is incident normally on a diffraction grating, as shown in Fig. 5.1.



**Fig. 5.1** (not to scale)

Spots of light are seen on a screen positioned parallel to the grating. The angle corresponding to each of the **second** order maxima is  $51^\circ$ . The number of lines per unit length on the diffraction grating is  $6.7 \times 10^5 \text{ m}^{-1}$ .

- (i) Determine the wavelength of the light.

wavelength = ..... m [2]

- (ii) State and explain the change, if any, to the distance between the second order maximum spots on the screen when the light from the laser is replaced by light of a shorter wavelength.

..... [1]

**42. M/J 19/P22/Q4**

booklet (scan + OCR)

**(a)** For a progressive water wave, state what is meant by:**(i)** *displacement*

..... [1]

**(ii)** *amplitude*.

.....[1]

**(b)** Two coherent waves X and Y meet at a point and superpose. The phase difference between the waves at the point is  $180^\circ$ . Wave X has an amplitude of 1.2 cm and intensity  $I$ . Wave Y has an amplitude of 3.6 cm.Calculate, in terms of  $I$ , the resultant intensity at the meeting point.

intensity = .....[2]

**(c) (i)** Monochromatic light is incident on a diffraction grating. Describe the diffraction of the light waves as they pass through the grating.

..... [2]

**(ii)** A parallel beam of light consists of two wavelengths 540 nm and 630 nm. The light is incident normally on a diffraction grating. Third-order diffraction maxima are produced for each of the two wavelengths. No higher orders are produced for either wavelength. Determine the smallest possible line spacing  $d$  of the diffraction grating. $d = \dots\dots\dots$  m [3]**(iii)** The beam of light in **(c)(ii)** is replaced by a beam of blue light incident on the same diffraction grating.

State and explain whether a third-order diffraction maximum is produced for this blue light.

..... [2]

**43. M/J 19/P21/Q5**

booklet (scan + OCR)



The sources emit coherent sound waves of wavelength  $0.85\text{ m}$ . A sound detector is moved parallel to the line  $S_1S_2$  from a point X to a point Y. Alternate positions of maximum loudness L and minimum loudness Q are detected, as illustrated in Fig. 5.2.

Distance  $S_1X$  is equal to distance  $S_2X$ . Distance  $S_2Y$  is  $7.40\text{ m}$ .

(i) Explain what is meant by *coherent* waves.

..... [1]

(ii) State the phase difference between the two waves arriving at the position of minimum loudness Q that is closest to point X.

phase difference = ..... ° [1]

(iii) Determine the distance  $S_1Y$ .

distance = ..... m [2]

#### 44. M/J 19/P23/Q5

booklet (scan + OCR)

A vertical tube of length  $0.60\text{ m}$  is open at both ends, as shown in Fig. 5.1.

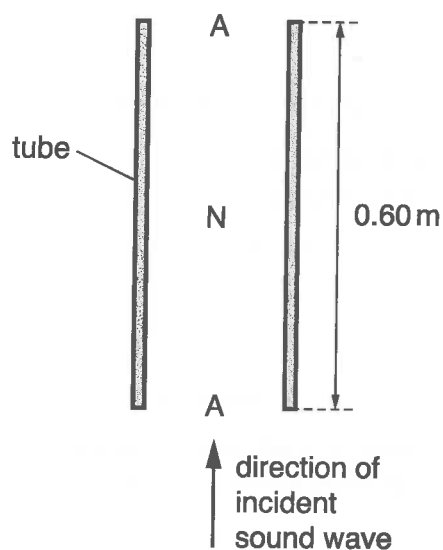


Fig. 5.1

An incident sinusoidal sound wave of a single frequency travels up the tube. A stationary wave is then formed in the air column in the tube with antinodes A at both ends and a node N at the midpoint.

(a) Explain how the stationary wave is formed from the incident sound wave.

.....

..... [2]

(b) On Fig. 5.2, sketch a graph to show the variation of the amplitude of the stationary wave with height  $h$  above the bottom of the tube.

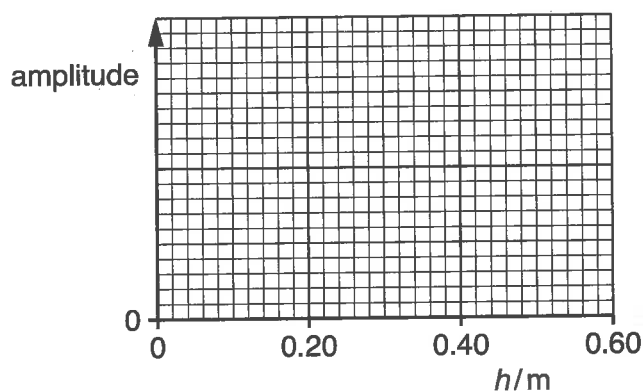


Fig. 5.2

[2]

(c) For the stationary wave, state:

- (i) the direction of the oscillations of an air particle at a height of 0.15 m above the bottom of the tube

..... [1]

- (ii) the phase difference between the oscillations of a particle at a height of 0.10 m and a particle at a height of 0.20 m above the bottom of the tube.

phase difference = ..... ° [1]

(d) The speed of the sound wave is  $340 \text{ m s}^{-1}$ .

Calculate the frequency of the sound wave.

frequency = ..... Hz [2]

(e) The frequency of the sound wave is gradually increased.

Determine the frequency of the wave when a stationary wave is next formed.

frequency = ..... Hz [1]

#### 45. MAR 19/P22/Q5

booklet (scan + OCR)

(a) By reference to two waves, state:

- (i) the principle of superposition

.....

..... [2]

- (ii) what is meant by *coherence*.

..... [1]

- (b) Two coherent waves P and Q meet at a point in phase and superpose. Wave P has an amplitude of 1.5 cm and intensity  $I$ . The resultant intensity at the point where the waves meet is  $3I$ .

Calculate the amplitude of wave Q.

amplitude = ..... cm [2]

- (c) The apparatus shown in Fig. 5.1 is used to produce an interference pattern on a screen.

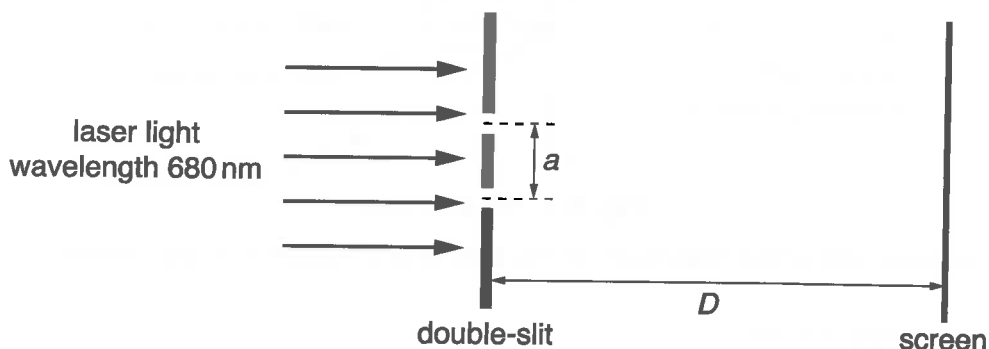


Fig. 5.1 (not to scale)

Light of wavelength 680 nm is incident on a double-slit. The slit separation is  $a$ . The separation between adjacent fringes is  $x$ . Fringes are viewed on a screen at distance  $D$  from the double-slit.

Distance  $D$  is varied from 2.0 m to 3.5 m. The variation with  $D$  of  $x$  is shown in Fig. 5.2.

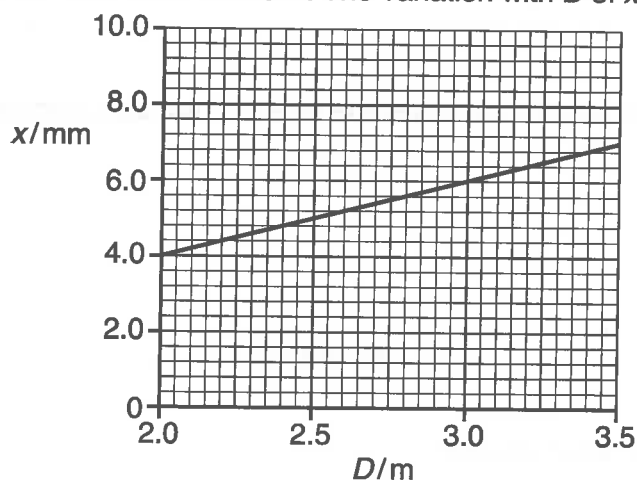


Fig. 5.2

- (i) Use Fig. 5.2 to determine the slit separation  $a$ .

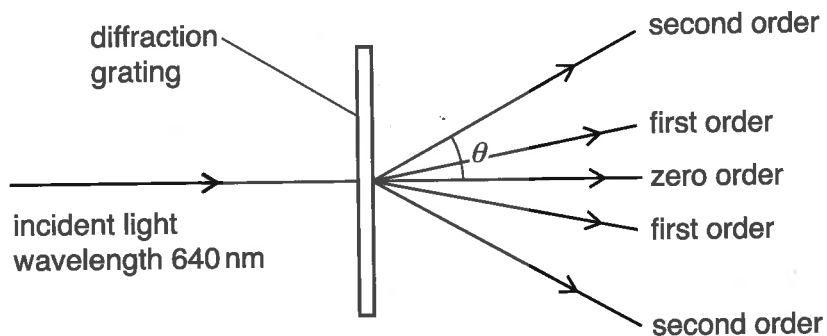
$a$  = ..... m [3]

- (ii) The laser is now replaced by another laser that emits light of a shorter wavelength. On Fig. 5.2, sketch a possible line to show the variation with  $D$  of  $x$  for the fringes that are now produced. [2]

**46. O/N 18/P22/Q5**

booklet (scan + OCR)

Red light of wavelength 640 nm is incident normally on a diffraction grating having a line spacing of  $1.7 \times 10^{-6}$  m, as shown in Fig. 5.1.

**Fig. 5.1** (not to scale)

The second order diffraction maximum of the light is at an angle  $\theta$  to the direction of the incident light.

**(a)** Show that angle  $\theta$  is  $49^\circ$ .

[3]

**(b)** Determine a different wavelength of **visible** light that will also produce a diffraction maximum at an angle of  $49^\circ$ .

wavelength = ..... m [2]

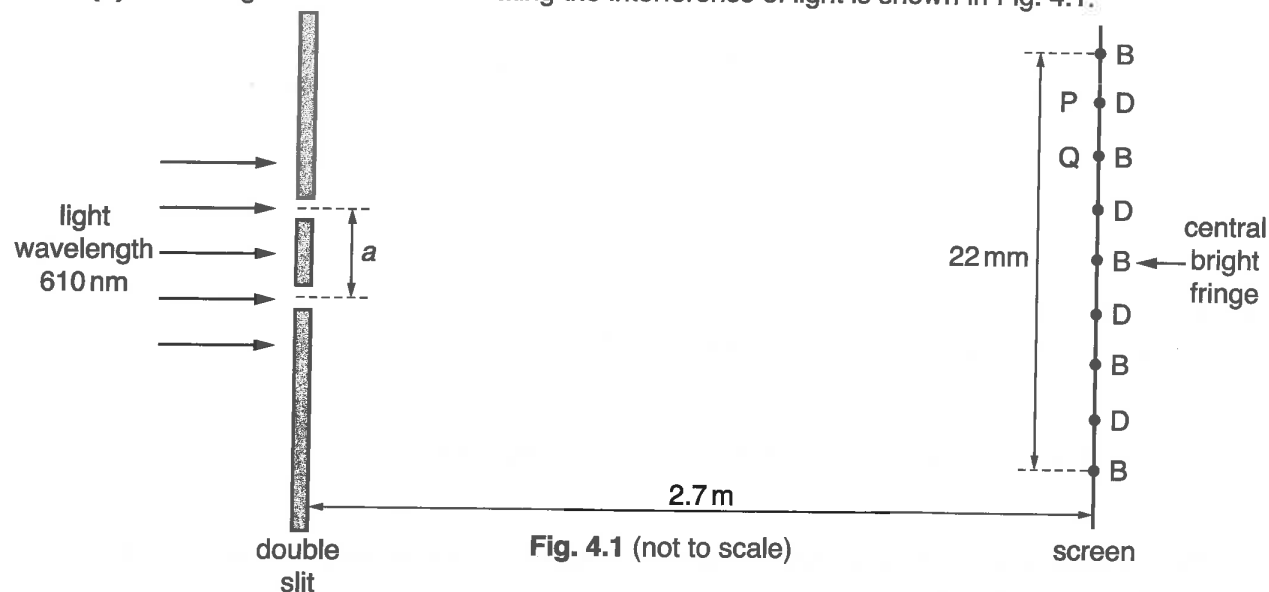
**47. O/N 18/P21/Q4**

booklet (scan + OCR)

**(a)** State the principle of superposition.

.....  
 ..... [2]

(b) An arrangement for demonstrating the interference of light is shown in Fig. 4.1.



The wavelength of the light is  $610\text{ nm}$ . The distance between the double slit and the screen is  $2.7\text{ m}$ .

An interference pattern of bright fringes and dark fringes is observed on the screen. The centres of the bright fringes are labelled B and centres of the dark fringes are labelled D. Point P is the centre of a particular dark fringe and point Q is the centre of a particular bright fringe, as shown in Fig. 4.1. The distance across five bright fringes is  $22\text{ mm}$ .

(i) The light waves leaving the two slits are coherent.

State what is meant by *coherent*.

..... [1]

(ii) 1. State the phase difference between the waves meeting at Q.

phase difference = ..... $^{\circ}$

2. Calculate the path difference, in nm, of the waves meeting at P.

path difference = ..... nm [2]

(iii) Determine the distance  $a$  between the two slits.

$a =$  ..... m [3]

(iv) A higher frequency of visible light is now used. State and explain the change to the separation of the fringes.

..... [1]

(v) The intensity of the light incident on the double slit is now increased without altering its frequency. Compare the appearance of the fringes after this change with their appearance before this change.

.....

..... [2]

**48. M/J 18/P22/Q4**

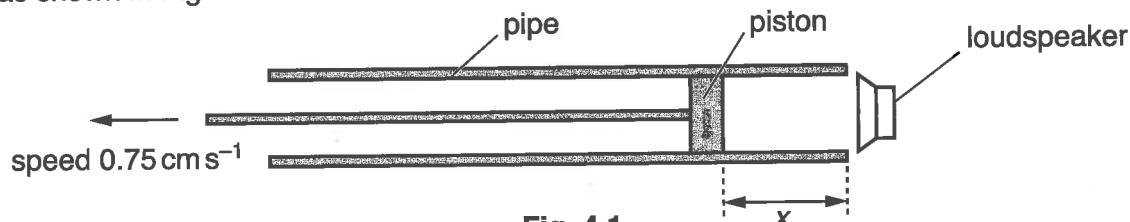
booklet (scan + OCR)

**(a) (i)** Define the *wavelength* of a progressive wave.

..... [1]

**(ii)** State what is meant by an *antinode* of a stationary wave.

..... [1]

**(b)** A loudspeaker producing sound of constant frequency is placed near the open end of a pipe, as shown in Fig. 4.1.**Fig. 4.1**

A movable piston is at distance  $x$  from the open end of the pipe. Distance  $x$  is increased from  $x = 0$  by moving the piston to the left with a constant speed of  $0.75 \text{ cm s}^{-1}$ .

The speed of the sound in the pipe is  $340 \text{ m s}^{-1}$ .

**(i)** A much louder sound is first heard when  $x = 4.5 \text{ cm}$ . Assume that there is an antinode of a stationary wave at the open end of the pipe.

Determine the frequency of the sound in the pipe.

frequency = ..... Hz [3]

**(ii)** After a time interval, a second much louder sound is heard. Calculate the time interval between the first louder sound and the second louder sound being heard.

time interval = ..... s [2]

**49. M/J 18/P21/Q5**

booklet (scan + OCR)

- (a) When monochromatic light is incident normally on a diffraction grating, the emergent light waves have been diffracted and are coherent.

Explain what is meant by

- (i) *diffracted waves*,

..... [1]

(ii) *coherent waves*.

- ..... [1]
- (b) Light consisting of only two wavelengths  $\lambda_1$  and  $\lambda_2$  is incident normally on a diffraction grating. The third order diffraction maximum of the light of wavelength  $\lambda_1$  and the fourth order diffraction maximum of the light of wavelength  $\lambda_2$  are at the same angle  $\theta$  to the direction of the incident light.

- (i) Show that the ratio  $\frac{\lambda_2}{\lambda_1}$  is 0.75.  
Explain your working.

[2]

- (ii) The difference between the two wavelengths is 170 nm.  
Determine wavelength  $\lambda_1$ .

$\lambda_1 = \dots\dots\dots$  nm [1]

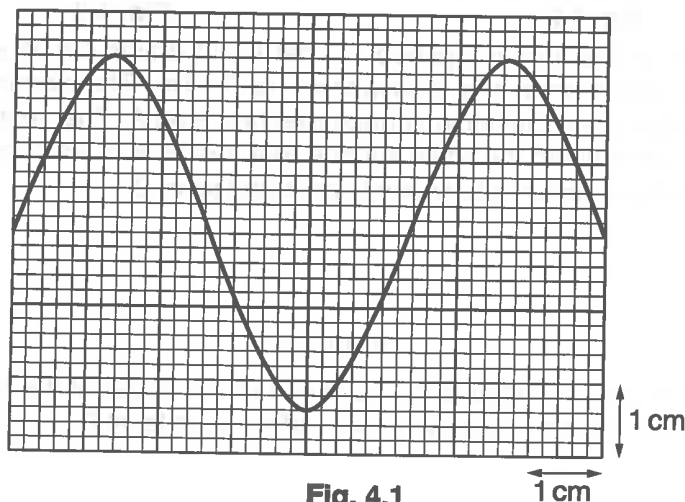
**50. MAR 18/P22/Q4**

booklet (scan + OCR)

- (a) State the conditions required for the formation of a stationary wave.

..... [2]

- (b) The sound from a loudspeaker is detected by a microphone that is connected to a cathode-ray oscilloscope (c.r.o.). Fig. 4.1 shows the trace on the screen of the c.r.o.

**Fig. 4.1**

In air, the sound wave has a speed of  $330 \text{ m s}^{-1}$  and a wavelength of  $0.18 \text{ m}$ .

(i) Calculate the frequency of the sound wave.

frequency = ..... Hz [2]

(ii) Determine the time-base setting, in  $\text{s cm}^{-1}$ , of the c.r.o.

time-base setting = .....  $\text{s cm}^{-1}$  [2]

(iii) The intensity of the sound from the loudspeaker is now halved. The wavelength of the sound is unchanged. Assume that the amplitude of the trace is proportional to the amplitude of the sound wave.

On Fig. 4.1, sketch the new trace shown on the screen of the c.r.o. [2]

(c) The loudspeaker in (b) is held above a vertical tube of liquid, as shown in Fig. 4.2.

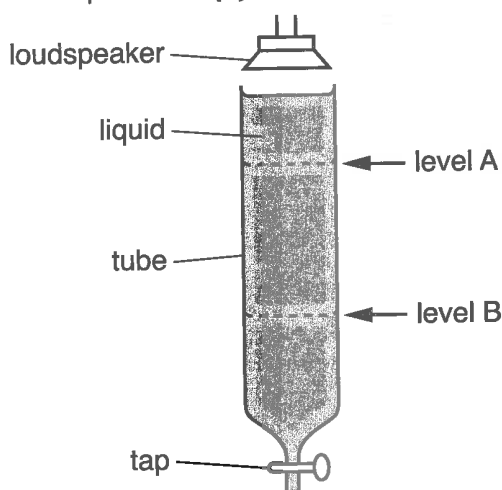


Fig. 4.2

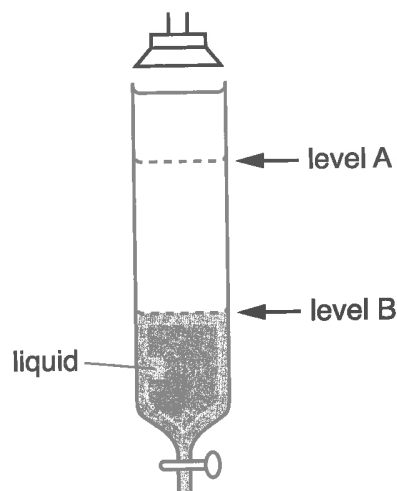


Fig. 4.3

A tap at the bottom of the tube is opened so that liquid drains out at a constant rate. The wavelength of the sound from the loudspeaker is  $0.18 \text{ m}$ . The sound that is heard first becomes much louder when the liquid surface reaches level A. The next time that the sound becomes much louder is when the liquid surface reaches level B, as shown in Fig. 4.3.

(i) Calculate the vertical distance between level A and level B.

distance = ..... m [1]

(ii) On Fig. 4.3, label with the letter N the positions of the nodes of the stationary wave that is formed in the air column when the liquid surface is at level B. [1]

- (iii) The mass of liquid leaving the tube per unit time is  $6.7 \text{ g s}^{-1}$ . The tube has an internal cross-sectional area of  $13 \text{ cm}^2$ . The density of the liquid is  $0.79 \text{ g cm}^{-3}$ . Calculate the time taken for the liquid to move from level A to level B.

time = ..... s [2]

### 51. O/N 17/P21/Q3

booklet (scan + OCR)

- (a) State the difference between a stationary wave and a progressive wave in terms of
- the energy transfer along the wave,  
..... [1]
  - the phase of two adjacent vibrating particles.  
..... [1]
- (b) A tube is open at both ends. A loudspeaker, emitting sound of a single frequency, is placed near one end of the tube, as shown in Fig. 3.1.

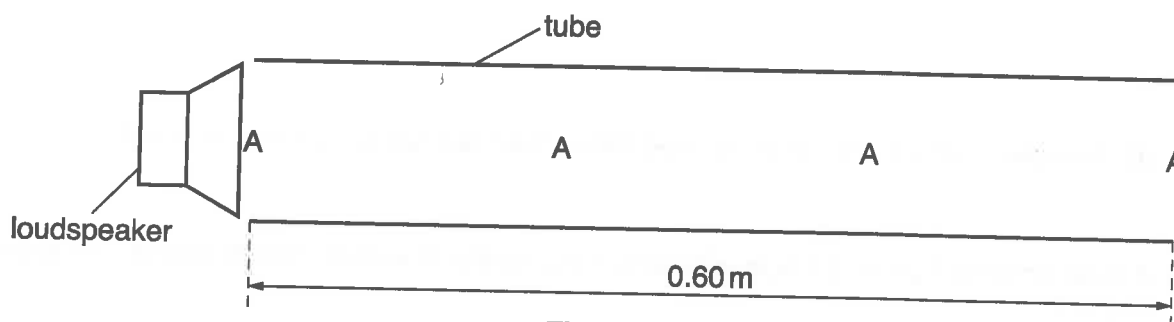


Fig. 3.1

The speed of the sound in the tube is  $340 \text{ m s}^{-1}$ . The length of the tube is  $0.60 \text{ m}$ . A stationary wave is formed with an antinode A at each end of the tube and two antinodes inside the tube.

- State what is meant by an *antinode* of the stationary wave.  
..... [1]
- State the distance between a node and an adjacent antinode.

distance = ..... m [1]

- Determine, for the sound in the tube,
  - the wavelength,

wavelength = ..... m [1]

## 2. the frequency.

frequency = ..... Hz [2]

- (iv) Determine the minimum frequency of the sound from the loudspeaker that produces a stationary wave in the tube.

minimum frequency = ..... Hz [2]

## 52. M/J 17/P22/Q6

booklet (scan + OCR)

- (a) Interference fringes may be observed using a light-emitting laser to illuminate a double slit. The double slit acts as two sources of light.

Explain

- (i) the part played by diffraction in the production of the fringes,

..... [2]

- (ii) the reason why a double slit is used rather than two separate sources of light.

..... [1]

- (b) A laser emitting light of a single wavelength is used to illuminate slits  $S_1$  and  $S_2$ , as shown in Fig. 6.1.

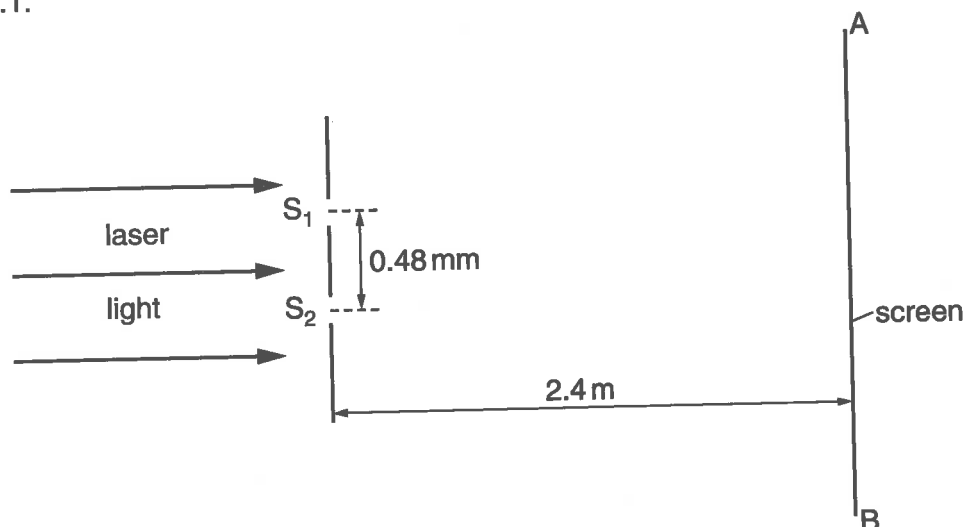
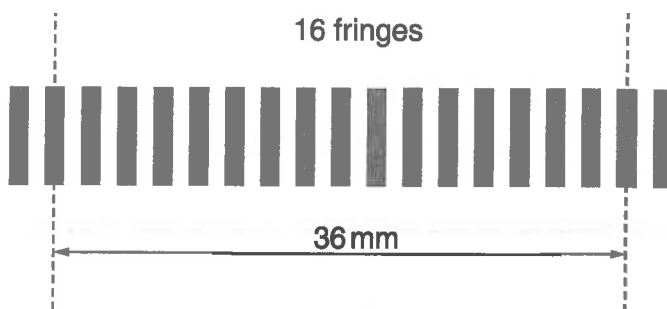


Fig. 6.1 (not to scale)

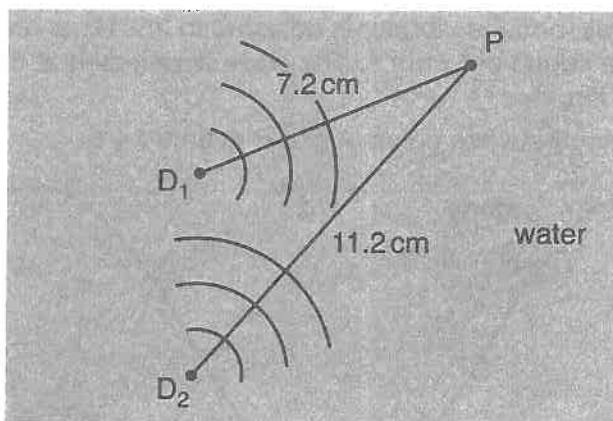
An interference pattern is observed on the screen AB. The separation of the slits is 0.48 mm. The slits are 2.4 m from AB. The distance on the screen across 16 fringes is 36 mm, as illustrated in Fig. 6.2.

**Fig. 6.2**

Calculate the wavelength of the light emitted by the laser.

wavelength = .....m [3]

- (c) Two dippers  $D_1$  and  $D_2$  are used to produce identical waves on the surface of water, as illustrated in Fig. 6.3.

**Fig. 6.3** (not to scale)

Point P is 7.2 cm from  $D_1$  and 11.2 cm from  $D_2$ .

The wavelength of the waves is 1.6 cm. The phase difference between the waves produced at  $D_1$  and  $D_2$  is zero.

- (i) State and explain what is observed at P.

.....

..... [2]

- (ii) State and explain the effect on the answer to (c)(i) if the apparatus is changed so that, separately,

1. the phase difference between the waves at  $D_1$  and at  $D_2$  is  $180^\circ$ ,

.....

2. the intensity of the wave from  $D_1$  is less than the intensity of that from  $D_2$ .

.....

..... [2]

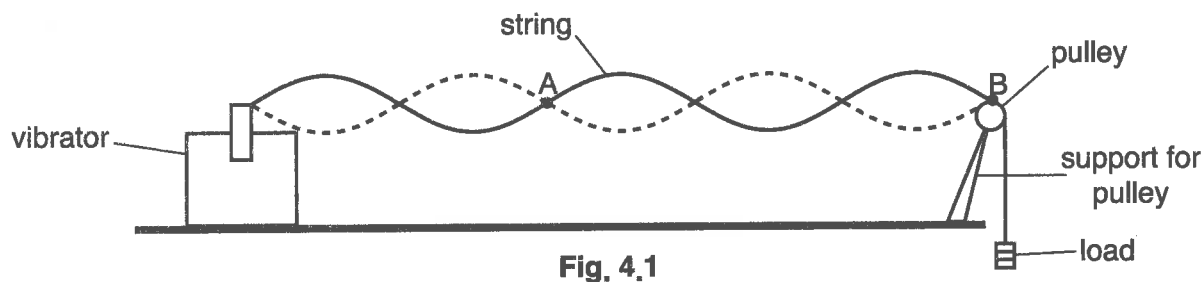
**53. M/J 17/P21/Q4**

booklet (scan + OCR)

**(a)** State the conditions required for the formation of stationary waves.

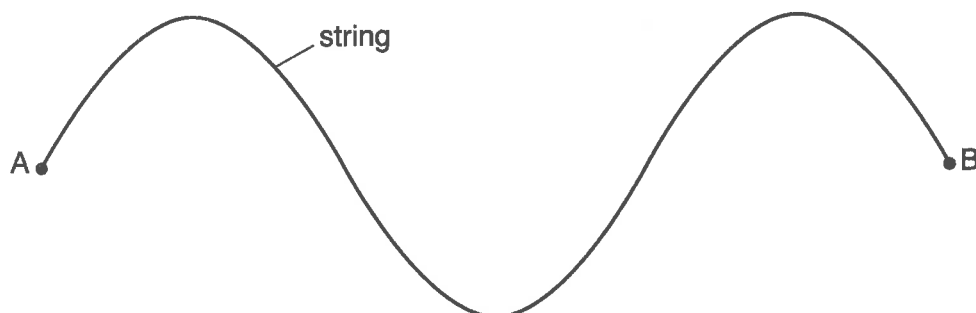
.....

..... [2]

**(b)** One end of a string is attached to a vibrator. The string is stretched by passing the other end over a pulley and attaching a load, as illustrated in Fig. 4.1.**Fig. 4.1**

The frequency of vibration of the vibrator is adjusted to 250 Hz and a transverse wave travels along the string with a speed of  $12 \text{ m s}^{-1}$ . The wave is reflected at the pulley and a stationary wave forms on the string.

Fig. 4.2 shows the string between points A and B at time  $t = t_1$ .

**Fig. 4.2**

At time  $t = t_1$  the string has maximum displacement.

**(i)** Calculate the distance AB.

distance = .....m [2]

**(ii)** On Fig. 4.2, sketch the position of the string between A and B at times

1.  $t = t_1 + 2.0 \text{ ms}$  (label this line P),
2.  $t = t_1 + 5.0 \text{ ms}$  (label this line Q).

**54. M/J 17/P23/Q5**

booklet (scan + OCR)

**(a)** A diffraction grating is used to determine the wavelength of light.**(i)** Describe the diffraction of light at a diffraction grating.

.....  
..... [2]

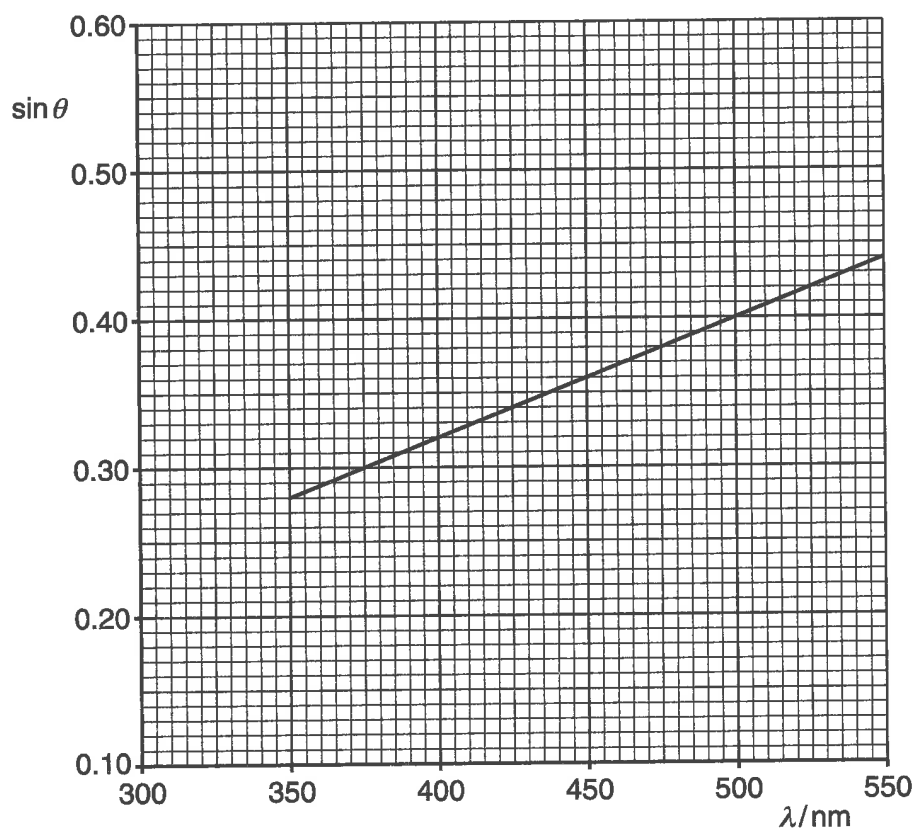
**(ii)** By reference to interference, explain**1.** the zero order maximum,

.....  
.....

**2.** the first order maximum.

.....  
..... [3]

**(b)** A diffraction grating is used with different wavelengths of light. The angle  $\theta$  of the second order maximum is measured for each wavelength. The variation with wavelength  $\lambda$  of  $\sin \theta$  is shown in Fig. 5.1.

**Fig. 5.1**

- (i) Determine the gradient of the line shown in Fig. 5.1.

gradient = ..... [2]

- (ii) Use the gradient determined in (i) to calculate the slit separation  $d$  of the diffraction grating.

$d = \dots\dots\dots \text{m}$  [2]

- (iii) On Fig. 5.1, sketch a line to show the results that would be obtained for the first order maxima. [1]

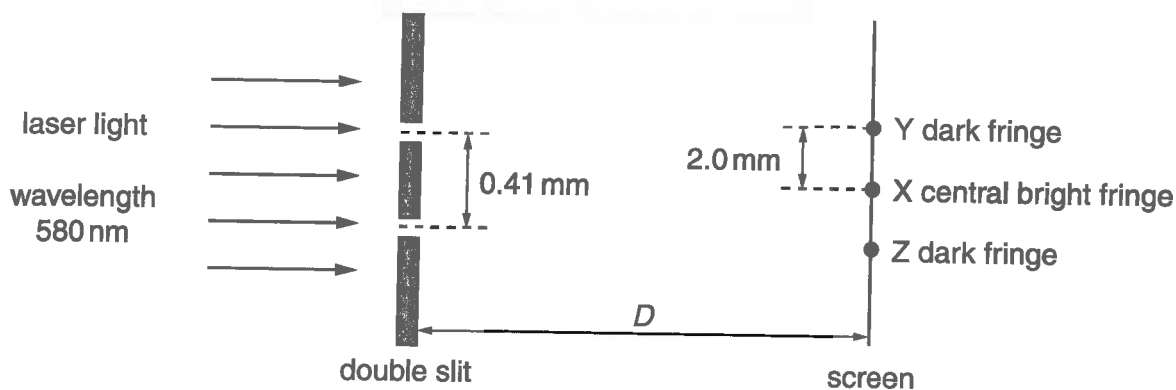
**55. O/N 16/P22/Q4**

booklet (scan + OCR)

**(a)** State what is meant by the *diffraction* of a wave.

.....

..... [2]

**(b)** An arrangement for demonstrating the interference of light is shown in Fig. 4.1.**Fig. 4.1** (not to scale)

The wavelength of the light from the laser is 580 nm. The separation of the slits is 0.41 mm. The perpendicular distance between the double slit and the screen is  $D$ .

Coherent light emerges from the slits and an interference pattern is observed on the screen. The central bright fringe is produced at point X. The closest dark fringes to point X are produced at points Y and Z. The distance XY is 2.0 mm.

**(i)** Explain why a bright fringe is produced at point X.

.....

..... [2]

**(ii)** State the difference in the distances, in nm, from each slit to point Y.

distance = ..... nm [1]

**(iii)** Calculate the distance  $D$ .

$D = \dots\dots\dots$  m [3]

**(iv)** The intensity of the light passing through the two slits was initially the same. The intensity of the light through **one** of the slits is now reduced. Compare the appearance of the fringes before and after the change of intensity.

.....

..... [2]

## Answers Section

### 1. M/J 26/P22/Q3/b(ii),c(ii)

|          |                                                                                                             |           |                                                                                                                                                                                    |
|----------|-------------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3(b)(ii) | $\lambda = 3.9 - 2.4$<br>$= 1.5$                                                                            | <b>C1</b> | The C1 marks may be credited independently of each other                                                                                                                           |
|          | $v = f\lambda$ or $v = \lambda / T$<br>$= 3 \times 1.5$ or $= 1.5 / 0.33$                                   | <b>C1</b> | Allow $c = f\lambda$<br>ECF $T$ from 3(b)(i)                                                                                                                                       |
|          | $= 4.5 \text{ m s}^{-1}$                                                                                    | <b>A1</b> |                                                                                                                                                                                    |
| 3(c)(ii) | The path difference (from A and B) is not a whole number of wavelengths / (The waves at P are) not in phase | <b>B1</b> | Allow '(waves are) out of phase' / 'phase difference not zero'<br>Not '(waves are) 180° out of phase / antiphase'<br>Allow 'not (complete) constructive interference (at P)' (BOD) |
|          | (The amplitude) decreases                                                                                   | <b>B1</b> | Allow 'it' to mean amplitude (unless it clearly refers to something else)<br>Allow 'less' / 'smaller' for decreases                                                                |

### 2. M/J 26/P23/Q3/a,b,c

|           |                                                                                                                                              |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3(a)      | (when two or more) waves meet / overlap (at a point)                                                                                         | <b>B1</b> | Allow 'waves intersect / merge'<br>Ignore 'waves superpose / interfere / combine / interact'<br>Allow answer given in the context of two waves forming a stationary wave. E.g. 'waves travelling in opposite directions'                                                                                                                                                                                                                                                                                                   |
|           | (resultant) displacement is sum of the individual displacements                                                                              | <b>B1</b> | Allow 'resultant wave' or 'total displacement' for 'resultant displacement'<br>Ignore just 'resultant displacement is the sum of the waves'<br>Ignore 'sum of displacement is the sum of the individual displacements' but <u>allow</u> 'sum of resultant displacement is the sum of the individual displacements'<br>Not 'amplitude' instead of 'displacement'<br><br>ignore 'waves of same speed / frequency / wavelength'<br>ignore the word 'algebraic' (as in 'algebraic sum')<br>ignore 'coherent / identical waves' |
| 3(b)(i)   | constant phase difference                                                                                                                    | <b>B1</b> | Ignore 'same frequency/wavelength/speed'<br>Ignore 'have the same phase difference'<br>Ignore 'phase relationship' for 'phase difference'<br>Ignore 'in phase'                                                                                                                                                                                                                                                                                                                                                             |
| 3(b)(ii)  | (the light) diffracts                                                                                                                        | <b>B1</b> | Ignore 'spreads'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3(b)(iii) | waves (from each slit) meet / superpose (at the screen)                                                                                      | <b>B1</b> | Allow 'overlap / superimpose / interfere / combine' for 'superpose'<br>Ignore 'interact / collide / coincide / intersect'                                                                                                                                                                                                                                                                                                                                                                                                  |
|           | waves have 0 path difference / 0 phase difference                                                                                            | <b>B1</b> | Allow 'in phase'<br>Ignore 360°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|           | constructive interference occurs / (resultant wave has) maximum amplitude                                                                    | <b>B1</b> | Ignore 'intensity is maximum'<br>Ignore references to nodes and antinodes                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3(b)(iv)  | $\lambda = ax / D$                                                                                                                           | <b>C1</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|           | $x = (560 \times 10^{-9} \times 0.43) / (1.4 \times 10^{-5})$<br>$= 0.017$                                                                   | <b>C1</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|           | distance = $0.017 / 2$<br>$= 0.0086 \text{ m}$                                                                                               | <b>A1</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3(c)(i)   | <i>either</i> : B marked above Y at same distance above Y as XY<br><i>or</i> : B marked below 0.43 m arrow, at a distance from X that is 2XY | <b>B1</b> | tolerance $\pm 3 \text{ mm}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3(c)(ii)  | D marked below X at distance below X equal to XY                                                                                             | <b>B1</b> | tolerance $\pm 3 \text{ mm}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

**3. M/J 26/P24/Q3/a,b,c,d**

|      |                                                                               |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|-------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3(a) | wave reflects at fixed point                                                  | <b>B1</b> | Ignore rebound / bounces back                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|      | incident and reflected waves superpose                                        | <b>B1</b> | Allow 'wave from (vibration) generator / initial / original wave' for 'incident wave'<br>Allow 'two progressive / travelling waves in opposite directions' for 'incident and reflected waves' (BOD), but ignore just 'different directions'<br>Allow 'overlap / superimpose / interfere / combine' for 'superpose'<br>Allow 'destructive / constructive interference' to imply 'superpose'<br>Ignore<br>'meet / interact / collide / coincide / intersect'<br>Ignore reference to phase difference<br>Ignore reference to coherence |
| 3(b) | cross vertically aligned with a node                                          | <b>A1</b> | within one dash length                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3(c) | $\lambda = 1.80 / 2$<br>$= 0.90$                                              | <b>C1</b> | The C marks are independent.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|      | $v = f\lambda$<br>$= 26 \times 0.90$                                          | <b>C1</b> | Allow use of $T = 1/f$ and $v = \lambda / T$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|      | $= 23 \text{ m s}^{-1}$                                                       | <b>A1</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3(d) | Line touches both fixed points and crosses the dotted line once in the middle | <b>B1</b> | Middle judged by eye below the 1.8 m label<br>Allow line to be $\pm 2 \text{ mm}$ from fixed point<br>ignore amplitude                                                                                                                                                                                                                                                                                                                                                                                                              |
|      | Line with sinusoidal shape from fixed point to fixed point                    | <b>B1</b> | Ignore a second correct solid or dashed line (candidate mirroring presentation in Fig 3.1).<br>Allow a sinusoid to include a 'half-loop' for the fundamental mode                                                                                                                                                                                                                                                                                                                                                                   |

**4. MAR 26/P22/Q5/b,c**

|           |                                                                                                         |           |
|-----------|---------------------------------------------------------------------------------------------------------|-----------|
| 5(b)(i)   | a cross vertically in line with a peak or trough on the string                                          | <b>B1</b> |
| 5(b)(ii)  | $(\lambda =) 0.96 \times 2 / 3$<br>$= 0.64$                                                             | <b>M1</b> |
|           | $v = f\lambda$                                                                                          | <b>C1</b> |
|           | $= 25 \times 0.96 \times 2 / 3 = 16 \text{ (m s}^{-1}\text{)}$                                          | <b>A1</b> |
| 5(b)(iii) | $\lambda = 16 / 30$ or $\lambda = 0.53$ or $\lambda = L / 1.8$ or $\lambda = 0.55L$                     | <b>B1</b> |
|           | $\lambda \neq 2L / n$ or $\lambda \neq 1.92 / n$ <u>and</u> (so) stationary wave not formed             | <b>B1</b> |
|           | OR<br>$f \neq vn / 2L$ or $f \neq 16n / 2L$ or $f \neq 8.3n$ <u>and</u> (so) stationary wave not formed | <b>B2</b> |
| 5(c)(i)   | $n = d \sin \theta / \lambda$                                                                           | <b>C1</b> |
|           | $= ((1 / 6.7 \times 10^6) \times \sin(37)) / 4.5 \times 10^{-7}$                                        | <b>C1</b> |
|           | $= 2$                                                                                                   | <b>A1</b> |
| 5(c)(ii)  | decreases                                                                                               | <b>B1</b> |

**5. O/N 25/P21/Q4/a,b**

|      |                                     |           |
|------|-------------------------------------|-----------|
| 4(a) | wave passing through gap / aperture | <b>B1</b> |
|      | (wave) spreads (out)                | <b>B1</b> |

|           |                                                                                              |           |
|-----------|----------------------------------------------------------------------------------------------|-----------|
| 4(b)(i)   | $n\lambda = d \sin \theta$                                                                   | <b>C1</b> |
|           | $\theta = \sin^{-1} [(540 \times 10^{-9}) / (5.0 \times 10^{-6})] = 6.2^\circ$               | <b>A1</b> |
| 4(b)(ii)  | $\theta = 12^\circ$                                                                          | <b>A1</b> |
| 4(b)(iii) | peaks / maxima shown at $\theta = 0, \pm 6^\circ$ and $\pm 12^\circ$                         | <b>B1</b> |
|           | peaks / maxima and zero intensity in between                                                 | <b>B1</b> |
|           | central peak / maxima at intensity $I_0$ and the other peaks all equal to or less than $I_0$ | <b>B1</b> |

**6. O/N 25/P22/Q4/a,b,c**

|      |                                                                                                                                    |           |
|------|------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 4(a) | light diffracts / spreads (at slit(s))                                                                                             | <b>B1</b> |
|      | light (from each slit) superposes / interferes (at screen)                                                                         | <b>B1</b> |
|      | when phase difference is 0 / path difference is $n\lambda$ (where $n$ is an integer) a bright fringe is formed                     | <b>B1</b> |
|      | or<br>when phase difference is $180^\circ$ / path difference is $(n+1)\lambda/2$ (where $n$ is an integer) a dark fringe is formed |           |
| 4(b) | $\lambda = ax / D$                                                                                                                 | <b>C1</b> |
|      | $\lambda = 1.0 \times 10^{-3} \times 3.3 \times 10^{-3} / 4.8$                                                                     | <b>C1</b> |
|      | $(= 6.875 \times 10^{-7} \text{ m})$                                                                                               |           |
|      | $f = v / \lambda$                                                                                                                  | <b>C1</b> |
|      | $= 3.0 \times 10^8 / 6.875 \times 10^{-7}$                                                                                         | <b>A1</b> |
|      | $= 4.4 \times 10^{14} \text{ Hz}$                                                                                                  |           |
| 4(c) | the light / beams (have different frequencies so) are not coherent / do not have constant phase difference                         | <b>B1</b> |

**7. O/N 25/P24/Q4/a,b(i,iv)**

|          |                                                                                          |           |
|----------|------------------------------------------------------------------------------------------|-----------|
| 4(a)     | when two (or more) waves meet                                                            | <b>M1</b> |
|          | (resultant) displacement equals the sum of the displacements of the (two separate) waves | <b>A1</b> |
| 4(b)(i)  | the incident and reflected waves superpose                                               | <b>B1</b> |
|          | (the waves superpose so that the resultant) amplitude is maximum at an antinode          | <b>B1</b> |
|          | (the waves superpose so that the resultant) amplitude is minimum / zero at a node        | <b>B1</b> |
| 4(b)(iv) | distance = $0.026 / 4$<br>$= 6.5 \times 10^{-3} \text{ m}$                               | <b>A1</b> |

**8. M/J 25/P21/Q4/a,b**

|          |                                                                                                                  |           |
|----------|------------------------------------------------------------------------------------------------------------------|-----------|
| 4(a)     | (when two or more) waves meet / overlap (at a point)                                                             | <b>B1</b> |
|          | (resultant) displacement is sum of the individual displacements                                                  | <b>B1</b> |
| 4(b)(i)  | Fringe width, $x = 3.2 \times 10^{-2} / 8$                                                                       | <b>C1</b> |
|          | $= 4.0 \times 10^{-3} \text{ (m)}$                                                                               |           |
|          | $D = ax / \lambda$                                                                                               | <b>C1</b> |
|          | $= (4.0 \times 10^{-3} \times 0.16 \times 10^{-3}) / 7.2 \times 10^{-7}$                                         |           |
|          | $= 0.89 \text{ m}$                                                                                               | <b>A1</b> |
| 4(b)(ii) | Curved line with a negative gradient of decreasing magnitude throughout, from slit separation 0.04 mm to 0.16 mm | <b>B1</b> |
|          | Line of negative gradient ending at (0.16, 0.4), from slit separation 0.04 mm                                    | <b>B1</b> |
|          | Line of negative gradient passing through (0.08, 0.8) and (0.04, 1.6)                                            | <b>B1</b> |

**9. M/J 25/P22/Q3/b(ii),c**

|          |                                                                                           |           |
|----------|-------------------------------------------------------------------------------------------|-----------|
| 3(b)(ii) | (X and Y have a <u>constant</u> phase difference (of $180^\circ$ ) so (they are) coherent | <b>B1</b> |
| 3(c)     | A single wave of amplitude 10.0 cm                                                        | <b>B1</b> |
|          | A single negative sine wave of wavelength 0.40 m                                          | <b>B1</b> |

**10. M/J 25/P23/Q6/a,b(ii,iii)**

|           |                                                                                                                       |           |
|-----------|-----------------------------------------------------------------------------------------------------------------------|-----------|
| 6(a)      | wave passes (through) an aperture and spreads<br><b>or</b><br>wave passes (by / through / around) an edge and spreads | <b>B1</b> |
| 6(b)(ii)  | $d = n\lambda / \sin \theta$                                                                                          | <b>C1</b> |
|           | $d = (2 \times 720 \times 10^{-9}) / \sin 26$                                                                         | <b>C1</b> |
|           | $(= 3.3 \times 10^{-6} \text{ m})$                                                                                    | <b>A1</b> |
|           | number of lines per m $= 1 / (3.3 \times 10^{-6})$<br>$= 3.0 \times 10^5 \text{ m}^{-1}$                              |           |
| 6(b)(iii) | $\lambda \times 3 = 720 \times 2$ <b>or</b> $\frac{1}{3.0 \times 10^5} \sin 26 = 3\lambda$                            | <b>C1</b> |
|           | $\lambda = 480 \text{ nm}$                                                                                            | <b>A1</b> |

**11. M/J 25/P24/Q4/b(ii,iii,iv)**

|           |                                                                                                                                                                                                       |             |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 4(b)(ii)  | (difference in path lengths, $XQ - YQ$ ) $= [72^2 + 30^2]^{\frac{1}{2}} - 72 = 6 \text{ (cm)}$                                                                                                        | <b>A1</b>   |
|           | Or calculate angle Q using tan and then cos to obtain XQ of 78<br>$\tan \theta = 30 / 72$ so $\theta = 22.6$<br>$XQ = 72 / \cos 22.6 = 78 \text{ (cm)}$<br>path difference $78 - 72 = 6 \text{ (cm)}$ | <b>(A1)</b> |
|           | Or calculate angle X using tan and then cos to obtain XQ of 78<br>$\tan \theta = 72 / 30$ so $\theta = 67.4$<br>$XQ = 72 / \sin 67.4 = 78 \text{ (cm)}$<br>path difference $78 - 72 = 6 \text{ (cm)}$ | <b>(A1)</b> |
| 4(b)(iii) | path difference $(= 6 \text{ cm} / 4 \text{ cm}) \lambda = 1.5\lambda$                                                                                                                                | <b>M1</b>   |
|           | (so) phase difference (at Q) $= 540^\circ$ <b>or</b> $180^\circ$                                                                                                                                      | <b>M1</b>   |
|           | (so) intensity is minimum                                                                                                                                                                             | <b>A1</b>   |
| 4(b)(iv)  | amplitude changes from maximum (at P) to minimum (at Q)                                                                                                                                               | <b>C1</b>   |
|           | amplitude changes from maximum (at P) to minimum to maximum to minimum (at Q)                                                                                                                         | <b>A1</b>   |

**12. MAR 25/P22/Q4/a,b**

|          |                                                                            |           |
|----------|----------------------------------------------------------------------------|-----------|
| 4(a)     | (micro)wave (from the transmitter) reflects (at metal/sheet)               | <b>B1</b> |
|          | The incident and reflected waves superpose                                 | <b>B1</b> |
|          | (resultant) amplitude is maximum at an antinode                            | <b>B1</b> |
|          | (resultant) amplitude is minimum/zero at a node                            | <b>B1</b> |
| 4(b)(i)  | $\lambda = c / f$<br>$= 3 \times 10^8 / 6.3 \times 10^9$                   | <b>C1</b> |
|          | $= 0.048 \text{ m}$                                                        | <b>A1</b> |
| 4(b)(ii) | distance PQ $= \frac{1}{4}\lambda$<br>$= 0.048 / 4$<br>$= 0.012 \text{ m}$ | <b>A1</b> |

|           |                                                                                                                                                                                                                                                                           |           |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 4(b)(iii) | (Distance QR is the) same (as PQ) <b>and</b> one of:<br><ul style="list-style-type: none"> <li>Distance (between maxima / minima) does not depend on intensity</li> <li>distance depends <b>only</b> on wavelength</li> <li>wavelength is unchanged / constant</li> </ul> | <b>B1</b> |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|

**13. O/N 24/P21/Q6/a,b(ii,iii)**

|           |                                                                                                            |           |
|-----------|------------------------------------------------------------------------------------------------------------|-----------|
| 6(a)(i)   | for maximum intensity, waves must be in phase (at detection)                                               | <b>B1</b> |
|           | (phase difference at source means) waves are not in phase / have a $90^\circ$ phase difference <b>at O</b> | <b>B1</b> |
| 6(a)(ii)  | point B labelled on the line PQ below O                                                                    | <b>M1</b> |
|           | distance OB = distance OA                                                                                  | <b>A1</b> |
| 6(b)(ii)  | $\Delta x = 0.012 / 4$<br>$= 0.0030 \text{ m}$                                                             | <b>A1</b> |
| 6(b)(iii) | $\lambda = ax / D$                                                                                         | <b>C1</b> |
|           | distance $= \lambda D / a = 0.012 \times 2.3 / 0.18$<br>$= 0.15 \text{ m}$                                 | <b>A1</b> |

**14. O/N 24/P22/Q4/b**

|           |                                                                                             |           |
|-----------|---------------------------------------------------------------------------------------------|-----------|
| 4(b)(i)   | A marked at the open end of the pipe                                                        | <b>B1</b> |
| 4(b)(ii)  | $f = v / \lambda$                                                                           | <b>C1</b> |
|           | $\lambda = 4 \times 4.5 \times 10^{-2}$                                                     | <b>C1</b> |
|           | $f = 340 / (4 \times 4.5 \times 10^{-2})$<br>$= 1900 \text{ Hz}$                            | <b>A1</b> |
| 4(b)(iii) | the node–antinode distance is longer<br><b>or</b><br>the wavelength (of the wave) is longer | <b>M1</b> |
|           | (the speed of sound is constant so) the frequency (of the wave) is lower                    | <b>A1</b> |

**15. O/N 24/P23/Q5/a(i,ii)**

|          |                                                                                                                                                                                                                                         |           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 5(a)(i)  | cross labelled N marked at the intersection of the solid and dashed lines or at X or Y<br><b>and</b><br>cross labelled A marked at a peak or a trough                                                                                   | <b>B1</b> |
| 5(a)(ii) | (XY is 1.5 wavelengths, so) wavelength $= 0.48 \times (2 / 3) = 0.32 \text{ m}$<br><b>or</b><br>(wavelength is twice node–node distance so) $\lambda / 2 = 0.48 / 3 = 0.16 \text{ m}$ and wavelength $= 0.16 \times 2 = 0.32 \text{ m}$ | <b>B1</b> |

**16. M/J 24/P21/Q5/a,b,c,d**

|      |                                                                                                          |           |
|------|----------------------------------------------------------------------------------------------------------|-----------|
| 5(a) | wave(s) (travel along string and) reflect at fixed point / wall / Q / end / vibration generator / P      | <b>B1</b> |
|      | incident and reflected waves superpose                                                                   | <b>B1</b> |
| 5(b) | 0.80 m                                                                                                   | <b>A1</b> |
| 5(c) | same wavelength as original throughout <b>and</b> passing through intersection of solid and dashed lines | <b>B1</b> |
|      | reflected in dashed line <b>and</b> of same amplitude                                                    | <b>B1</b> |
| 5(d) | $180^\circ$                                                                                              | <b>A1</b> |

**17. M/J 24/P22/Q6/a,b(ii)**

|          |                                                                                    |           |
|----------|------------------------------------------------------------------------------------|-----------|
| 6(a)     | $n\lambda = d \sin \theta$                                                         | <b>C1</b> |
|          | $n = (3.8 \times 10^{-6} \times \sin 90^\circ) / (520 \times 10^{-9})$             | <b>C1</b> |
|          | ( = 7.3)                                                                           |           |
|          | number of bright fringes formed = 15 (given as an integer)                         | <b>A1</b> |
| 6(b)(ii) | (red light has) longer wavelength                                                  | <b>M1</b> |
|          | (so) number (of bright fringes formed) is less / fewer (bright fringes are formed) | <b>A1</b> |

**18. M/J 24/P23/Q4/d**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 4(d) | resultant displacement is the sum of the displacements of the waves (from $S_1$ and $S_2$ )<br><b>or</b><br>waves (from $S_1$ and $S_2$ ) <u>superpose</u> (at P)                                                                                                                                                                                                                                                                                                            | <b>B1</b> |
|      | Any one point from:<br><ul style="list-style-type: none"> <li>(at P) the waves (from the two sources) (always) destructively interfere</li> <li>(at P) the waves have a path difference that is (always) an odd number of half-wavelengths / their path difference is one and a half wavelengths</li> <li>(at P) the waves have a phase difference that is (always) <math>180^\circ</math> / they are in antiphase / crest of one wave meets trough of other wave</li> </ul> | <b>B1</b> |
|      | amplitude (of the resultant wave) is zero (at all times)                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>B1</b> |

**19. MAR 24/P22/Q6/a,b**

|           |                                                                                                                                                                                                                                                                                                                                                                                    |           |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 6(a)(i)   | Any three from:<br><ul style="list-style-type: none"> <li>Light diffracts at the (two) slits.</li> <li>Light (from each slit) meets / superposes (at the screen).</li> <li>When the phase difference is 0 (degrees) a bright fringe / (intensity) maximum is formed.</li> <li>When the phase difference is 180 (degrees) a dark fringe / (intensity) minimum is formed.</li> </ul> | <b>B3</b> |
| 6(a)(ii)  | $\lambda = ax / D$                                                                                                                                                                                                                                                                                                                                                                 | <b>C1</b> |
|           | $\lambda = 1.2 \times 10^{-3} \times (10.2 \times 10^{-3} / 6) / 3.1$                                                                                                                                                                                                                                                                                                              | <b>C1</b> |
|           | $\lambda = 6.6 \times 10^{-7} \text{ m}$                                                                                                                                                                                                                                                                                                                                           | <b>A1</b> |
| 6(a)(iii) | (new fringe separation will be) smaller                                                                                                                                                                                                                                                                                                                                            | <b>B1</b> |
| 6(b)(i)   | A cross at the intersection of the string and the mean position line.                                                                                                                                                                                                                                                                                                              | <b>B1</b> |
| 6(b)(ii)  | 0                                                                                                                                                                                                                                                                                                                                                                                  | <b>A1</b> |
| 6(b)(iii) | $180^\circ$                                                                                                                                                                                                                                                                                                                                                                        | <b>A1</b> |

**20. O/N 23/P21/Q4/a,b,c,d**

|         |                                                                                                                                                              |           |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 4(a)    | (when two or more) waves meet / overlap (at a point)                                                                                                         | <b>B1</b> |
|         | (resultant) displacement is the sum of the individual displacements                                                                                          | <b>B1</b> |
| 4(b)    | $\lambda = ax / D$                                                                                                                                           | <b>C1</b> |
|         | $\lambda = [(0.65 \times 10^{-3}) \times (1.7 \times 10^{-3})] / 1.9$                                                                                        | <b>C1</b> |
|         | $\lambda = 5.8 \times 10^{-7} \text{ m}$                                                                                                                     | <b>A1</b> |
| 4(c)(i) | waves are out of phase by a quarter of a cycle / period<br><b>or</b><br>when one wave has maximum/minimum displacement, the other wave has zero displacement | <b>B1</b> |

|          |                                                                                                                                                                                                                                                                                   |           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 4(c)(ii) | (statement 2 is) not correct (because)<br><br>waves do not have phase difference of $180^\circ$ / not in antiphase<br><b>or</b><br>one wave has some displacement when other has no displacement<br><b>or</b><br>the displacements of the waves are not always equal and opposite | <b>B1</b> |
| 4(d)     | more diffraction (of light/waves by slits)<br><b>or</b><br>light/waves are more spread (by slits)<br><b>or</b><br>light/waves (from slits) have less intensity                                                                                                                    | <b>B1</b> |
|          | more (bright/dark) fringes<br><b>or</b><br>bright fringes are less bright / are dimmer / have lower intensity<br><b>or</b><br>no change to fringe spacing/separation/width                                                                                                        | <b>B1</b> |

**21. O/N 23/P22/Q5/b,c**

|          |                                                                                                                               |           |
|----------|-------------------------------------------------------------------------------------------------------------------------------|-----------|
| 5(b)     | wave passes (through) an aperture<br><b>or</b><br>wave passes (by / through / around) an edge                                 | <b>B1</b> |
|          | wave spreads (into geometrical shadow)                                                                                        | <b>B1</b> |
| 5(c)(i)  | $n\lambda = d \sin \theta$                                                                                                    | <b>C1</b> |
|          | $d = (3 \times 4.3 \times 10^{-7}) / \sin 68^\circ$<br><br>$= 1.4 \times 10^{-6} \text{ m}$                                   | <b>A1</b> |
| 5(c)(ii) | $1.4 \times 10^{-6} \times \sin 68^\circ = 2 \times \lambda$<br><b>or</b><br>$3 \times 4.3 \times 10^{-7} = 2 \times \lambda$ | <b>C1</b> |
|          | $\lambda = 6.5 \times 10^{-7} \text{ m}$                                                                                      | <b>A1</b> |

**22. O/N 23/P23/Q5/a,b(i)**

|          |                                                                                                                                                                          |           |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 5(a)(i)  | cross labelled Y drawn:<br><br>at any position where wavefronts cross<br><b>or</b><br>centrally in a 'diamond' shape formed between any adjacent wavefronts from A and B | <b>B1</b> |
| 5(a)(ii) | cross labelled Z drawn on a wavefront from one source at a point midway between adjacent wavefronts from the other source                                                | <b>B1</b> |
| 5(b)(i)  | $\lambda = ax / D$                                                                                                                                                       | <b>C1</b> |
|          | $a = (2.9 \times 10^{-5} \times 140) / (1.2 \times 10^{-2})$                                                                                                             | <b>C1</b> |
|          | $= 0.34 \text{ m}$                                                                                                                                                       | <b>A1</b> |

**23. M/J 23/P21/Q5**

- (a) (i) infrared [1]
- (ii)  $v = f\lambda$  or  $c = f\lambda$   
 $f = 3.0 \times 10^8 / 8.4 \times 10^{-6}$   
 $= 3.6 \times 10^{13} \text{ (Hz)}$   
 $= 36 \text{ THz}$  [2]
- (b) (i) constant phase difference (between the waves) (with time) [1]
- (ii)  $\lambda = ax / D$   
 $x = 22 / 8$  or  $2.75 \text{ (mm)}$  or  $22 \times 10^{-3} / 8$  or  $2.75 \times 10^{-3} \text{ (m)}$   
 $a = (6.2 \times 10^{-7} \times 2.8) / (22 \times 10^{-3} / 8)$   
 $= 6.3 \times 10^{-4} \text{ m}$  [3]
- (c) (i) difference in distances  $= 6.2 \times 10^{-7} / 2$  [1]  
 $= 3.1 \times 10^{-7} \text{ m}$
- (ii) phase difference  $= 360^\circ$  [1]

**24. MAR 23/P22/Q5**

- (a) (i) period or  $T = 1 / 5000 (= 2 \times 10^{-4} \text{ s})$   
time-base setting  $= 1.5 \times 2 \times 10^{-4} / 6.0$  or  $2 \times 10^{-4} / 4.0$   
 $= 5 \times 10^{-5} \text{ s cm}^{-1}$  [2]
- (ii) new trace drawn with same period as original trace  
new trace drawn with amplitude greater than 1.0 cm  
new trace drawn with amplitude of 1.7 cm [3]
- (b) (i) path difference (from slits to P) is zero or phase difference (between waves at P) is zero (so constructive interference) [2]
- (ii)  $\lambda = ax / D$   
 $D = (3.6 \times 10^{-4} \times 4.0 \times 10^{-3}) / 630 \times 10^{-9}$   
 $= 2.3 \text{ m}$  [2]
- (c) upward sloping straight line starting from a non-zero value of  $x$  at  $\lambda = 400 \text{ nm}$  [1]

**25. M/J 22/P22/Q5**

- (a) constant phase difference (between the waves) [1]
- (b) (i) path difference  $= 1.5 \times 660$   
 $= 990 \text{ nm}$  [1]
- (ii) phase difference  $= 360^\circ \times 1.5$   
 $= 540^\circ$  [1]
- (c)  $\lambda = ax / D$   
 $x = (660 \times 10^{-9} \times 1.8) / 0.44 \times 10^{-3}$   
 $= 2.7 \times 10^{-3} \text{ m}$  [3]
- (d) bright fringes are brighter  
no change to dark fringes  
no change to (fringe) separation / (fringe) spacing [3]
- (e) (blue light has) shorter wavelength  
(so) decrease (slit) separation

**26. M/J 22/P23/Q5****(a) (i) power = intensity  $\times$  area**

$$= 1.3 \times 10^3 \times (\pi \times 0.055^2)$$

$$= 12 \text{ W}$$

**[2]****(ii) intensity = power / area**

$$= 12 / (\pi \times 0.0015^2)$$

$$= 1.7 \times 10^6 \text{ W m}^{-2}$$

**[1]****(b) (i) ( $\lambda =$ )  $v/f$  or  $c/f$** 

$$(\lambda =) 3.0 \times 10^8 / 3.7 \times 10^{15} = 8.1 \times 10^{-8} \text{ (m)}$$

**[2]****(ii) ultraviolet****[1]****(iii)  $d \sin \theta = n\lambda$  or  $(1/N) \times \sin \theta = n\lambda$** 

$$d = 1/2400 \times 10^3 \text{ (m)}$$

$$= 4.2 \times 10^{-7} \text{ (m)}$$

**or**

$$N = 2400 \times 10^3 \text{ (m}^{-1}\text{)}$$

$$n = 4.2 \times 10^{-7} \times \sin 90^\circ / 8.1 \times 10^{-8} \text{ or } \sin 90^\circ / 2400 \times 10^3 \times 8.1 \times 10^{-8}$$

$$n = 5.2 \text{ or } 5.1$$

**or**when  $n = 5$ ,  $\theta = 76.4^\circ$  and when  $n = 6$ ,  $\sin \theta > 1$ (so)  $n = 5$ 

$$\text{number of maxima} = (2 \times 5) + 1$$

$$= 11$$

**[4]****(iv) the wavelength has increased**

(so) number of maxima decreases

**[2]****27. MAR 22/P22/Q5****(a) (i) (two) waves travelling (at same speed) in opposite directions overlap**  
waves (of the same type) have same frequency/wavelength**[2]****(ii) phase difference = 0****[1]****(b) (i)  $f_o = f_s v / v - v_s$** 

$$543 = f \times 334 / (334 - 13)$$

$$f = 522 \text{ Hz}$$

**[2]****(ii) (the speed is) decreasing****[1]****(c) (i)  $I \propto A^2$** 

$$I_T / I_0 = \cos^2 20^\circ \text{ or } A_T / A_0 = \cos 20^\circ$$

$$\text{ratio} = 0.94$$

**[3]****(ii) angle =  $140^\circ$** **[1]**

(so) (much) louder sound can be heard

**[1]**

**28. O/N 21/P22/Q5**

(a) a (much) louder sound can be heard

(b)  $v = f\lambda$ 

$$\lambda = 340 / 530$$

$$(\lambda = 0.64 \text{ m})$$

$$\text{height} = 0.64 / 4$$

$$= 0.16 \text{ m}$$

[3]

(c) distance =  $0.64 / 2$  or  $0.16 \times 2$  or  $(\frac{3}{4} \times 0.64 - \frac{1}{4} \times 0.64)$ 

$$= 0.32 \text{ m}$$

[1]

... maximum displacement (of a point/particle on string/wave)

[1]

**29. O/N 21/P23/Q5**

(a) maximum displacement (of a point/particle on string/wave)

(b)  $v = \lambda / T$ 

or

$$v = f\lambda \text{ and } f = 1/T$$

$$T = 690 \times 10^{-9} / 3.00 \times 10^8$$

$$= 2.3 \times 10^{-15} \text{ s}$$

[3]

(c) (i)  $\lambda = ax / D$ 

$$G = x / D$$

$$(\text{so}) a = \lambda / G$$

[2]

(ii) straight line from origin always below printed line  
line is half the height of printed line at maximum  $D$ 

[2]

**30. M/J 21/P22/Q4**

(a) time for one oscillation/vibration/cycle

or

time between adjacent wavefronts (passing the same point)

or

shortest time between two wavefronts (passing the same point)

[1]

(b) (when two or more) waves meet/overlap (at a point)

(resultant) displacement is sum of the individual displacements

[2]

(c) (i) microwave(s)

[1]

(ii)  $v = \lambda / T$ 

or

$$v = f\lambda \text{ and } f = 1/T$$

$$T = 0.040 / 3.00 \times 10^8$$

$$= 1.33 \times 10^{-10} \text{ (s)}$$

$$= 1.33 \times 10^{-10} / 10^{-12} \text{ (ps)}$$

$$= 130 \text{ ps}$$

[3]

$$\text{(iii)} (1.380 - 1.240) / 0.040 = 3.5$$

or

$$1.380 / 0.040 - 1.240 / 0.040 = 3.5$$

[1]

$$\text{(iv)} \text{phase difference} = 1260^\circ \text{ or } 180^\circ$$

[1]

(v) (always) zero

[1]

(vi) increase in distance between (adjacent intensity) maxima/minima

[1]

**31. M/J 21/P21/Q4**

(a) distance moved by wavefront/energy during one cycle/oscillation/period (of source)

or

minimum distance between two wavefronts

- or  
distance between two adjacent wavefronts [1]
- (b)  $v = \lambda / T$   
or  
 $v = f\lambda$  and  $f = 1 / T$   
 $T = 460 \times 10^{-9} / 3.00 \times 10^8$   
 $= 1.5 \times 10^{-15} \text{ s}$  [3]
- (c) waves pass through/enter the slit(s)  
waves spread (into geometric shadow) [2]
- (d) (i)  $n\lambda = d \sin \theta$   
 $G = \sin \theta / \lambda$   
 $d = 4 / G$  [2]
- (ii) straight line from 400 nm to 700 nm that is always below printed line  
straight line has smaller gradient than printed line and is 5 small squares high at wavelength of 700 nm [2]

**32. M/J 21/P23/Q4**

- (a) (when two or more) waves meet/overlap (at a point)  
(resultant) displacement is sum of the individual displacements [2]
- (b) intensity  $\propto$  amplitude<sup>2</sup>  
maximum intensity =  $9I$  [2]
- (c) (i)  $x = \lambda D / a$   
 $= (550 \times 10^{-9} \times 1.2) / (0.35 \times 10^{-3})$   
 $= 1.9 \times 10^{-3} \text{ m}$  [3]
- (ii) red light has longer wavelength (than 550nm) so distance (between fringes) increases [1]

**33. MAR 21/P22/Q4**

- (a) (two or more) waves meet (at a point)  
(resultant) displacement is the sum of the individual displacements [2]
- (b) (i) it is a (wave) reflector / it reflects (the wave) [1]
- (ii)  $v = f\lambda$  or  $c = f\lambda$   
 $f = 3.0 \times 10^8 / 0.040$   
 $= 7.5 \times 10^9 \text{ (Hz)}$   
 $= 7.5 \times 10^9 / 10^9 \text{ (GHz)}$   
 $= 7.5 \text{ GHz}$  [3]
- (iii) 1 distance = 0.020 m  
2 number = 5 [2]

**34. O/N 20/P22/Q5**

- (a)  $v = f\lambda$  or  $c = f\lambda$   
 $f = 3.0 \times 10^8 / 0.040$   
 $= 7.5 \times 10^9 \text{ (Hz)}$   
 $= 7.5 \text{ GHz}$  [3]
- (b) (i) path difference = 0.020 m [1]
- (ii) phase difference =  $180^\circ$  [1]
- (c) (intensity) increases  
(intensity) increases by a factor of 4 [2]
- (d) (i) minimum moves closer to the maximum  
or  
decrease in separation of maximum and minimum [1]
- (ii) the maximum and minimum exchange places  
or  
the maximum becomes a minimum and the minimum becomes a maximum [1]

**35. O/N 20/P23/Q5**

- (a) (i)  $T = 2.0 \times 10^{-5} \times 6.0$  ( $= 1.2 \times 10^{-4} \text{ s}$ )  
 $f = 1 / (2.0 \times 10^{-5} \times 6.0)$   
 $= 8300 \text{ Hz}$  [2]
- (ii) new trace shows the same period  
 new trace shows amplitude of 10 small squares [2]
- (iii) (trace is a) vertical line [1]
- (b) (i)  $n\lambda = d \sin \theta$   
 $\lambda = (3.4 \times 10^{-6} \times \sin 16^\circ) / 2$   
 $= 4.7 \times 10^{-7} \text{ m}$  [2]
- (ii)  $n = 3.4 \times 10^{-6} (\times \sin 90^\circ) / 4.7 \times 10^{-7}$  or  $2 (\times \sin 90^\circ) / \sin 16^\circ$   
 $(= 7.2 \text{ or } 7.3)$   
 highest order = 7 [2]

**36. M/J 20/P22/Q4**

- (a) progressive waves transfer energy  
 or  
 stationary waves do not transfer energy [1]
- (b) (i) 0.32 m [1]
- (ii)  $v = \lambda / T$   
 or  
 $v = f\lambda$  and  $f = 1 / T$   
 $v = 0.32 / 0.020$  or  $50 \times 0.32$   
 $= 16 \text{ m s}^{-1}$  [2]
- (iii)  $450^\circ$  or  $90^\circ$  [1]
- (iv) (P has) maximum downward displacement at 0.005 s  
 returns to original position/point (at 0.010 s) [2]
- (c) (i) (position where) zero amplitude [1]
- (ii) 2 [1]
- (iii)  $180^\circ$  [1]
- (iv) string drawn between X and Y with one antinode midway along the string [1]

**37. M/J 20/P21/Q4**

- (a) (i) vibrations (of particles) are parallel to direction of energy propagation [1]
- (ii) waves meet/overlap (at a point)  
 (resultant) displacement is sum of individual displacements [2]
- (b) (i)  $\lambda = ax / D$   
 $= (3.7 \times 10^{-4} \times 4.3 \times 10^{-3}) / 2.3$   
 $= 6.9 \times 10^{-7} \text{ (m)}$   
 $= 690 \text{ nm}$  [3]
- (ii) • no change to fringe separation/fringe width/number of fringes  
 • bright fringes are darker  
 • dark fringes are brighter  
*Any two marking points, 1 mark each* [2]

**38. MAR 20/P22/Q4**

- (a) (i) distance moved by wavefront / energy during one cycle / vibration / oscillation / period (of source)  
 or  
minimum distance between two wavefronts  
 or  
 distance between two adjacent wavefronts [1]
- (ii) maximum displacement (of particle / point on wave) [1]

- (b) (i) 1 light / waves spread (at each slit) [1]  
 2 constant phase difference (between light / waves) [1]  
 (ii)  $n\lambda = d \sin \theta$   
 $d = 3 \times 650 \times 10^{-9} / \sin 34^\circ$   
 $d = 3.5 \times 10^{-6} \text{ m}$  [3]  
 (iii) wavelength of blue light is shorter (than 650 nm / red light)  
 so angle (between third order diffraction maxima) decreases [2]

**39. O/N 19/P22/Q5**

- (a) distance moved by wavefront/energy during one cycle/oscillation/period (of source)  
 or  
minimum distance between two wavefronts  
 or  
 distance between two adjacent wavefronts [1]  
 (b) ( $T=$ )  $2.0 \times 2.5$  (= 5.0 ms) or  $2.0 \times 2.5 \times 10^{-3}$  (=  $5.0 \times 10^{-3} \text{ s}$ )  
 $f = 1 / (5.0 \times 10^{-3})$   
 = 200 Hz [2]  
 (c) (i) (path difference =)  $8.0 + (20.8^2 - 8.0^2)^{0.5} - 20.8 = 6.4 \text{ (m)}$  [1]  
 (ii) • path difference =  $4\lambda$   
 • waves (meet at C) in phase  
 • constructive interference (of waves)  
*any two points, one mark each* [2]  
 (iii)  $v = 200 \times 1.6$   
 =  $320 \text{ (m s}^{-1}\text{)}$   
 $\Delta t = 6.4 / 320$  or  $27.2 / 320 - 20.8 / 320$   
 =  $0.020 \text{ s}$  [2]  
 (iv)  $3\lambda = 6.4$   
 $\lambda = 2.1 \text{ m}$  [1]

**40. O/N 19/P21/Q5**

- (a) (i) the dippers are connected to the same vibrator/motor [1]  
 (ii) (the overlapping waves have) similar/same amplitude [1]  
 (b) any means of 'freezing' the pattern e.g. use a stroboscope/strobe [1]  
 (c)  $vT = \lambda$   
 or  
 $v = f\lambda$  and  $f = 1 / T$   
 $T = 0.060 / 0.40$   
 =  $0.15 \text{ s}$  [2]  
 (d) (i) path difference =  $3.0 \text{ cm}$  [1]  
 (ii) phase difference =  $180^\circ$  [1]  
 (e) line drawn joining points where only maxima are observed (i.e. through points where wavefronts intersect) of length at least  $4 \text{ cm}$  [1]

**41. O/N 19/P23/Q5**

- (a) (i) (coherence means) constant phase difference (between waves) [1]  
 (ii) (interference is) the sum/addition/combination of the displacements of overlapping/meeting waves [1]  
 (b) (i)  $n\lambda = d \sin \theta$   
 $\lambda = \sin 51^\circ / (2 \times 6.7 \times 10^5)$   
 =  $5.8 \times 10^{-7} \text{ m}$  [2]  
 (ii) smaller angle (corresponding to second order maxima  
 and so) shorter distance (between second order maxima spots) [1]

**42. M/J 19/P22/Q4**

- (a) (i) distance (in a specified direction of particle/point on wave) from the equilibrium position [1]  
 (ii) the maximum distance (of particle/point on wave) from the equilibrium position  
 or  
 the maximum displacement (of particle/point on wave) [1]
- (b)  $I \propto A^2$   
 $I_R / I = (3.6 - 1.2)^2 / (1.2)^2$   
 resultant intensity =  $4.0I$  [2]
- (c) (i) as wave(s) pass through the slit(s)  
 wave(s) spread (into geometric shadow) [2]
- (ii)  $n\lambda = d \sin \theta$   
 $3\lambda = d \sin 90^\circ$  or  $3\lambda = d$   
 $d = 3 \times 630 \times 10^{-9}$   
 $= 1.9 \times 10^{-6} \text{ m}$  [3]
- (iii) wavelength of blue light is shorter (than 540 nm/630 nm/wavelengths of original light)  
 (so) third order diffraction maximum is produced [2]

**43. M/J 19/P21/Q5**

- (a) (i) 1.  $N\lambda$   
 2.  $N/f$  [2]  
 (ii)  $v$  (= distance / time) =  $N\lambda / (N/f)$  so  $v = f\lambda$  [1]
- (b)  $T = 4.0 \times 0.20 = 0.80 \text{ (ms)}$  or  $8.0 \times 10^{-4} \text{ (s)}$   
 $f = 1 / 8.0 \times 10^{-4}$   
 $= 1300 \text{ Hz}$  [2]
- (c) (i) constant phase difference (between the waves) [1]  
 (ii)  $180^\circ$  [1]  
 (iii) path difference =  $2\lambda$  or  $S_1Y - S_2Y = 2\lambda$   
 distance =  $7.40 + (0.85 \times 2)$   
 $= 9.1 \text{ m}$  [2]

**44. M/J 19/P23/Q5**

- (a) (incident) wave reflects at end/top of tube  
 (incident) wave and reflected wave interfere/superpose [2]
- (b) line has maximum value of amplitude at  $h = 0$  and  $h = 0.60 \text{ m}$  only  
 line has minimum/zero value of amplitude at  $h = 0.30 \text{ m}$  only [2]
- (c) (i) vertical/along length of tube/along axis of tube [1]  
 (ii) phase difference = 0 [1]
- (d)  $v = f\lambda$   
 $v = 340 / (2 \times 0.60)$   
 $= 280 \text{ Hz}$  [2]
- (e)  $f = 340 / 0.60$   
 $= 570 \text{ Hz}$  [1]

**45. MAR 19/P22/Q5**

- (a) (i) (two) waves meet/overlap (at a point)  
 (resultant) displacement is sum of the displacement of each wave [2]  
 (ii) constant phase difference (between the waves) [1]
- (b)  $I \propto A^2$   
 $3I / I = (A + 1.5)^2 / 1.5^2$   
 $A = 1.1 \text{ cm}$  [2]

(c) (i)  $\lambda = ax / D$

e.g.  $a = 680 \times 10^{-9} \times 2.0 / 4.0 \times 10^{-3}$

$a = 3.4 \times 10^{-4} \text{ m}$

[3]

(ii) straight line from positive value on x-axis and always below 'old' line

straight line with a smaller positive gradient than 'old' line

[2]

**46. O/N 18/P22/Q5**

(a)  $n\lambda = d \sin \theta$

$\lambda = 640 \times 10^{-9} \text{ (m)}$

$2 \times 640 \times 10^{-9} = 1.7 \times 10^{-6} \times \sin \theta$  so  $\theta = 49^\circ$

[3]

(b)  $2 \times 640 \times 10^{-9} = 3 \times \lambda$

or

$1.7 \times 10^{-6} \times \sin 49^\circ = 3 \times \lambda$

$\lambda = 4.3 \times 10^{-7} \text{ m}$

[2]

**47. O/N 18/P21/Q4**

(a) when (two or more) waves meet (at a point)

(resultant) displacement is the sum of the individual displacements

[2]

(b) (i) constant phase difference (between the waves)

[1]

(ii) 1. phase difference =  $360^\circ$  or 0

2. path difference =  $1.5\lambda$

$= 1.5 \times 610$

$= 920 \text{ nm}$

[2]

(iii)  $\lambda = ax / D$

$x = 22/4 (= 5.5 \text{ mm})$  or  $22 \times 10^{-3}/4 (= 5.5 \times 10^{-3} \text{ m})$

$a = (610 \times 10^{-9} \times 2.7) / (5.5 \times 10^{-3})$

$= 3.0 \times 10^{-4} \text{ m}$

[3]

(iv) shorter wavelength and (so) separation decreases

[1]

(v) • no change to fringe separation/fringe width/number of fringes

• bright fringes are brighter

• dark fringes are unchanged

Any two of the above three points, 1 mark each.

[2]

**48. M/J 18/P22/Q4**

(a) (i) distance moved by wavefront/energy during one cycle/oscillation/period (of source)

or

minimum distance between two wavefronts

or

distance between two adjacent wavefronts

[1]

(ii) (position where) maximum amplitude

[1]

(b) (i)  $\lambda = 4 \times 0.045$

$(= 0.18 \text{ (m) or } 18 \text{ (cm)})$

$v = f\lambda$

$f = 340 / 0.18$

$= 1900 \text{ Hz}$

[3]

(ii) distance =  $\lambda/2$

$$= 0.09(\text{m}) \text{ or } 9(\text{cm})$$

$$\text{time} = 0.09 / 0.0075$$

$$= 12 \text{ s}$$

or

$$t = 4.5 / 0.75 \text{ and } t = 13.5 / 0.75$$

$$\text{time} = 18 - 6$$

$$= 12 \text{ s}$$

[2]

#### 49. M/J 18/P21/Q5

- (a) (i) waves spread at (each) slit/gap [1]  
 (ii) constant phase difference (between (each of) the waves) [1]

- (b) (i)  $n\lambda = d \sin \theta$   
 $d \sin \theta$  is the same and  $3\lambda_1 = 4\lambda_2$  so  $\lambda_2 / \lambda_1 = 0.75$  [2]  
 (ii)  $\lambda_2 / \lambda_1 = 0.75$  and  $\lambda_1 - \lambda_2 = 170$   
 $\lambda_1 = 680 \text{ nm}$  [1]

#### 50. MAR 18/P22/Q4

- (a) (two) waves (travelling at same speed) in opposite directions overlap  
 (waves are same type and) have same frequency / wavelength [2]

- (b) (i)  $v = f\lambda$   
 $f = 330 / 0.18$   
 $= 1800 \text{ Hz (1830 Hz)}$  [2]

- (ii)  $T = 1 / 1800 (= 5.5 \times 10^{-4})$   
 time-base setting  $= (1.5 \times 5.5 \times 10^{-4}) / 8.0$  or  $1 / (1800 \times 5.3)$   
 $= 1.0 \times 10^{-4} \text{ s cm}^{-1}$  [2]

- (iii) waveform drawn with same period as original waveform  
 waveform drawn with amplitude of 1.7 cm [2]

- (c) (i) distance  $= \lambda / 2 = 0.18 / 2$   
 $= 0.090 \text{ m}$  [1]

- (ii) letter N shown at level B and at level A and not anywhere else. [1]

- (iii)  $m = \rho A x$   
 $= 0.79 \times 13 \times 9.0 (= 92.4)$  or  $790 \times 13 \times 10^{-4} \times 0.090 (= 0.0924)$   
 $t = 92.4 / 6.7$  or  $0.0924 / 0.0067$   
 $= 14 \text{ s}$  [2]

#### 51. O/N 17/P21/Q3

- (a) (i) in a stationary wave energy is not transferred  
 or  
 in a progressive wave energy is transferred [1]

- (ii) in a stationary wave (adjacent) particles are in phase  
 or  
 in a progressive wave (adjacent) particles are out of phase/have a phase difference/not in phase [1]

- (b) (i) (position where) maximum amplitude [1]

- (ii) distance  $= 0.10 \text{ m}$  [1]

- (iii) 1.  $\lambda = 0.60 / 1.5$   
 $= 0.40 \text{ m}$  [1]

2.  $v = f\lambda$   
 $f = 340 / 0.40$

$$= 850 \text{ Hz}$$

(iv)  $\lambda = 2 \times 0.60$  or  $\lambda = 3 \times 0.40$  or  $f = 850/3$   
 $f = 280$  (283) Hz

[2]

[2]

**52. M/J 17/P22/Q6**

(a) (i) waves at (each) slit/aperture spread

(into the geometric shadow) wave(s) overlap/superpose/sum/meet/intersect

[2]

(ii) there is not a constant phase difference/coherence (for two separate light source(s))  
 or

waves/light from the double slit are coherent/have a constant phase difference

[1]

(b)  $x = \lambda D / a$

$$\lambda = (36 \times 10^{-3} \times 0.48 \times 10^{-3}) / (16 \times 2.4)$$

$$= 4.5 \times 10^{-7} \text{ m}$$

[3]

(c) (i) no movement of the water/water is flat/no ripples/disturbance

the path difference is  $2.5\lambda$  or the phase difference is  $900^\circ$  or  $5\pi$  rad

[2]

(ii) 1. surface/water/P vibrates/ripples  
 and

as (waves from the two dippers) arrive in phase

2. surface/water/P vibrates/ripples  
 and

as amplitudes/displacements are no longer equal/do not cancel

[2]

**53. M/J 17/P21/Q4**

(a) (two) waves travelling (at same speed) in opposite directions overlap  
 waves (are same type and) have same frequency/wavelength

[2]

(b) (i)  $\lambda = 12 / 250$  (= 0.048 m)

$$\text{distance} = 1.5 \times 0.048$$

$$= 0.072 \text{ m}$$

[2]

(ii)  $T = 1 / 250$

$$= 0.004 \text{ (s) or } 4 \text{ (ms)}$$

1. curve drawn is mirror image of that in Fig. 4.2 and labelled P

2. horizontal line drawn between A and B and labelled Q

[3]

**54. M/J 17/P23/Q5**

(a) (i) waves at the elements/slits

waves spread (into the geometric shadow)

[2]

(ii) 1. waves (from each element/slit) overlap/meet/superpose  
 with a phase difference/path difference of zero

2. phase difference is  $360^\circ$ /path difference of  $\lambda$

[3]

(b) (i) e.g. gradient =  $(0.40 - 0.32) / [(500 - 400) \times 10^{-9}]$   
 $= 8(.0) \times 10^5$

[2]

(ii)  $d \sin \theta = n\lambda$

$$d = n / \text{gradient}$$

$$= 2 / 8.0 \times 10^5 = 2.5 \times 10^{-6} \text{ m}$$

[2]

(iii) straight line drawn with lower gradient (about  $\frac{1}{2}$ ) and all points lower

[1]

**55. O/N 16/P22/Q4**

- (a)** wave incident on/passes by or through an aperture/edge  
wave spreads (into geometrical shadow)

[2]

- (b) (i)** waves (from slits) overlap (at point X)  
path difference (from slits to X) is zero/  
phase difference (between the two waves) is zero  
(so constructive interference gives bright fringe)

[2]

- (ii)** difference in distances =  $\lambda/2 = 580/2$   
= 290 nm

[1]

- (iii)**  $\lambda = ax/D$

$$D = [0.41 \times 10^{-3} \times (2 \times 2.0 \times 10^{-3})] / 580 \times 10^{-9}$$

$$= 2.8 \text{ m}$$

[3]

- (iv)** same separation/fringe width/number of fringes  
bright fringe(s)/central bright fringe/(fringe at) X less bright  
dark fringe(s)/(fringe at) Y/(fringe at) Z brighter  
contrast between fringes decreases

*Any two of the above four points, 1 mark each*

[2]